Responsible Sourcing? Evidence from Costa Rica*

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Abstract

Multinational enterprises (MNEs) increasingly impose "Responsible Sourcing" (RS) standards on their suppliers, including requirements on worker compensation, benefits and working conditions. Are these policies just "hot air" or do they impact suppliers and their workers? What is the welfare incidence of RS in sourcing countries? To answer these questions, we combine the near-universe of RS rollouts by MNEs with subsidiaries in Costa Rica with firm-to-firm transactions and matched employer-employee microdata. We find that RS rollouts lead to reductions in sales and employment at exposed suppliers, salary increases for their low-wage workers and a decrease in their low-wage employment share. To rationalize these effects and study their implications in general equilibrium, we develop a simple open-economy model of RS. We show that the welfare effect of RS is ambiguous, depending on the interplay between an export tax (+) and a consumption tax (−), and that RS has distributional implications within worker types. Combining model and evidence for counterfactual analysis, we find that RS has delivered significant gains for low-wage workers at exposed suppliers, but led to adverse indirect effects on wages and the price index for the broader low-wage workforce.

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1 Introduction

In response to calls by policymakers and the general public for multinational enterprises (MNEs) to adopt more equitable production practices in low- and middle-income countries, MNEs increasingly impose "Responsible Sourcing" (RS) requirements on their suppliers worldwide (e.g., ILO, 2016). RS requirements mainly take the form of "Supplier Codes of Conduct" and include compulsory standards on working conditions (such as wage floors, guaranteed benefits, maximum working hours, paid leave and safety standards), other production practices (such as worker representation and environmental standards) and enforcement provisions (such as third-party auditing). Despite MNEs' growing adoption of RS policies, there is relatively limited theoretical work or empirical evidence on the economic incidence of these policies and their effectiveness in improving worker welfare in sourcing countries. In this paper, we combine data and theory to shed light on these questions. Our analysis focuses on an empirical setting where RS policies primarily affect working conditions, leaving other potential impacts of RS—such as those on the environment—for future research.

We build a unique new database that allows us to track the rollout of RS requirements by MNEs and trace their effects on suppliers and workers in a sourcing country. Our empirical context is Costa Rica (CR), a middle-income country that hosts hundreds of foreign MNE subsidiaries sourcing inputs locally across a diverse range of industries. We merge several administrative datasets covering the 2008–2019 period, including firm-to-firm transactions, matched employer-employee relationships, exports and imports, corporate tax returns, and foreign ownership, in addition to ORBIS data on the global outcomes of MNEs with subsidiaries in CR. We combine these data with a novel dataset constructed through a comprehensive double-blind search of all RS rollouts among 481 MNEs with subsidiaries sourcing in CR. We identify 238 RS rollouts by 169 MNEs targeted at improving labor standards at their suppliers over the period 2000–2019. The data confirm the growing reach of RS: by the end of our sample in 2019, 43% of all sales by private domestic firms in CR were made by suppliers to MNE subsidiaries subject to active RS codes.

Using text analyses of the RS policies, we document that labor-related requirements are the primary focus of RS policies in our sample, with most policies including requirements on wages and compensation, health and safety, and prohibitions on forced labor and discrimination. Compliance with these requirements is monitored by the MNE buyers and third-party auditors, and failure to comply is met with penalties: two thirds of RS policies in our sample specify penalties (such as contract termination). We also find that a baseline requirement, included in all RS codes in our sample, is compliance with existing local labor laws and regulations—though about one third of policies mandate supplier standards that go beyond these local laws. In a

context where local legal standards may be only weakly enforced by public regulators, even the least ambitious RS codes can have meaningful implications for suppliers when subjected to private enforcement by MNE buyers.

To provide evidence on the implications of RS, we then use this database to implement an event-study design that tracks changes in outcomes among suppliers selling to MNEs, comparing those whose MNE buyer rolled out its first RS code to those whose MNE buyers have yet to do so. We present alternative specifications, including using global MNE rollouts of RS as instruments for local RS adoption in CR. As supplier exposure to RS is based on positive sales to the MNE pre-rollout, we also include controls for the timeline of sales events to any MNE, or match suppliers based on pre-rollout MNE sales events, to address concerns that RS exposure correlates with positive sales shocks pre-rollout (positive pre-trends). We find that RS policies lead to a reduction in total sales and employment at exposed suppliers. Four years after the RS rollout, their sales decline by about 7.5% and employment by 6%. These declines are larger among suppliers in sectors with smaller firms and those exposed to RS policies with a greater emphasis on worker compensation, health and safety standards, and compliance and enforcement provisions. Sales and input demand of the MNE—both at the local subsidiary level and globally—do not display discernible pre-trends and remain largely unaffected by RS. Using the firm-to-firm transactions data, we find that exposed suppliers experience sales declines to both non-RS buyers and the RS-MNE.

Using the matched employer-employee data, we find that RS rollouts lead to a roughly 2% increase in the average monthly earnings of exposed workers at the supplier, driven by a 5.3% increase among those in the bottom quartile of initial earnings. The employment of low-wage workers declines at exposed suppliers, both in absolute and relative terms. We also find suggestive evidence that some workplace amenities increase, including a reduction in time spent on accident leave and an increase in the duration of maternity leave. Taken together, these findings suggest that RS requirements are, on average, not just "hot air", with effects on exposed suppliers and workers that indicate increases in labor-related costs concentrated among initially low-wage workers. By design, however, this reduced-form evidence would be insufficient to assess the implications of RS for domestic workers, as it only captures relative effects on exposed suppliers and workers in the wake of individual RS rollout events.

To rationalize these effects and derive welfare implications in general equilibrium (GE), we develop a parsimonious open-economy model. Heterogeneous firms in the sourcing country produce final goods for the domestic market and intermediate inputs for foreign-owned MNE subsidiaries in the country. Suppliers to RS-active MNEs face an increase in labor costs for low-wage workers but not for high-wage workers whose working conditions are unlikely to change

under RS. The RS standards apply at the firm level, to all of a supplier's production, including sales to non-RS buyers. We allow these cost increases to be only partially passed through to the MNE. A lower pass-through reflects greater buyer market power of the MNE. Since MNEs may adopt RS policies in response to consumer pressure, we allow MNEs to experience a demand increase after implementing RS practices. In this setting, we show that the adoption of RS has an *a priori* ambiguous impact on workers' welfare in the sourcing country. If all the output from RS-affected suppliers is exported and RS costs are fully borne by the RS-MNEs, RS policies function like an export tax, improving welfare through a classic terms-of-trade effect. However, if MNEs have buyer market power over their suppliers, RS costs are in part borne by the suppliers rather than acting as an export tax for foreigners, thereby limiting the welfare gain. Furthermore, as suppliers produce for both the export and domestic markets, RS policies "leak" into the domestic price index. This reduces welfare, as RS then acts akin to a consumption tax. Finally, when RS is accompanied by a positive demand shock for the MNE's output, its welfare effect becomes more favorable. In terms of distributional implications, RS generates symmetric welfare effects in GE for low- and high-wage workers at the aggregate level. However, there are meaningful distributional effects within worker groups. Low-wage workers at exposed suppliers directly benefit from improved working conditions under RS, while non-exposed low-wage workers experience only indirect GE adjustments to their wages and price index.

In the final part, we confront the model's comparative statics with the empirical evidence to quantify the model and conduct counterfactual analyses. Combining model with evidence, we find that RS rollouts on average led to a labor-cost increase of approximately 12.5% among low-wage workers, which was mostly passed through to the MNE (about 85%). We find no discernible demand boost for MNE output due to RS rollouts. Armed with the quantified model, we compare a counterfactual equilibrium without RS to one featuring the extent of RS policies observed at the end of our sample in 2019. We estimate that RS policies had positive but modest effects on aggregate welfare (+0.1%). These aggregate effects mask significant heterogeneity within worker groups: the 40% of low-wage workers employed at exposed MNE suppliers before rollout experience significant welfare gains (+6.5%), while the remaining majority of low-wage workers at non-exposed firms face welfare losses (-6%) due to adverse GE effects on their wages and the leakage of RS into the domestic price index. These findings underscore a trade-off involved in RS policies. On one hand, RS can bring meaningful benefits to workers at targeted suppliers. On the other hand, it imposes additional costs on suppliers, and its widespread adoption by MNEs can generate indirect adverse effects on non-targeted workers.

Our analysis contributes to a small but growing empirical literature on the effects of MNE sourcing policies on supplier and worker outcomes. Harrison and Scorse (2010) study the effect

of anti-sweatshop campaigns against MNE contractors in Indonesia's textile, footwear, and apparel sectors. Using a difference-in-differences design across sectors and regions, they find that campaigns led to higher wages, lower profits and firm exits. More recently, Boudreau (2024) uses a randomized controlled trial to study the impact of occupational safety and health committees at apparel producers in Bangladesh, and Amengual and Distelhorst (2020) study compliance with Gap Inc.'s labor code of conduct. Both studies find that RS requirements improve compliance with the law as well as workers' health and safety.[1] The paper also relates to the literature on the implications of "fair trade" certification (e.g., Dragusanu et al., 2022, De Janvry et al., 2015, Podhorsky, 2013, 2015). This literature has emphasized the role of fair trade in redistributing agricultural profits from imperfectly competitive intermediary wholesalers to farmers in developing countries (e.g., Dragusanu et al., 2022, Podhorsky, 2015). In our setting, RS requirements are chosen and implemented by the MNEs within their own supply chains. More recently, Macchiavello and Miquel-Florensa (2019) study a "buyer-driven" quality and sustainability upgrading program among coffee farmers in Colombia. Using a spatial regression discontinuity design, they find that eligible farmers improved the quality of their coffee and the program led to significant income gains.

Our analysis also relates to the minimum wage literature,[2] as in our model RS is akin to a targeted wage premium applied to firms supplying RS-MNEs. One focus of this literature has been to measure (often elusive) employment effects. In our context, we find that RS rollouts have significant employment effects among exposed workers. Theoretically, there are two distinct features of our setup. First, RS policies only apply to a selected set of firms, leading to excess labor supply among RS suppliers under a rationing equilibrium in the labor market. This leads to an increase in wage inequality (while a uniform minimum wage typically compresses the wage distribution, see, e.g., Lee, 1999, Autor et al., 2016) and gives rise to potential misallocation across firms. Second, we study RS in an open economy, while the focus of the minimum wage literature is typically on closed economies. A key finding in our setup is that RS may yield positive welfare gains through GE terms-of-trade effects that are specific to the open-economy context. This also connects to insights from an earlier literature on countries' labor standards and international

[1] Bossavie et al. (2023) study the effects of improvements in Bangladeshi labor regulations after the 2013 Rana Plaza factory collapse. They find that both working conditions and wages improved. Using Myanmar's opening to trade, Tanaka (2020) finds that exporting has positive effects on working conditions, including health and safety. On the theory side, Koenig et al. (2025) study the geography of NGO campaigns against unethical practices in a model of international trade. Herkenhoff and Krautheim (2022) introduce cost savings from unsafe practices in a model of global sourcing with incomplete contracts. Im and McLaren (2023a,b) propose a model of how globalization affects host countries' incentives to set labor standards. Datta and Machin (2024) examine the effects of UK local governments' procurement policies requiring suppliers to pay workers "living wages." They find that suppliers fully absorb the wage costs through employment reorganization and pay policy reform.

[2] Recent studies and reviews include Cengiz et al. (2019), Harasztosi and Lindner (2019), Dube (2019), Manning (2021).

trade: Brown et al. (1993) show that labor standards imposed on all firms may increase the welfare of exporting countries through a terms-of-trade effect.

The paper proceeds as follows. Section 2 presents the data and policy context, Section 3 the empirical evidence, Section 4 the theory and Section 5 the quantitative analysis. Section 6 concludes.

2 Data and context

Our analysis builds on three main databases. First, we integrate several administrative datasets that track all formal firm-, firm-to-firm-, and worker-by-firm-level outcomes in Costa Rica (CR). In these datasets, we are able to identify MNE subsidiaries in CR and trace their linkages to domestic suppliers and their workers. Second, we link them with external datasets that provide information about the global operations of these MNEs and events that may influence RS policies, such as NGO campaigns and leadership changes. Third, we build a new dataset that tracks RS policy rollouts and their contents among MNEs with subsidiaries in CR. We discuss each of these building blocks in turn and then describe the policy context of the empirical analysis.

2.1 Administrative data from Costa Rica

Firm-to-firm transaction data The dataset tracks the near-universe of formal firm-to-firm relationships in CR from 2008 to 2019.[3] This information is collected by the Ministry of Finance through the D-151 tax form. Each year, firms must report the tax identifiers of all their suppliers and buyers with whom they conduct transactions of at least 2.5 million Costa Rican colones (around 5,000 U.S. dollars), along with the total transaction amounts. We use this data to identify the suppliers exposed to a new RS policy of an MNE subsidiary in CR and to estimate its effects on various types of sales and sale transactions.

Matched employer-employee data Based on data from the Social Security Administration, we construct a panel of employment records for all formal workers in CR between 2006 and 2019. For each of the 1.9 million workers observed at least once, the data includes the employer identifier (consistent across firm-level datasets), monthly labor earnings, employment status (full-time or part-time), gender, occupation, and records of leave due to workplace accidents and maternity.

Other firm-level data We then incorporate yearly corporate income tax returns from the Ministry of Finance for all taxpaying firms in CR from 2008 to 2019. These returns include typical balance sheet variables such as total sales and employment, as well as the firm's primary 4-digit sector (out of a total of 375 4-digit sectors in CR). Additionally, we incorporate data from CR customs declarations on yearly firm-level imports and exports.

[3] See Alfaro-Ureña et al. (2022) and Alfaro-Ureña et al. (2021) for additional description of the administrative datasets.

MNE sample The analysis sample of MNEs consists of the 481 foreign-owned firms in CR that have average annual domestic input purchases exceeding 1 million U.S. dollars and average annual employment of more than 50 workers (averaged across all years the subsidiary has operated in CR).[4] These 481 subsidiaries account for 73% of the domestic input purchases and 79% of the employment by foreign-owned firms in the country (Appendix Table A1).

2.2 Global data on MNEs

Orbis data We link MNE subsidiaries in CR to historical Orbis panel data for their MNE groups. To measure global MNE sales, we use the consolidated turnover of the global ultimate owner of the subsidiaries in CR, as reported in Orbis' ownership module. This approach allows us to observe the global sales of 203 out of the 481 MNEs of interest from 2008 to 2019.[5]

Sigwatch data The consultancy firm Sigwatch collects comprehensive data on international consumer-facing NGO campaigns targeting MNE production practices, including their sourcing practices (Hatte and Koenig, 2020).[6] Between 2010 and 2020, Sigwatch recorded an average of over 10,000 NGO campaigns annually. After matching MNEs using their names and ISIN identifiers, we found that 190 out of the 481 MNEs of interest were targeted by these campaigns. We use this data to explore whether negative NGO campaigns against MNEs tend to precede their RS rollouts.

BoardEx data We combine data from BoardEx and Orbis to track leadership changes at the MNE group level. Both datasets provide information on the identity of all current and former executives in key leadership positions within the MNEs of interest, along with their appointment and resignation dates. From these combined datasets, we identify that 144 of the 481 MNEs of interest underwent leadership changes. We leverage this data to examine whether RS rollouts are systematically associated with prior leadership changes at the MNE level.

2.3 Responsible Sourcing (RS) policies

Data construction and analysis sample We build a database that tracks the RS policies implemented by the 481 MNEs with subsidiaries in CR we describe above. To do so, we employ a double-blind search process conducted by two independent research teams. Their findings are then cross-checked and consolidated into a single comprehensive database. For each MNE, we collect information on corporate social responsibility and supplier requirements (typically called a "Supplier Code of Conduct") by analyzing all publicly available company reports, press releases (both in CR and internationally), and corporate filings. This information is primarily found on

[4] We consider an MNE subsidiary as a firm with positive MNE ownership. Most subsidiaries are almost fully owned by the MNE (median ownership share is 95%).

[5] The matching rate with Orbis sales data is limited by its varying geographical coverage, which tends to be lower outside of Europe and North America.

[6] We thank Pamina Koenig for making the Sigwatch data available to us.

the websites of the MNE groups. For each policy, we record the year of implementation and note whether it applied to suppliers worldwide or was specific to CR or the Central American region. We search for RS rollout events without date restrictions, with the earliest event found in 2000. In our analysis, we focus specifically on RS codes that include provisions related to labor conditions, broadly defined.[7] The full search yields 238 RS policy rollouts related to labor conditions, implemented by 169 distinct MNEs. Of these 169 MNEs, 104 trigger a first-time RS exposure event for domestic suppliers in CR over the period of our estimation sample 2009–2019, as we discuss in more detail in Section 3. 89% of these 104 RS rollouts apply to all MNE suppliers globally.[8] Appendix Table A3 reports descriptive statistics across the sequence of sampling steps.

By combining the firm-to-firm transaction data with the data on RS rollouts by MNEs, we can measure the fraction of total production by domestic (non-MNE) firms in CR that are subject to active RS supplier codes with labor provisions.[9] This fraction rose from 30% in 2008 to 43% in 2019. The growing reach of RS in domestic production indicates that its effects could extend to firms and workers not directly affected. This motivates our analysis that combines partial-equilibrium comparative statics of exposed vs. non-exposed firms, as estimated in the data, with a model to investigate implications in GE.

Looking at descriptive statistics about MNEs implementing RS policies and their suppliers in CR, we see that subsidiaries of MNEs with RS policies are larger than MNE subsidiaries without RS and are more likely to be headquartered in the U.S. or Europe (Appendix Table A4). Exposed suppliers—defined as those supplying to MNE subsidiaries in the year before their first RS rollout—employ, on average, 18 workers. These suppliers span all sectors in the economy: 13% operate in manufacturing, 28% in retail and wholesale trade, 11% in agriculture and mining and 48% in services (Appendix Table A5).

Content of the RS policies We collect the code-of-conduct documents for each RS policy in our analysis sample. We use OpenAI's GPT-5 model, a large language model (LLM), to analyze the text of these documents and grade each document on several dimensions. Based on the extent of labor-related content in RS policy documents, labor standards emerge as the primary focus

[7] In practice, 68% of all RS policies we identified fall into this category. When we focus on first-time RS rollouts by MNEs, 81% include labor provisions. To further investigate the role of labor provisions, we run a placebo exercise that estimates the effects of RS codes without labor requirements and find no effects.

[8] Appendix Table A2 includes the full list of 169 MNEs with RS rollouts, with an asterisk (*) indicating the 104 MNE rollouts that trigger first-time RS exposure in our sample. Of these 104 RS rollouts, 103 involved the introduction of a new supplier code of conduct, while one involved launching a recurring workshop for suppliers. Of the remaining 65 MNEs that we exclude from the analysis sample (together with their suppliers), 56 implemented RS policies before 2009 (the first year when we can compute exposure of suppliers using the firm-to-firm transaction data), while 9 had only suppliers that already had experienced prior RS rollouts.

[9] The numerator is the sum of the total sales of all domestic firms that supply to an MNE subsidiary with an active RS code in a given year. The denominator is the sum of the total sales of all domestic firms in the same year.

of the typical RS policy in our sample.[10] Using a 1-to-10 scale (with 1 indicating no reference to labor standards and 10 indicating exclusive focus on them), the LLM assigns an average (median) labor-focus score of 6 (7) to documents in the sample (Appendix Table A6).[11] Next, we examine the most commonly included labor requirements. Panel A in Figure 1 reports the share of policies that explicitly require each of eight broad categories of labor practices: health & safety (95%), no discrimination (91%), no child/forced labor (89%), no harassment (79%), wages & compensation (76%), freedom of association (64%), work-hour caps (29%), and rest days (15%).[12]

A natural question is whether the RS documents merely offer a set of vague values and aspirations or actually impose specific and mandatory standards on suppliers. Panel B in Figure 1 reports the distribution of RS policies by overall stringency, defined as the average of three dimensions, each scored from 1 to 10: specificity of requirements, monitoring provisions, and consequences for non-compliance.[13] There is notable heterogeneity, with overall stringency scores ranging from 1.7 to 8.7, with a mean of 4.8 and a standard deviation of 1.5 (Appendix Table A8). Panel C reports the distribution of policies across three categories: those requiring only compliance with CR labor laws and regulations ("Local Only"), those setting standards that may exceed local laws ("May Exceed"), and those explicitly exceeding local minimums ("Explicitly Exceed"). 26% of policies are "Local Only", 44% "May Exceed" local laws, and 30% "Explicitly Exceed" local minimums. In particular, 41% of policies appear stricter than local health and safety standards and 38% stricter than local wage and compensation standards (Appendix Table A9). Finally, Panel D shows that two thirds of policy documents explicitly mention penalties (such as contract termination) faced by suppliers who fail to comply with the RS labor rules.

An example of an RS event in our sample is Ericsson's initial rollout of a Supplier Code of Conduct in 2009 (Ericsson, 2019). Based on its 4,193 words, the LLM scores Ericsson's policy as 8/10 in focus on labor conditions. It describes requirements for suppliers regarding their workers, including pay (minimum compliance with applicable laws or industry standards, whichever is higher), premium overtime pay, a maximum 48-hour workweek, freedom of association (e.g.,

[10] In a survey of 800 MNEs, The Economist (2017) finds that the most frequent requirements relate to labor conditions and compensation (workplace safety, working-hour limits, living wages, compensation for injury/sickness, and maternity/paternity and sick leave). The second most common category is environmental practices, such as waste reduction and recycling, pollution and climate change.

[11] For comparison, the equivalent focus score for environmental provisions is 3. In Section 3, we return to the potential role that environmental provisions may play in shaping the effects of RS rollouts in our estimation sample.

[12] Appendix Table A6 reports summary statistics on the number of "wages & compensation" commitments included in RS policies; Appendix Table A7 further disaggregates the "wages & compensation" and "no child/forced labor" categories.

[13] The scoring guide defines each scale as follows. Specificity: 1–3 for vague principles only (e.g., "fair wages" or "safe workplace"), 4–6 for some specific requirements but many left vague, and 7–10 for concrete, measurable standards (e.g., specific hours or procedures). Monitoring: 1–3 when no monitoring is mentioned, 4–6 for a general mention of audits or inspections, and 7–10 for detailed audit procedures. Consequences: 1–3 when no consequences are mentioned, 4–6 for vague consequences, and 7–10 for explicit consequences (e.g., contract termination).

union membership), and adoption of an Occupational Health and Safety (OHS) management system. The code is a mandatory component of all Ericsson supplier agreements. Verification of compliance includes regular supplier reporting and on-site visits by third-party auditors. Breaches, failure to take corrective actions, lack of remedies, or non-cooperation constitute grounds for termination of the contract by Ericsson.

2.4 Policy context

As shown in Figure 1, about a third of RS policies in our sample impose expectations that are more demanding than local labor laws and regulations—the remaining two thirds either require compliance with local laws or have requirements that may exceed local labor standards. If local laws and regulations are already well enforced, RS rollouts may be partially redundant. Conversely, if local laws and regulations are poorly enforced, even the least ambitious RS codes can have meaningful implications for suppliers, since private enforcement of RS may substitute for weak public oversight. Our empirical context resembles this latter scenario, as we outline below. Limited compliance with existing regulations in a context of weak state enforcement is not unique to CR, but has been found to be prevalent among low- and middle-income countries (e.g., Harrison et al., 2003, Ronconi, 2019).

On paper, CR's labor standards are high. The statutory minimum wage for low-skill occupations is approximately 70% of the median wage, a higher ratio than in any OECD country (OECD, 2017). Moreover, Social Security contributions, at 36.5% of gross salaries, exceed the OECD average of 27.2%. Employer contributions are particularly high at 26.3%, compared to the OECD average of 17.7% (OECD, 2016). This pattern aligns with the observation that low- and middle-income countries frequently have stricter employment laws and regulations than high-income economies (World Bank, 2020).

In practice, state enforcement of labor standards in CR has been weak. Inspectors employed by the Ministry of Labor are responsible for investigating violations related to minimum wages, Social Security contributions, occupational safety standards, insurance payments, maternity leave, holidays, overtime pay, working-time regulations, and health and safety standards. In 2015, CR employed 92 such labor inspectors (0.4 per 10,000 employees), reportedly operating with limited resources (Gindling et al., 2015). According to the Ministry of Labor, staffing levels are insufficient to inspect even 10% of firms, leaving especially small and medium enterprises largely unregulated (La Nación, 2009). For instance, household survey data from 2017 show that a quarter of formal private-sector employees earned less than the legal minimum wage (OECD, 2017). Enforcement of RS requirements by MNEs is likely to be, on average, significantly more effective than the *status quo* of state regulators, due to both MNE resources and their

incentives not to be caught misrepresenting sourcing practices. In line with the content analysis of the RS codes above,[14] a survey of 800 MNEs (The Economist, 2017) documents that they combine quarterly supplier reports with regular on-site inspections to ensure compliance, mostly contracted to third-party auditors. The survey also reveals that over one-third of MNE RS policies are more demanding than government laws and regulations, with this proportion being higher among larger MNEs.

The many detailed assessment and compliance provisions included in RS codes could still, in principle, just be "hot air". Next, we provide empirical evidence on the average and heterogeneous effects of RS rollouts to investigate these questions.

3 Effects of RS rollouts on suppliers and workers

We use the data described above to provide evidence on the effects of RS rollouts by MNEs on domestic suppliers and their workers in CR.

3.1 Empirical strategy

Supplier-level specifications We estimate event-study specifications of the following form:

$$y_{ist} = \alpha_i + \gamma_{st} + \sum_{\eta=k_l}^{\eta=k_u} \beta_\eta I(\text{Years since RS}_{it} = \eta) + \epsilon_{ist}\,, \tag{1}$$

where y_{ist} is an outcome (e.g., log firm sales or log employment) of firm i from the sample of firms that are suppliers to MNE subsidiaries in CR at some point during the period 2008–2019. The subscript s indexes the 375 4-digit sectors in CR and t indexes years. α_i are firm fixed effects and γ_{st} are sector-by-year fixed effects. The term $\sum_{\eta=k_l}^{\eta=k_u} \beta_\eta I(\text{Years since RS}_{it} = \eta)$ represents the event-study design for first-time RS exposure: $I(\cdot)$ is an indicator function and η indexes the number of years before or after the rollout of an RS policy by the MNE that firm i is exposed to.

In line with the theory we develop in Section 4, exposed (or "treated") suppliers are those selling to an MNE in the year before the first RS policy rollout of that MNE (at $\eta = -1$). Given our sample period (2008–2019), treated suppliers must be exposed to an RS rollout that occurred between 2009 and 2019.[15] Moreover, treated suppliers are restricted to those that have not sold to any RS-active MNE before the rollout event. The baseline estimation sample thus includes first-time treated MNE suppliers and never-treated MNE suppliers (firms that are not exposed between 2009 and 2019 and have not sold to an RS-active MNE). To adjust for firm-level autocorrelation

[14] E.g., Walmart's "Responsible Sourcing Audit and Assessment Expectations and Guidance" requires suppliers to take responsibility for "choosing an appropriate third-party audit program from the list of Walmart-approved programs, arranging, paying for, obtaining and cooperating in the audit, submitting the valid audit report to Walmart within the specified timelines" (Walmart, 2024).

[15] 85% of treated suppliers experience a single exposure to one of the RS policies rolled out between 2009 and 2019. Below, we also report very similar results after excluding suppliers with multiple treatments.

across years, we cluster the standard errors of $\hat{\beta}_\eta$ at the supplier level. In the results below, we estimate and report the event studies both before and after pooling the point estimate for $\eta \geq 4$ and $\eta \leq -4$ side by side (i.e., showing the point estimates for $\eta = 4$ and $\eta = -4$ in a specification without pooling, and the point estimates for $\eta \geq 4$ and $\eta \leq -4$ in a separate specification with pooling to document longer-run effects).[16]

The primary identification concern for estimating the β_η coefficients is that the timing of the RS rollouts may not be as good as random from the perspective of CR suppliers. For instance, MNEs may have rolled out their RS policies during periods when CR suppliers faced contemporaneous productivity shocks. We address this concern in several ways. First, to mitigate concerns about different time trends across firm types, we restrict the estimation sample to CR firms that we observe supplying MNE subsidiaries in CR at least once between 2008 and 2019. Second, we assess the likelihood of potential confounding shocks that may have preceded the MNE's RS rollout decision using the event-study design to estimate the β_η coefficients both before and after the rollout. A related concern could be that firms exposed to RS earlier (e.g., around 2010) may experience systematically different dynamic effects (timelines of treatment effects) compared to firms exposed in later years (e.g., 2015–2019). In addition to the standard two-way fixed effects event-study estimates, we thus report estimates from the procedure developed by Sun and Abraham (2021), estimating the weighted average timeline of effects across treatment cohorts, with weights equal to the cohort's sample size.

Supplier sales to MNEs can fluctuate for various reasons, such that defining exposure to RS based on a positive MNE sales event in year $\eta = -1$ may in part capture particularly successful ("lucky") periods for the exposed suppliers. This lumpy nature of sales could create positive pre-trends. To address this concern, we estimate the specification in equation (1) both before and after including an additional set of event-study indicators, $\sum_{\tilde{\eta}=k_l}^{\tilde{\eta}=k_u} \delta_{\tilde{\eta}} I(\text{Years since MNE}_{it} = \tilde{\eta})$, where the $\tilde{\eta}$ years mirror the RS convention (the year of positive MNE sales happens at $\tilde{\eta} = -1$), and the MNE sales event-study dummy switches to 1 for all CR suppliers with positive sales to any MNE subsidiary in CR. So in event year $\eta = -1$ (one year before the RS policy was rolled out), the RS event-study dummy, $I(\text{Years since RS}_{it} = -1)$, equals 1 among exposed suppliers, while among non-exposed suppliers in the control group the event-study dummy for MNE sales events, $I(\text{Years since MNE}_{it} = -1)$, equals 1 among those suppliers who record positive sales to any other MNE. When including these additional event-study terms in equation (1), we thus estimate the event-study effects of RS rollouts among exposed CR suppliers, while controlling for the full timeline of potential effects that may stem from having had positive sales to any MNE.

[16] We include all periods η observed during the sample period (i.e., $k_l = -11$ and $k_u = 10$), except for the omitted period at $\eta = -1$. Appendix Figure B1 provides the fraction of treated suppliers covered across the η timeline.

As an alternative approach, we estimate the event study after matching exposed suppliers to suppliers in the control group with positive MNE sales in the same year as exposed suppliers' $\eta = -1$. Finally, we present the event-study estimates using only global RS rollouts (decided by the MNE headquarters to apply to all suppliers globally) as instruments for rollouts affecting CR suppliers. Employing global RS rollout decisions as an IV addresses the concern that rollouts may have been timed to coincide with contemporaneous shocks to the CR suppliers (not captured by pre-trends).

Worker-level specifications To estimate the effects of RS rollouts at the worker level, we use the matched employer-employee data to estimate event-study specifications of the following form:

$$y_{ijst} = \alpha_{ij} + \gamma_{st} + \sum_{\eta=k_l}^{\eta=k_u} \beta_\eta I(\text{Years since RS}_{jt} = \eta) + \epsilon_{ijst}\,, \tag{2}$$

where α_{ij} now indicate fixed effects for worker (i)-by-firm (j) pairs. γ_{st} and $\sum_{\eta=k_l}^{\eta=k_u} \beta_\eta I(\text{Years since RS}_{jt} = \eta)$ remain defined as the sector-by-year fixed effects of the employing firm j and the firm j-level RS event-study terms. As above, we estimate this specification before and after including the full timeline of effects of having sold to any MNE, using the IV specification and the estimation procedure by Sun and Abraham (2021). We cluster the standard errors at the worker level (denoted by i here). For outcome y_{ijst}, we focus on log monthly earnings, calculated as annual labor earnings divided by months of employment. The estimation sample includes all workers who have ever worked at suppliers of MNEs. Given the α_{ij} fixed effects, the β_η estimates capture potential changes in the labor earnings of continuing workers at exposed suppliers, relative to continuing workers at non-exposed suppliers.[17]

MNE-level specifications To investigate the MNE-level context of RS rollout decisions, which matters for the interpretation of the supplier effects, we estimate event studies with MNE outcomes as the dependent variable in equation (1). Firm i is now a subsidiary in CR of an MNE rolling out RS for the first time, and the event-study timeline $I(\text{Years since RS}_{it} = \eta)$ tracks MNE subsidiary outcomes in the years before and after this initial RS rollout. The estimation sample includes all 481 MNEs. The MNE-level specifications allow us to determine whether RS rollouts are preceded by positive or negative MNE-level trends or followed by significant effects on MNE output and input demand. We also match the MNE subsidiaries in our sample to panel data from Orbis on the global outcomes of their parent MNE groups. This allows us to check if the total sales of the MNE subsidiary in CR respond differently relative to global MNE sales, shedding light on potential substitution across sourcing countries. We cluster the standard errors at the MNE level

[17] We do not separately observe monthly hours worked, but we only include full-time employees.

in these specifications.

To investigate whether RS rollout decisions could be correlated with other MNE-level shocks that may not be apparent in the estimated pre-trends, we use the BoardEx database to test whether RS rollouts occur after changes in various leadership positions within the MNE. Finally, we assess a remaining concern that RS rollout decisions may coincide with shocks to the likelihood of negative publicity events, which could violate the (post-treatment) parallel trends assumption between RS-MNEs and MNEs sourcing from suppliers in the control group. To this end, we make use of the matched Sigwatch database on NGO campaigns targeting MNEs. We also revisit and evaluate the implications of violations of the MNE-level parallel trends assumption as part of the theory and counterfactual analysis.

Transaction-level specifications To estimate the effect of RS on the intensive margin of sales to the MNE, we estimate event-study specifications at the firm-to-firm transaction level. We estimate the same specification as in equation (2), where y_{ijst} is the log transaction amount sold by supplier i to buyer j. α_{ij} is thus supplier-by-buyer fixed effects. Given the bilateral nature of the transaction data, we also include both supplier i's sector-by-year and buyer j's sector-by-year fixed effects. Our estimation sample includes sales transactions for the same sample of suppliers described above. For exposed suppliers, we include only transactions with the MNEs that later become RS-active (the MNE triggering their event) to avoid capturing the effects of RS on the sales to other non-RS buyers. For non-exposed suppliers, we keep all sales transactions with MNEs. All other aspects of the regression remain the same as in the analogous worker-level specifications. We also use the transaction-level database to compute suppliers' total sales to non-RS-active buyers in each period, and estimate specification (1) with this outcome on the left-hand side.

3.2 Results

Suppliers We first investigate the effect of RS on total supplier sales and employment. For each outcome, columns 1 and 5 of Table 1 present the two-way fixed effect specification with firm and sector-by-year fixed effects. Columns 2 and 6 add the flexible controls for having sold to any MNE in the run-up to RS exposure. Columns 3 and 7 present the same specification as in columns 2 and 6, but are estimated using the method proposed by Sun and Abraham (2021). Columns 4 and 8 present the same specification as in columns 2 and 6, but with the treatment event dummies instrumented using only RS rollouts that were global in nature. Each specification reports both the baseline event-study point estimates without pooling effects at $\eta = -4$ or $\eta = 4$, as well as the point estimates for $\eta \leq -4$ and $\eta \geq 4$ from a separate event-study regression where we pool the longer-term effects (while keeping all else unchanged).[18] Panels A and B of Figure 2 plot the

[18] Appendix Figure B1 includes point estimates across wider event-study timelines.

respective point estimates for the IV specification, corresponding to columns 4 and 8 in Table 1. In the remainder of the analysis, we focus on event-study figures with point estimates from the IV specifications, but we present the companion tables with all specifications in Appendix C.[19]

According to the IV specification, the total sales of exposed suppliers decrease by, on average, 7.4% 4 years after the first exposure to an RS rollout and by 10% when pooling 4 years or more after the event. This is accompanied by a fall in employment of 5.8% 4 years after and 9.6% after 4 years and beyond.[20] For both outcomes, the two-way fixed effects, Sun-and-Abraham and IV specifications in columns 2–4 (6–8) of Table 1 yield similar point estimates. This suggests that heterogeneity in dynamic adjustments across treatment cohorts or the MNEs targeting the timing of RS rollouts to CR-specific shocks are unlikely confounders in our empirical setting.[21] The concern of mechanical positive pre-trends and negative post-trends is apparent in column 1, where we do not control for having sold to any MNE in event year $\eta = -1$. After we include the event timeline for having sold to any MNE, pre-trends disappear. Appendix Figure B5 presents similar event-study estimates from an alternative matching approach. Instead of including flexible controls for MNE sales histories, we match exposed suppliers to their nearest control suppliers in the same 4-digit sector based on pre-event size and conditioning on positive sales to an MNE in the same event year as the exposed firm's $\eta = -1$.[22]

Heterogeneity To further explore these effects, we study heterogeneity in the supplier sales response by characteristics of the supplier, RS policy and the MNEs. In Panels A–F of Figure 3, we find that the negative effect of RS exposure on supplier sales is larger for: policies requiring above-minimum-wage pay (Panel A);[23] suppliers in sectors with smaller firms (Panel B); RS policies with a stronger overall focus on labor-related provisions (higher labor scores discussed in the previous section) (Panel C); and RS policies with a higher share of word counts related to compensation (Panel D), workforce health and labor safety standards (Panel E), and compliance and enforcement (Panel F). We also find that the negative sales effects are concentrated among

[19] Appendix Table C13 reports a sensitivity analysis to violations of parallel trends following Rambachan and Roth (2023) for every event-study estimate in the paper whose baseline $\eta \geq 4$ estimate is statistically significant at the 5% level.

[20] Appendix Figure B2 shows that the worker separation rate at exposed suppliers rises after RS exposure. The magnitude is in line with the employment effect.

[21] Appendix Figure B3 presents very similar results after excluding suppliers with more than one RS exposure event. We also examine whether some of the negative effects could be attributed to exposed suppliers splitting their production into separate entities. Appendix Figure B4 shows that groups of workers from exposed suppliers are not more likely to move jointly to new entities.

[22] We also assess the potential concern of spillover effects on the control group. The average RS policy in our sample exposes 20 suppliers, which span multiple input sectors. Appendix Figure B6 reports very similar results after excluding RS policies (and their exposed suppliers) that expose more than 100, 150 or 200 suppliers. The figure also reports similar results after replacing the 4-digit sector-by-year with 2-digit sector-by-year fixed effects.

[23] The average exposed supplier has 15% of worker-months at or below the minimum wage in the social security database before RS exposure.

suppliers in services, and are more pronounced for suppliers linked to MNEs headquartered in countries with higher average firm management scores (Appendix Figure B7). The corresponding heterogeneity analyses for supplier-level employment and worker-level earnings by worker group mirror the heterogeneity of the sales effects (Appendix Figures B8, B9, and B7). In line with the policy context described in Section 2, these findings suggest a setting in which RS policies affect domestic suppliers through labor-related provisions that vary in stringency and enforcement, with stronger effects on suppliers in less-regulated segments of the domestic economy.

Workers We use specification (2) to estimate the worker-level effects of RS rollouts at exposed suppliers. Figure 4 plots the event-study coefficients from the IV specification. Panel A of Figure 4 presents the results including all workers who, at some point during the sample period, were employed by a supplier to an MNE. Panel B of Figure 4 breaks up the average effect into three worker groups: those initially in the bottom quartile of monthly earnings, the top quartile of monthly earnings and the middle group comprising workers between these two quartiles.[24]

As shown in Panel A of Figure 4, average workers' earnings increase by 1.7% 4 years after first-time RS exposure and 1.9% 4 years or more post-rollout. In Panel B of Figure 4, we find that the effect is concentrated among workers in the bottom quartile of initial earnings, whose monthly earnings increase by, on average, 5.4% 4 years after the rollout and by 5.3% 4 years or more post-rollout. We find insignificant and close to zero point estimates for the effect of RS on the initially high-earning workers and a weaker effect on workers in the middle group.[25]

In line with these effects, we find that the supplier-level employment reduction is driven by workers in the bottom quartile of initial earnings, with a decline (-8% for $\eta \geq 4$) in their relative employment share (Panels C and D of Figure 4). Given the possibility of employing zero workers classified in the low-wage group, we also use PPML estimation to account for the extensive margin. The reduction in relative employment grows to -14.1% 4 years or more post-rollout (column 5 in Appendix Table C1).

MNEs Next, we study the effect of RS rollouts on the MNEs themselves. Figure 5 presents the MNE-level event-study results, showing the impact of RS on the sales and employment of the MNE subsidiary in CR (Panel A) and the subsidiary's share of global MNE sales or employment (Panel B).[26] We find no discernible effect on the total sales or employment of the MNE subsidiary in CR,

[24] To categorize workers, we use their average monthly earnings during the first year they appear in the data (starting in 2006). For workers observed for the first time after 2006, we adjust their initial monthly earnings for inflation using the annual consumer price index in CR.

[25] When defining exposure at the worker level and replacing worker-by-firm fixed effects with worker fixed effects in (2), we find an attenuated earnings effect for low-earning workers (Appendix Figure B10). This would be expected if separated workers get hired at their previous non-RS market wage.

[26] Given concerns about transfer pricing by MNEs, effects on employment serve as a useful comparison.

nor on its sales and employment as a share of global MNE sales and employment.[27] Moreover, we find no evidence of significant pre-trends at the MNE level. In Panel C, we investigate whether RS rollout decisions by MNEs are preceded by or coincide with changes in MNE leadership positions, as they could potentially cause a break in trends even if pre-trends are parallel in the data. Using the BoardEx and Orbis data described in Section 2, we find no evidence to support this concern across different types of leadership positions.

The results in Panels A–C of Figure 5 suggest that RS rollout decisions may be motivated by longer-term considerations, such as insurance against rare tail events due to negative publicity campaigns by prominent NGOs in the future. Since the timing of such forward-looking investments by the MNE is plausibly unrelated to the short- to medium-term shocks to MNE sales within the roughly 4-year post-rollout timeframe that we are able to capture in our event studies, this scenario would not violate the parallel trends assumption in our setting. But a plausible remaining concern is that RS rollout decisions may coincide with an unexpected increase in the probability of negative publicity events in the short term (relative to MNEs in the control group). This would imply that, in the absence of RS rollouts, treated MNEs could have faced more negative changes in output and input demand compared to the control group—introducing a negative bias in the statistically insignificant results from Panel A of Figure 5.

We address this concern in three ways. First, we use the Sigwatch data to investigate whether RS rollouts by MNEs are preceded by, or coincide with, negative NGO campaigns. Panel D presents estimation results of an event study with the indicator for RS rollout decisions on the left-hand side and an event-study timeline for negative NGO campaigns targeted at the MNE on the right. We find no evidence that NGO campaigns precede or coincide with RS rollout decisions. Second, we verify the extent to which negative publicity campaigns appear to be rare ("tail") events in the data. We use the Sigwatch data to compute the incidence of negative campaigns by prominent NGOs across MNE-year observations in our database. The fraction of all MNE-year observations in our database with such adverse campaigns is about 3%. Since not every NGO campaign meaningfully affects MNE outcomes, this is likely an upper bound for the arrival rate of negative RS-related events. The rarity of such publicity shocks adds some further reassurance on the *ex ante* likelihood of confounding negative MNE sales shocks in the immediate aftermath of RS rollouts.

Third, we revisit this concern as part of both the theory and quantification. As part of the counterfactual analysis, we verify the robustness of our results to this potentially remaining

[27] While the point estimates in Panels A and B are close to zero 4 years post-rollout and beyond, the confidence intervals are relatively large. We will also address the possibility of positive MNE-level sales effects as part of both the theory and the quantitative analysis.

negative MNE-level bias. Specifically, such bias would mean that RS has additional positive effects on MNE sales that the empirical evidence in Figure 5 may have missed. Through the lens of our model, such positive effects of RS on MNE sales and input demand are rationalized by positive shocks to MNE demand from consumers due to MNEs' RS announcements. We can thus explore the sensitivity of the counterfactual analysis to this concern by allowing for larger RS-induced MNE demand shocks relative to our baseline estimation in Section 5, which is based on our estimation in Figure 5.

Transactions The transaction-level version of the specification in equation (2) estimates the effects of RS rollouts by MNEs j on the average transaction sales amount among their continuing suppliers i. Panel A of Figure 6 plots the event-study coefficients from the IV specification. We find that, on average, the compliers' sales to the RS-MNE decline by 5.4% 4 years post-rollout and by 7.6% 4 years or more post-rollout. Panel B of Figure 6 uses the transaction data to estimate the effect of RS exposure on supplier sales to other (non-RS) buyers. Sales to other buyers decline significantly, falling, on average, by 6% 4 years post-exposure and 9.7% 4 years or more. These findings are consistent with the text-based analysis of RS codes of conduct described above, where RS standards are set at the firm level, affecting all of a supplier's production.

The transaction-level evidence adds information about the extent to which the effect on suppliers' total sales in Figure 2 is driven by suppliers adopting the RS standards vs. ceasing to sell to the MNE buyer on the extensive margin. From the results above, suppliers who continue selling to the MNE post-rollout experience an average decline of sales to the MNE of about 7.6% and a 9.7% reduction among domestic buyers. The average pre-rollout shares of sales between those two markets are 25% and 75%, respectively, implying that, to a first order, the average effect on total sales among continuing MNE suppliers is about 9.2%. Comparing this effect among continuing suppliers to the estimated effect on total sales of exposed suppliers (-10%), thus, points to a limited role for the extensive margin in driving the overall effect on suppliers. A loss of the average pre-rollout sales share to the MNE when dropping the relationship would imply that 5% of exposed suppliers decide to stop selling to the MNE post-rollout.[28]

In line with the evidence pointing to a limited extensive margin, Appendix Figure B11 presents a statistical zero for the effect on the number of sellers to the MNE.[29] The model we propose in Section 4 formalizes the different margins of adjustment, and Section 5 combines the model's comparative statics with the empirical evidence to quantify the underlying parameters.

[28] $-.1 = (1-\alpha) \times (-.092) + \alpha \times (-.25)$, where α is the share of suppliers who decide to stop selling to the MNE.

[29] The figure also documents that there is no entry margin: we find no evidence that the number of new sellers to the MNE is increasing post-rollout.

Workplace amenities and environmental practices While workplace amenities are typically hard to measure, our data also allow us to explore the effects of RS rollouts on some of them (Appendix B.1). We find suggestive evidence that some workplace amenities increase, including a reduction in time spent on accident leave and an increase in the duration of maternity leave. We also find no effects on measures of supplier environmental practices or in a placebo specification using codes of conduct without labor-related standards, suggesting that the effects of RS rollouts in our context are primarily driven by the labor-related provisions that are the main focus of RS codes in our estimation sample. We return to these questions as part of the model and counterfactual analysis, where we consider alternative assumptions about the nature of the RS-induced increase in supplier costs.

Benchmarking RS effects against MNE exposure To put these effects in perspective, we compare them with the existing evidence on the firm- and worker-level effects of joining MNE supply chains. Using the same CR microdata, Alfaro-Ureña et al. (2022) find that first-time suppliers to MNEs experience increases in total sales of about 33% and in employment of 26% over a four-year period. The RS-induced declines that we document here among existing suppliers to MNEs—7 to 10% in sales and 6 to 10% in employment—would amount to roughly one quarter to one third of those supplier-level gains. On the worker side, we find that RS raises continuing workers' earnings by about 1.7% on average, and by 5.4% for initially low-earning workers. These findings suggest that RS redistributes part, but not all, of the supplier-level gains from joining MNE supply chains to workers—and to initially lower-paid workers in particular.

4 A simple open-economy model of Responsible Sourcing

The empirical analysis suggests that MNE rollouts of RS codes are, on average, not just "hot air." Instead, the effects on exposed suppliers and workers are qualitatively consistent with an increase in labor-related costs that are concentrated among initially low-wage workers. To rationalize these effects and shed light on their welfare implications in a setting where RS affects a meaningful share of domestic production, this section develops a parsimonious open-economy model. There are two countries: Home (CR in our analysis) and Foreign (the rest of the world). MNEs are headquartered in Foreign. Heterogeneous local firms in Home can supply intermediate inputs to an MNE, as well as produce final goods for their domestic market. MNEs may implement RS policies—potentially in response to demand from Foreign consumers—that raise labor costs for their suppliers. Appendix D provides more detailed model derivations.

4.1 Setup

The final demand by consumers in country $i \in \{H, F\}$ (for Home, Foreign) is CES across goods ω with elasticity of substitution $\sigma > 1$:

$$U_i = \left(\int_{\Omega_i} d_{\omega,i}^{\frac{1}{\sigma}} q_{\omega,i}^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}, \tag{3}$$

where Ω_i is the set of goods for final consumption in country i, $q_{\omega,i}$ denotes consumption of ω and $d_{\omega,i}$ is a demand shifter for ω. Final goods are produced by either "MNEs" or "firms."

Firms A fixed mass of local firms in country i can produce final goods for the domestic market, and firms in Home can produce intermediate inputs for Foreign MNEs. Firms have heterogeneous productivities z following a Pareto distribution with parameter $\theta > \sigma - 1$ and minimum $\underline{z}_i$.[30] They use labor as the sole factor of production, combining the output of two types of workers (low- and high-wage workers, indexed by $t \in \{l, h\}$) as follows:

$$\ell = \left[\alpha_i^l \left(\ell^l \right)^{\frac{\rho-1}{\rho}} + \alpha_i^h \left(\ell^h \right)^{\frac{\rho-1}{\rho}} \right]^{\frac{\rho}{\rho-1}}, \tag{4}$$

with the corresponding labor cost index denoted as W. In equation (4), ℓ^t denotes type-t labor and α_i^t its relative demand in country i. Firms pay a fixed cost to produce and then have constant marginal costs. The labor required to produce q_ω units of final variety and m_ω units of intermediate inputs for MNE x are, respectively:

$$\ell_\omega = \frac{q_\omega}{z} + f_{ii} \text{ for } \omega \in \Omega_i \quad \text{and} \quad \ell_\omega = \frac{m_\omega}{z} + f_M \text{ for } \omega \in \Omega_x, \tag{5}$$

where f_{ii} (f_M, respectively) is the fixed cost to produce a final variety in country i (an MNE-specific intermediate input, respectively). Fixed costs in production lead to firm selection, à la Melitz (2003). We assume that fixed costs are ordered such that Home firms supplying MNEs are more productive than firms only serving the domestic market, as in the data.

MNEs There is a fixed mass of (homogeneous) foreign MNEs. Each MNE x produces through a subsidiary at Home that combines local labor ℓ (as in equation (4)) with local intermediate inputs. The CES production function of each MNE is:

$$M = \left(\nu^\ell \ell^{\frac{\sigma-1}{\sigma}} + \nu^m \int_{\Omega_x} m_\omega^{\frac{\sigma-1}{\sigma}} d\omega \right)^{\frac{\sigma}{\sigma-1}}, \tag{6}$$

[30] The productivity distribution in country i is $G_i(z) = 1 - (z/\underline{z}_i)^{-\theta}$.

where m_ω measures the use of intermediate input $\omega \in \Omega_x$ and ν^ℓ, ν^m are input shifters. The corresponding CES cost index is denoted R. Output M is then imported by the headquarters, subject to an iceberg trade cost ϱ. For parsimony we use the same σ for the MNE's elasticity of substitution between labor and intermediates and for the consumer's elasticity of substitution across goods. Appendix D relaxes this assumption.

Trade and market structure Trade patterns are simple: only MNE subsidiaries export from Home to Foreign, while only non-MNE firms export from Foreign to Home. Product markets are monopolistically competitive in the baseline. Workers collectively own domestic firms, with profits distributed proportionally to labor income.

4.2 Responsible Sourcing (RS)

An MNE may decide to implement RS, imposing higher labor standards on its suppliers.

RS wage premium We model the RS standards as higher monetary wages for workers: suppliers to MNEs with RS policies must pay their workers at a premium compared to non-RS firms. In line with Sections 2 and 3, we assume these wage increases are binding only for low-wage workers and they apply to all firm production. Formally, the wage $w^t_{H,r}$ offered to type-t workers by RS vs. non-RS firms (indicated by $r \in \{R, N\}$) is:

$$w^t_{H,R} = \tau^t_R w^t_{H,N} \quad \text{for } t = h, l, \tag{7}$$

where $\tau^l_R > 1$ for low-wage workers and $\tau^h_R = 1$ for high-wage workers. In the model, RS is thus akin to a targeted minimum wage, applied only in firms that sell intermediate inputs to RS-MNEs.[31] As RS firms pay a higher wage than the prevailing market wage at non-RS firms, there is excess labor supply for RS firms, and the labor market is under a rationing equilibrium. RS suppliers hire along their labor demand curve. Workers not hired by RS firms are employed by non-RS firms at a wage $w^t_{H,N}$, which adjusts in general equilibrium to ensure labor market clearing.

RS demand shock One may wonder why MNEs would implement policies that increase their suppliers' costs. The literature on Corporate Social Responsibility (CSR) (e.g., Bénabou and Tirole, 2010) typically considers two views on why firms engage in costly CSR activities.[32] These activities could reflect the preferences of altruistic managers. Alternatively, they could be a profit-enhancing

[31] Writing the RS premium as proportional is without loss of generality for our purposes. An additive premium, $w^l_{H,R} = w^l_{H,N} + c^l_R$, leaves the supplier's problem unchanged, with $\tau^l_R = 1 + c^l_R / w^l_{H,N}$. The welfare expressions we derive below depend on the policy only through the equilibrium log change in the wage paid by RS firms relative to that paid by non-RS firms. This is the object that we measure as part of the quantification in Section 5.

[32] See also Besley and Ghatak (2007), Campbell (2007), Hart and Zingales (2017), Eichholtz et al. (2010), Fioretti (2020), Kotchen (2006).

response to demands from consumers, employees, or investors. Our model incorporates this latter possibility in a reduced-form way, by allowing MNE demand to depend positively on RS implementation. Concretely, we assume that demand d_x for MNE x in (3) is shifted by $d_R \geq 1$ for RS-MNEs, while $d_N = 1$ for non-RS MNEs. In this scenario, MNEs implementing RS face a trade-off between the negative profit impact of higher production costs and the RS-induced increase in demand for their output.

Pass-through of RS costs to MNE input prices Another important question is who ultimately bears the cost of RS: the supplier or the MNE? Under monopolistic competition, prices would be set at a constant markup over marginal costs, so any labor cost increase for the supplier would be fully passed through to the MNE. In contrast, if the MNE has buyer power, it may be that little or none of this cost increase is passed through to the MNE. We capture a range of possible pass-through rates with a reduced-form parameter $\beta \in [0,1]$, assumed to be constant across MNEs. When $\beta = 0$, Home suppliers bear the full cost increase from RS policies; when $\beta = 1$, RS cost increases are fully passed through to the input price paid by the MNE.

4.3 Comparative statics: effects of RS

How do RS rollouts impact supplier-level outcomes? As in the data, we assume that some MNEs implement RS while others do not. Using hat notation, $\hat{y} = d\log y$, to denote log changes, we compute the first-order effect of an RS policy summarized by $\left(\hat{\tau}_R^l, \hat{d}_R, \beta\right)$. In line with the empirical analysis in Section 3, we compare changes in outcomes of firms "exposed" to RS through their MNE buyer—i.e., firms that sell to the MNE before it implements RS—against changes in outcomes of *ex-ante* similar firms that sell to MNEs that do not adopt RS. As the expressions below show, these comparisons difference out any GE effects of RS on wages at control firms, to which we turn as part of the GE welfare expressions below and quantify in the counterfactual analysis in Section 5.

Effect of RS on suppliers and workers We start by computing the theoretical effect of RS policies on relative wages and employment of exposed suppliers, corresponding to the estimates in Figure 4. The wage of low-wage workers at RS-exposed firms increases relative to those at non-RS firms by:

$$\hat{w}_{H,R}^l - \hat{w}_{H,N}^l = \hat{\tau}_R^l. \tag{8}$$

The relative impact of RS on the marginal cost of suppliers is thus:

$$\hat{W}_{H,R} - \hat{W}_{H,N} = \chi_H^l \hat{\tau}_R^l, \tag{9}$$

where χ_H^l denotes the share of low-wage workers in the supplier wage bill before RS.[33] The change in relative labor costs at RS suppliers induces a reallocation of workers (Panel D of Figure 4), subject to the elasticity of substitution (ρ) between worker types: $\hat{\ell}_{H,R}^l - \hat{\ell}_{H,R}^h - \left(\hat{\ell}_{H,N}^l - \hat{\ell}_{H,N}^h\right) = -\rho\hat{\tau}_R^l$.

Next, we turn to the effect of RS on firm sales. The effect on exposed-supplier total sales in Panel A of Figure 2 is the sum of effects across different markets supplied by the exposed firm. First, consider the *domestic sales* of exposed suppliers. Given CES demand (3) and monopolistic competition in the final goods market, firms price at a constant markup over marginal costs and the RS cost increase, $\chi_H^l\hat{\tau}_R^l$ from (9), is fully passed through to domestic prices. The impact of RS on domestic sales of exposed suppliers—corresponding to estimates in Panel B of Figure 6—is thus:

$$\hat{y}_{HH,R} - \hat{y}_{HH,N} = (1-\sigma)\,\chi_H^l\hat{\tau}_R^l < 0. \tag{10}$$

Second, RS impacts the sales of *intermediate inputs to the MNE*, both on the intensive and extensive margins. These effects differ from the effect of RS on final goods sales above, for three reasons. RS-MNEs may benefit from a demand boost, $\hat{d}_R$, due to RS, which increases their demand for inputs. MNEs may also exert buyer power, so that the cost of RS to their suppliers is only partially passed through to the input price paid by the MNEs, by a factor $\beta \in (0,1)$. Combining these two channels, the intensive-margin impact of RS on suppliers' sales to the MNE—corresponding to estimates in Panel A of Figure 6—is muted compared to the impact on domestic sales in (10):

$$\hat{y}_{HM,R} - \hat{y}_{HM,N} = (1-\sigma)\,\beta\chi_H^l\hat{\tau}_R^l + \hat{d}_R. \tag{11}$$

The third effect of RS on sales to the MNE operates on the extensive margin.[34] Only firms above a certain productivity level z^* select into the MNE market before RS. RS raises this threshold by increasing costs, although this effect may be offset if RS also stimulates demand. Specifically:

$$\hat{z}_{HM,R}^* - \hat{z}_{HM,N}^* = \frac{\sigma}{\sigma-1}\chi_H^l\hat{\tau}_R^l - \frac{1}{\sigma-1}\hat{d}_R. \tag{12}$$

Combining the intensive- and extensive-margin responses from equations (11) and (12) yields

[33] Specifically, $\chi_H^l = \frac{w_{H,N}^l \ell_{H,N}^l}{w_{H,N}^l \ell_{H,N}^l + w_{H,N}^h \ell_{H,N}^h}$.

[34] Extensive-margin effects enter to the first order in the effect on exposed firms' sales to the MNE and the effect on sales of the MNE (see below). They become second order in the effects on exposed firms' wages, relative employment and domestic sales.

the following relative change in sales of exposed suppliers to the MNE:

$$\hat{Y}_{HM,R} - \hat{Y}_{HM,N} = \Big[\underbrace{\beta\,(1-\sigma)}_{\text{intensive}<0} + \underbrace{\frac{\sigma\,(\sigma-1-\theta)}{\sigma-1}}_{\text{extensive}<0}\Big]\,\chi_H^l\hat{\tau}_R^l \;+\; \underbrace{\frac{\theta}{\sigma-1}}_{\text{int+ext}>0}\,\hat{d}_R. \tag{13}$$

The extensive-margin response is mediated by the shape parameter θ of the Pareto firm productivity distribution, which determines how a shift in the selection threshold (12) translates into a change in the mass of suppliers to the RS-MNEs.

Combining all sales effects above, the effect of RS on the total sales of exposed suppliers ($Y_{Htot} = Y_{HH} + Y_{HM}$)—corresponding to the estimates in Panel A of Figure 2 – is the sum of (10) and (13), appropriately weighted:

$$\hat{Y}_{Htot,R} - \hat{Y}_{Htot,N} = \underbrace{\Big[1-\sigma-\zeta\sigma\frac{\theta-\sigma+1}{\sigma-1} + (1-\beta)\,\zeta\,(\sigma-1)\Big]}_{<0}\,\chi_H^l\hat{\tau}_R^l + \underbrace{\zeta\frac{\theta}{\sigma-1}\hat{d}_R}_{>0}, \tag{14}$$

where $\zeta \equiv \frac{Y_{HM}}{Y_{Htot}}$ is the share of supplier sales that correspond to sales to the MNE prior to RS.

Effect of RS on the sales of the MNE Finally, we examine the effect of RS on the MNE subsidiary sales, estimated in Panel A of Figure 5. RS increases the production cost of the MNE by increasing its intermediate input costs (each supplier becomes more expensive, and there may be fewer suppliers, through the extensive-margin effect). This cost increase is proportional to the cost share of intermediate inputs, Ξ. While this cost increase reduces MNE sales, it may be partially offset by the direct demand boost from RS. On net, the MNE subsidiary sales change as follows:

$$\widehat{R_R M_R} - \widehat{R_N M_N} = \Big[\underbrace{\beta\,(1-\sigma)}_{\text{supplier int.}<0}\;\underbrace{-\frac{\sigma\,(\theta-\sigma+1)}{\sigma-1}}_{\text{supplier ext.}<0}\Big]\Xi\chi_H^l\hat{\tau}_R^l + \Big[1+\Xi\frac{\theta-\sigma+1}{\sigma-1}\Big]\hat{d}_R. \tag{15}$$

In Section 5, we quantify the model by confronting these comparative statics with the empirical evidence.

4.4 Welfare implications of RS

The overall desirability of RS policies from the perspective of Home workers is a priori unclear. On one hand, RS policies have a mechanical positive effect, as some Home workers experience wage increases, paid for—at least in part—by Foreign MNEs. On the other hand, RS policies impose increases in production costs that may leak into domestic prices, and GE effects on all workers may offset the direct gains. To investigate this interplay of forces in GE, we derive the first-order effect on Home's welfare of all MNEs implementing an RS policy $\left(\hat{\tau}_R^l, \hat{d}_R, \beta\right)$. To derive closed-form expressions and gain intuition, we assume here that MNE subsidiaries at Home do

not directly employ local labor ($\nu^\ell = 0$), and take the limit $\theta \to (\sigma - 1)$ (as in Burstein and Vogel, 2017). In Section 5, we also quantify more general versions of the model.

We first analyze the welfare impact of RS policies at Home on average over both worker types. Given utility (3), average welfare is: $U_H = \frac{X_H}{P_H}$, where X_H is total expenditure and P_H is the ideal price index. The first-order effect of RS on U_H can be expressed as:

$$\hat{U}_H \;=\; W^{tax}\beta\chi_H^l\hat{\tau}_R^l \;-\; W^{tax}\Lambda\chi_H^l\hat{\tau}_R^l \;+\; W^d\lambda_{FH}\hat{d}_R, \tag{16}$$

with strictly positive sufficient statistics $W^{tax} = \frac{\lambda_{HH}\lambda_{FH}\sigma}{1+(\lambda_{FF}+\lambda_{HH})(\sigma-1)}$ and $W^d = \frac{\lambda_{FF}}{1+(\lambda_{FF}+\lambda_{HH})(\sigma-1)}$. In these expressions, $\lambda_{kk'}$ denote trade shares as is standard in the literature on international trade (with λ_{kk} denoting the share of trade with country $k \in \{H, F\}$ itself), while Λ represents the share of expenditure on domestic goods that is spent on goods produced by RS-compliant firms.[35] Λ thus measures the degree of "leakage" of RS policies into the domestic price index.

Readers familiar with the trade policy literature will recognize that the sufficient statistic W^{tax} is equivalent to the first-order effect of a unilateral export tax on Home exports. If RS policies did not leak into the domestic market ($\Lambda = 0$) and had no demand benefit ($\hat{d}_R = 0$), they would be equivalent to an export tax from an efficiency perspective. The term $\beta\chi_H^l\hat{\tau}_R^l$ represents the size of this pseudo-export tax. In our setting, rather than being an actual tax, it measures the price premium imposed by RS on exports of intermediate inputs. The mechanism mirrors that of an export tax: by raising the price of exports, RS gives rise to a classic GE terms-of-trade effect that increases Home welfare. Since the price premium of exports increases with β, the corresponding welfare gains do too. In the limit where $\beta = 0$, there is no increase in export costs and no change in export quantity, causing the first term in (16) to disappear.

The second term in expression (16) also has a simple interpretation in terms of taxes. Through its leakage into the domestic market, RS inflates domestic goods prices, effectively acting, from an efficiency perspective, as a consumption tax $\chi_H^l\hat{\tau}_R^l$ imposed on a share Λ of domestic consumption. With RS, as with a consumption tax, the cost of these goods is inflated by a premium, with the proceeds of this premium rebated to Home. Since a consumption tax is equivalent to an export subsidy, it has a negative welfare impact of $-\Lambda W^{tax}\chi_H^l\hat{\tau}_R^l$ on Home workers. This term dampens the welfare-enhancing export tax effect of RS from the first term and reverses it if $\Lambda > \beta$.

Finally, the third term in expression (16) is unambiguously positive. The sufficient statistic W^d measures the effect of a positive demand shock impacting all of Home's production. Since this demand boost applies only to output sold to MNEs implementing RS, the resulting welfare

[35] Specifically, we define $\lambda_{ii} = \int_{\Omega_{ii}} \left(\frac{\sigma}{\sigma-1}\frac{W_i}{z}\right)^{1-\sigma} dG_i(z) / P_i^{1-\sigma}$ and $\lambda_{ji} = 1 - \lambda_{ii}$ for $i \neq j$, where $\Omega_{ii'}$ denotes the set of varieties produced in i and marketed in i'. $\Lambda = \int_{\Omega_{HH,R}} \left(\frac{\sigma}{\sigma-1}\frac{W_H}{z}\right)^{1-\sigma} dG_H(z) / \int_{\Omega_{HH}} \left(\frac{\sigma}{\sigma-1}\frac{W_H}{z}\right)^{1-\sigma} dG_H(z)$, where $\Omega_{HH,R}$ represents the subset of final varieties produced by firms impacted by RS.

gains are scaled by the share of Home production affected by it (λ_{FH}).

Distributional implications The model also allows us to investigate the heterogeneous effects of RS policies across workers. First, we show that in GE, low- and high-wage workers experience on average the same welfare gains (or losses) in this simple setup ($\hat{U}_H^l = \hat{U}_H^h$). Intuitively, GE forces adjust the wages for low-wage workers at non-RS firms to restore the labor market equilibrium after RS. Since the relative supply of high- and low-wage workers remains unchanged after RS, their relative labor payments also remain unchanged. As both worker types face the same change in consumption prices, they experience the same welfare change.

Next, we examine the effect of RS on exposed and non-exposed workers within the low-wage worker group. Similar to our definition of exposed firms, exposed workers are those employed at MNE suppliers before the MNE rolls out an RS policy. We denote $\hat{U}_H^{l,E}$ (resp. $\hat{U}_H^{l,NE}$) the per-capita welfare gains of exposed (resp. non-exposed) low-wage workers. The policy has stark distributional effects within worker types in GE. We find that there is a positive gap in welfare effects between exposed and non-exposed low-wage workers, corresponding to the direct benefit of the policy for exposed low-wage workers, i.e., the RS wage premium $\hat{\tau}_R^l$ (17). We also find a typically negative welfare effect of RS for non-exposed low-wage workers (18):

$$\hat{U}_H^{l,E} - \hat{U}_H^{l,NE} = \hat{\tau}_R^l > 0, \tag{17}$$

$$\hat{U}_H^{l,NE} = \left[\underbrace{(\beta - \Lambda) W^{tax} \chi_H^l - (\lambda_{FH} + \Lambda \lambda_{HH})}_{<0} \right] \hat{\tau}_R^l + \lambda_{FH} W^d \hat{d}_R. \tag{18}$$

On net, exposed low-wage workers may benefit from RS even with limited pass-through (β) and leakage into the local price index (Λ)—and they always do when aggregate welfare gains in (16) are positive. In contrast, RS typically has negative overall labor market consequences for non-exposed low-wage workers due to (i) negative GE effects on wages, as aggregate demand for low-wage workers declines, and (ii) rising consumer prices, due to leakage of RS costs into the domestic price index. The only source of welfare gains for non-exposed low-wage workers is the demand shock which, through GE effects, raises wages for all Home workers.

4.5 Alternative mechanisms and model extensions

RS as providing amenities or creating red tape Our model treats RS-related costs incurred by suppliers as directly paid to their workers. While this assumption provides transparency in the model, it may lack realism. First, as shown in Appendix Figure B12, RS suppliers may incur additional costs to improve workplace safety and other amenities, which benefit workers but are not directly paid to them. Second, some RS expenses may be "red tape" costs that do not provide

any direct benefit to workers.

In the first case, one can show that, under fairly weak assumptions, the welfare effects of RS in the sourcing country remain the same as in equations (16)–(18). To start, the welfare equivalence is straightforward if we assume that an improvement in amenities that costs a dollar to the firm is valued at a dollar by the worker. But this holds also more generally: whenever firms choose a mix of direct wage payments and costly amenities to attract workers, they minimize costs by equating the marginal benefit of a dollar spent on each. Therefore, workers value a marginal dollar spent on amenities the same as an additional dollar in earnings.[36]

In the second case, some of the RS costs may not benefit workers (such that the equivalence above does not hold). To capture this possibility, we assume that firms' labor costs remain as in equation (7), but only part $\tilde{\tau}_R^l < \tau_R^l$ of the premium is actually passed on to workers. The remainder of the RS premium is an iceberg-type wedge, effectively lost to the economy. In this case, the overall welfare impact of RS (omitting here the demand shock, $\hat{d}_R$, in (16) for simplicity) becomes:

$$\hat{U}_H = W^{tax}\left(\beta - \Lambda\right)\chi_H^l \hat{\tilde{\tau}}_R^l - W^{wedge}\chi_H^l(\hat{\tau}_R^l - \hat{\tilde{\tau}}_R^l). \tag{19}$$

The first term mirrors the first two terms in the baseline welfare effect (16), but it is scaled down to reflect only the portion of the cost rebated to workers ($\hat{\tilde{\tau}}_R^l < \hat{\tau}_R^l$). The new term corresponds to the pure wedge, which is negative.[37] Unsurprisingly, pure "red tape" costs reduce welfare. The overall welfare effect of RS is strictly lower than in the baseline case (16). We will quantify this alternative scenario for different values of $\tilde{\tau}_R^l < \tau_R^l$ in Section 5.

Full model and extensions Our baseline setup is intentionally parsimonious to highlight the key forces at play and yield closed-form, interpretable expressions for the welfare effects of RS. However, some realistic features omitted above may matter for the quantitative analysis that follows. We develop a less stylized version of the model in Appendix D, which incorporates realistic trade patterns (not restricting exporting solely to MNE subsidiaries) and allows us to compute welfare when MNE subsidiaries use domestic labor in production and the extensive margin due to Melitz-type selection is left unrestricted (allowing θ to differ from $\sigma - 1$). The augmented model requires numerical analysis. In Section 5, we compare the quantitative results from the simple baseline model in (16) and the augmented version. In Appendix D, we also derive the effect of RS under several model extensions. Some of these extensions expand the scope

[36] Formally, assume that a firm can hire a worker if $U(w,a) \geq \bar{u}$, where $U(w,a)$ represents the worker's utility over direct labor payments w and amenities a, with $U_w, U_a > 0$, and $\bar{u}$ captures the worker's outside option. Normalize amenities such that providing a units of amenities costs the firm a per worker. Then, by cost minimization, the firm optimally chooses a compensation schedule (w,a) satisfying $U_w = U_a$.

[37] Specifically, $W^{wedge} = \left[\frac{(1-\lambda_{HH})(\sigma-1)\lambda_{FF}\beta + \Lambda\lambda_{HH}(\sigma+\lambda_{FF}(\sigma-1))}{1+(\lambda_{FF}+\lambda_{HH})(\sigma-1)}\right]$.

of RS—considering that RS might enhance the productivity of impacted suppliers or apply to workers at the MNE subsidiary itself alongside those at suppliers. Other extensions concern the model environment. Specifically, we consider a labor market with unemployment and one with monopsony power among suppliers, as well as a more flexible demand system and a multi-country extension of the two-country setup described above. We show how our baseline results extend to these cases.

5 Model quantification and counterfactuals

In this section, we rationalize the effects of RS policies on firms and workers measured in Section 3 through the lens of our model and quantify the corresponding welfare implications. We first estimate the model's key parameters using the event-study estimates. Armed with the quantified model, we then proceed to the counterfactual analysis.

5.1 Scope and limitations

Quantifying counterfactuals in GE requires assumptions. While the model of Section 4 makes these assumptions transparent, some—though standard—are arguably strong in our context. In this light, we view the quantitative analysis as a first exploration of the aggregate and distributional implications posed by the widespread adoption of RS standards by MNEs, while fully acknowledging that our quantification does not provide definitive answers. Here, we discuss briefly several potentially important features that our analysis abstracts from, highlighting promising avenues for future research on these topics.

Our model is static, while RS can operate in a dynamic context characterized by repeated supplier relationships (e.g., Boudreau et al., 2023). These relationships may involve potential rent-sharing and relationship-specific investments that the model abstracts from.[38] Longer-term effects of RS may also come through the additional margin of firm entry and exit. In addition, the model abstracts from labor market distortions, which are often central to minimum-wage policies and could potentially serve as a policy motivation for RS. Although Appendix D discusses extensions of the analysis with unemployment and monopsony power of suppliers, a systematic investigation of how labor market distortions (e.g., search and matching frictions) may interact with RS is left for future research.

Moreover, in the model, RS impacts both fixed and variable costs proportionately through a labor cost increase. In reality, RS may impose additional fixed costs beyond those we consider. For instance, given the focus on labor-related RS provisions in our empirical context in Sections 2 and 3, the model does not examine the costs (and benefits) of stricter environmental standards

[38] In reference to Antràs (2020), our model follows a "broad/traditional" approach to MNE-supplier relationships and stops short of modeling the "narrow/relational" aspects.

that often accompany RS and could impact suppliers and welfare in other contexts. Finally, our calibration below relies on the average estimated treatment effects on suppliers and workers, thereby abstracting from the significant heterogeneity documented in Sections 2 and 3 across different types of suppliers, MNEs and RS codes of conduct. Related to this, we define the model's low-wage worker type in terms of the bottom 25% of initial earnings, whereas the true variations in worker types and RS effects are not binary.

5.2 Model quantification

We confront the model's comparative statics with the event-study estimates to quantify the average characteristics of the RS policies in our sample. Specifically, we estimate the extent of the RS-induced cost increase $\left(\hat{\tau}_R^l\right)$, the fraction of this cost increase passed through to the MNE (β), and the RS-induced demand shock $\left(\hat{d}_R\right)$. We also estimate the shape parameter of the firm productivity distribution (θ), which governs supplier responses on the extensive margin.

There are two natural approaches to estimate the size of the RS cost shock $\hat{\tau}_R^l$. Using the first-order effect in equation (8), we could directly measure the effect of RS on the labor earnings of initially low-wage workers. Using the estimate for $\eta \geq 4$ in Panel B of Figure 4, we would obtain $\hat{\tau}_R^l = 5.3\%$. Alternatively, we can infer the size of the cost shock from the sales responses of suppliers. Specifically, from equation (10), $\chi_H^l \hat{\tau}_R^l$ is revealed by the decline in the sales to domestic (non-RS) buyers of exposed suppliers following RS, conditional on a value of the elasticity of substitution σ. This second approach has advantages. As discussed above, direct payments to workers likely represent only part of the RS-imposed cost to suppliers, due to improved work amenities or costly red tape.[39] While an earnings-based estimate may underestimate the RS cost shock, the sales responses of exposed vs. non-exposed suppliers capture all RS cost increases combined. We adopt this second approach and compare it with the one based on worker earnings. Combining the event-study estimate corresponding to equation (10) (-0.097 at $\eta \geq 4$, s.e. 0.036 from Panel B of Figure 6) with the median cost share of low-wage workers at suppliers in the data $\left(\chi_H^l = 0.19\right)$ and $\sigma = 5.03$ (from Alfaro-Ureña et al., 2022),[40] we infer the average cost shock on low-wage workers to be $\hat{\tau}_R^l = 12.5\%$. Comparing this estimate to the 5.3% increase in the monthly earnings of exposed low-wage workers suggests that, indeed, not all RS-induced labor cost increases at suppliers are reflected in worker earnings.[41]

The remaining parameters must be estimated jointly using additional event-study moments

[39] If suppliers formalize (previously unobserved) informal work relationships to comply with RS codes, our earnings regressions based on formal employer-employee microdata and worker-by-firm fixed effects would also not fully capture the cost increase.

[40] Alfaro-Ureña et al. (2022) estimate σ in a way consistent with our model and with the same firm-level CR microdata. Their 5.03 estimate is in the mid-range of existing estimates in the literature (for a review, see Hottman et al., 2016).

[41] Cost increases due to amenities could, in principle, be larger than this difference, if higher salaries reflect RS-induced productivity gains. See Appendix D for related model extensions.

that directly map to our theory. The change in compliers' sales to the RS-MNE in equation (11), compared to the decline in sales to other (non-RS) buyers of exposed suppliers in equation (10), informs us about β and $\hat{d}_R$.[42] The sales response to the RS-MNE (-0.076 at $\eta \geq 4$, s.e. 0.035 in Panel A of Figure 6) is slightly less negative than the sales response to other (non-RS) buyers (-0.097, s.e. 0.036), suggesting a mitigating role for $\beta < 1$ and $\hat{d}_R > 0$. We need another moment to disentangle β and $\hat{d}_R$. We use the effect of RS on the sales of the MNE subsidiary in equation (15), which we estimate to be statistically zero (Panel A of Figure 5).[43] Finally, comparing the total sales response of exposed suppliers in equation (14) (-0.100 at $\eta \geq 4$, s.e. 0.028, see Panel A of Figure 2) to the intensive margin effect sheds light on the extensive margin response governed by θ.

Formally, to estimate $\left(\hat{d}_R, \beta, \theta\right)$, we invert this system of three equations on supplier and MNE sales, (14), (11) and (15), in the three unknowns.[44] We find that the cost pass-through is not complete, but high ($\beta = 0.85$). We find little evidence of a discernible positive demand shift for the MNE output ($\hat{d}_R = 0.006$). Intuitively, given the relatively small cost shock that RS policies represent for the MNEs (see footnote 43), the negligible effect on MNE subsidiary sales can be rationalized without much of a positive shift in MNE output demand. Finally, we estimate a shape parameter of the domestic productivity distribution of $\theta = 5.16$, which lies in the mid-range of existing estimates in international trade and macroeconomics (e.g., Melitz and Redding, 2014).

Additional moments The welfare expressions in (16)–(18) require three additional data moments. The first is Λ, which measures the share of domestically sold output affected by RS. As discussed in Section 2, we measure this to be 43% in the 2019 transaction data. Second, we need to quantify the share of RS-affected exports relative to total production in CR, $\lambda_{FH} = 1 - \lambda_{HH}$. To do so, we link the customs microdata using firm identifiers to our other firm-level microdata. We find that the share of CR's exports sold by RS-active CR firms and MNE affiliates relative to total domestic production in 2019 is $\lambda_{FH} = 0.075$.[45] Finally, for the share of Foreign expenditure on RS-affected exports from CR, $\lambda_{HF} = 1 - \lambda_{FF}$, we use a value close to zero.[46]

[42] A more direct estimation of the extent of cost pass-through (β) could be based on additional information on product prices (or unit values) alongside sales values. Unfortunately, the CR firm-to-firm transaction data do not include prices (or unit values) during our sample period.

[43] Our model suggests three forces that attenuate the effect of RS on the MNEs: (i) MNE subsidiaries are affected by RS costs only in proportion to their cost share spent on local suppliers ($\Xi = 0.14$); (ii) cost increases at suppliers may not be fully passed through to the MNEs ($\beta < 1$); and (iii) MNEs may experience demand increases due to RS ($\hat{d}_R > 0$).

[44] We also need to calibrate the average sales share to the RS-MNE by exposed firms before RS, which we measure in the transaction data ($\zeta = 0.25$).

[45] Our calibration corrects for the fact that our baseline model features trade patterns that are simpler than reality. Specifically, our measurement of Λ and λ_{FH} recognizes that not all MNEs in CR implement RS, not all exports in CR are done by MNEs, and not all MNE output is exported.

[46] We use 0.00001, and verify that none of the counterfactuals are sensitive to the precise value (e.g., scaling this value up or down by a factor of 100).

5.3 Counterfactual analysis

Armed with the quantified model, we proceed to investigate the welfare implications of RS policies. We consider a counterfactual that compares an equilibrium without RS to one corresponding to the extent of RS rollouts that we observe in CR at the end of the sample in 2019.

Welfare impact of RS in CR We quantify the aggregate and distributional implications in equations (16)–(18). The quantification suggests that RS in CR has had a positive, but minor, aggregate impact on welfare, both for initially low-wage and high-wage workers (+0.1%). The welfare expression (16) provides intuition as to why this occurs in our context. First, we find no evidence of discernible demand boosts from RS, so the last (positive) term of the welfare expression is close to zero. We can therefore focus on the first two terms driven by the export vs. consumption tax effects. The export tax effect is mitigated by the extent of cost pass-through from suppliers to prices paid by the MNE (β) and the extent of leakage of RS to the domestic market (Λ). If pass-through were complete ($\beta = 1$) and all the RS-affected production were exported ($\Lambda = 0$), then the representative worker in CR would fully benefit from the RS-induced improvement in CR's terms of trade. In our context, however, we find that the pass-through is incomplete ($\beta = 0.85$) and the leakage is significant ($\Lambda = 0.43$). Therefore, while the gains from the RS export tax remain positive (since $\beta > \Lambda$), they are more than halved. In addition, the terms-of-trade effect is scaled down by the cost share of the affected low-wage workers in suppliers' production ($\chi^l_H = 0.19$) and their estimated RS-induced increase in labor cost ($\hat{\tau}^l_R = 12.5\%$). These estimates imply a relatively small aggregate export-tax equivalent of about 2.4% (that is mitigated by both β and Λ).

These aggregate effects, however, mask significant heterogeneity within worker types. About 40% of all CR low-wage workers are "exposed" to RS in our counterfactual, i.e., in the initial equilibrium, they are employed at suppliers selling inputs to MNEs that end up adopting RS policies in the counterfactual. The quantification in (17) and (18) suggests these exposed low-wage workers, as a group, experience sizable welfare gains (+6.5%), while the remaining majority of low-wage workers in CR experience significant real income losses (-6%) due to adverse GE effects on their wages and an increase in the domestic price index. To assess the variability of these results, we also bootstrap the counterfactual analysis. We randomly draw, with replacement, individual suppliers from the estimation sample for the event-study estimates that we use to quantify the model above. We thus create 500 estimation samples, record the event-study estimates, quantify the model and perform the counterfactual analysis. The resulting 95% confidence intervals of the welfare effects indicate gains of 4.9 to 11.7% among exposed low-wage workers (compared to +6.5% above), and losses of -10.3 to -4.2% among the non-exposed low-wage workers (-6%

above).

Additional counterfactual results We presented above the welfare estimates based on the sufficient statistics in expressions (16)–(18), which we derived in a very stylized model—trade patterns are simple, parameters are such that the extensive margin is muted, and MNEs do not use domestic labor in production. We were careful, however, to measure moments like the leakage of RS to the domestic market (Λ) and the share of RS-affected exports in Home production (λ_{FH}) to prevent overstating the importance of RS. To get a more complete picture, we also compute (numerically) the welfare gains from RS in the "full model" described in Section 4.5 and Appendix D. We find that both the aggregate (0.1%) and distributional implications of RS are similar to the baseline findings, with exposed (non-exposed) low-wage workers experiencing 7.9% (4.1%) welfare gains (losses). This similarity suggests that the careful measurement of moments in the stylized model largely compensates for its simplifying assumptions.[47]

We also revisit the empirical concern that the estimated effects of RS on MNE output demand (explored in Figure 5) may be subject to negative bias due to violations of the parallel trends assumption (post-RS rollout in the absence of RS). In the model, such violations could be rationalized through an underestimation of the change in MNE output demand due to announcing RS practices ($\hat{d}_R$ in equation (15)). To account for this potential downward bias, we quantify the aggregate and distributional welfare implications of RS after assuming a range of higher RS-induced MNE demand effects relative to the close to zero estimate in our baseline model quantification above. This analysis reveals two implications of a more pronounced positive demand effect: the aggregate welfare gain increases (from 0.1 to 0.5%, as $\hat{d}_R$ increases from 0 to 50%), and the distributional implications within low-wage workers remain similar (with a 7% gain among exposed workers and a 5.5% loss among non-exposed, see Appendix Figure E1).

Finally, Appendix E presents counterfactual analyses of the welfare implications of RS in CR under alternative parameter values and model extensions. These include allowing for elastic labor supply and unemployment (Appendix Figure E3) and attributing part of the RS cost increase to "red tape" (following the discussion in Section 4.5). Unsurprisingly, following insights from equation (19), the baseline aggregate and distributional effects of RS in (16)–(18) decrease roughly proportionately with the fraction of the RS cost increase that is captured—through direct pay or workplace amenities—by domestic workers (Appendix Figure E2). We also present the quantified welfare effects across ranges of alternative parameter values for σ, β, $\hat{\tau}_R^l$, Λ and the cost share of affected workers (χ_H^l) in Appendix Figures E4–E8. For compactness, we do not discuss these additional counterfactuals here and instead refer the reader to Appendix E. These additional

[47] We also find that the estimated θ is not far from the value imposed in the baseline model that restricts the extensive margin ($\theta \to (\sigma - 1)$).

results serve as a reference to assess the sensitivity of our findings in the current setting of CR, and to explore how the impacts of RS may vary across different empirical settings.

6 Conclusion

Our analysis combines a new collection of microdata with a simple model to study the effects of MNE RS policies on sourcing countries. Empirically, we find that RS policies are, on average, not just "hot air": the sales and employment of RS-exposed MNE suppliers decline, while the labor earnings of their initially low-wage workers increase. This evidence is consistent with RS policies raising labor-related production costs, particularly for workers at the bottom of the initial earnings distribution. Relative to the existing body of work on the firm-level gains from joining multinational supply chains, our findings suggest that RS policies act to redistribute part of the supplier-level gains to workers at exposed suppliers, and initially lower-paid workers in particular. On its own, however, this reduced-form evidence would be insufficient to evaluate the welfare implications of RS.

The model highlights that the welfare implications of widespread adoption of RS policies are *a priori* ambiguous in sourcing countries, and that RS gives rise to distributional effects within worker types. Akin to an export tax, RS policies applied to production destined for export can improve the terms of trade of the sourcing country and generate welfare gains. However, these gains can be dampened or even reversed in environments where the RS cost pass-through to MNE buyers is incomplete and where RS-related cost increases leak into domestic production. Additional gains arise when RS increases demand for the MNE's output due to consumer preferences in favor of RS. In the quantification, we find that RS led to positive, but minor, aggregate welfare gains in CR. These aggregate implications, however, conceal meaningful distributional effects: low-wage workers at exposed suppliers experience sizable welfare gains, while the (non-exposed) majority of low-wage workers face losses due to adverse equilibrium effects on wages and prices.

Overall, our findings highlight a trade-off involved in RS policies. On one hand, RS can meaningfully benefit workers at targeted suppliers. On the other hand, RS imposes additional costs on suppliers, and its widespread adoption by MNEs may give rise to unintended indirect effects on the non-targeted workforce. This counterfactual analysis is based on a simple and stylized framework that emphasizes the basic supply-and-demand forces at work. And while we formalize several model extensions and sensitivity analyses, it is clear that there is ample space for additional work on quantifying the aggregate implications of RS across sourcing countries. Important dimensions for future contributions that we abstract from in this paper include richer analyses of the labor markets, dynamic firm responses including new entrants adapting to RS and relational contracts between MNEs and suppliers. In addition, as our evidence and

model highlight, the effects of RS are likely to vary across countries depending on labor market institutions, enforcement capacity, and which RS provisions are most binding locally, including environmental provisions. Understanding how these factors mediate the effects of RS remains an important avenue for future research.

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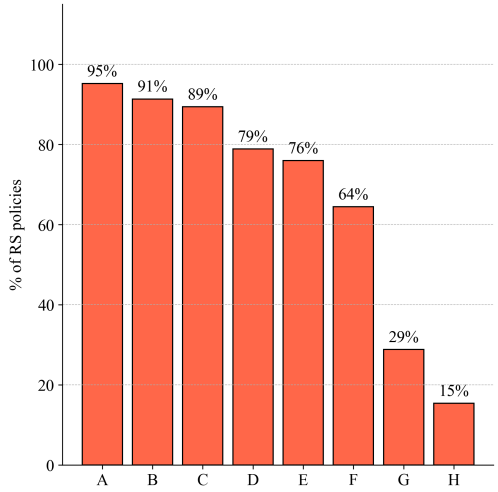
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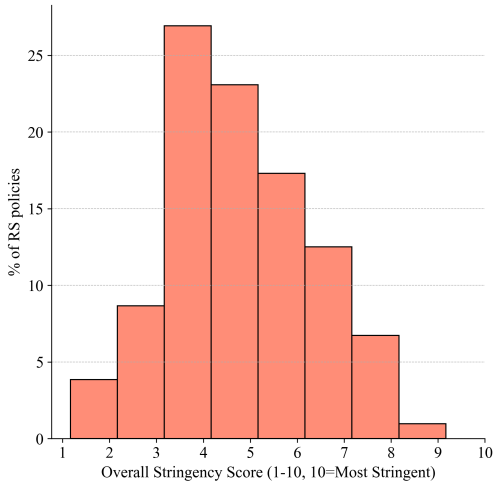
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Figures

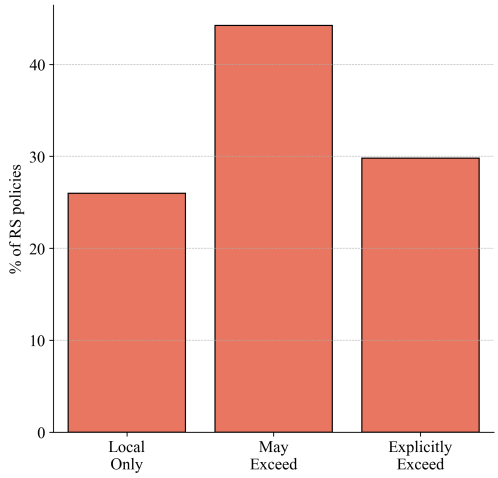
Figure 1: Descriptive Analysis of the RS Policy Documents



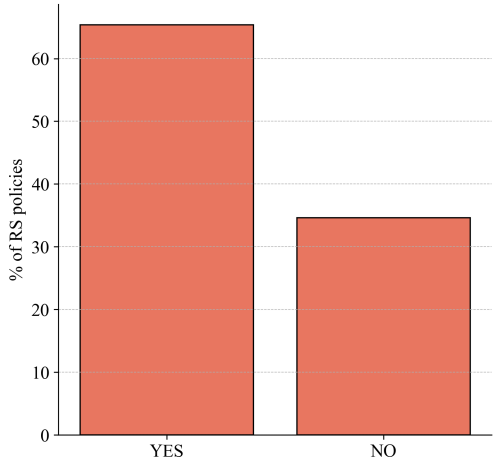
(A) Binary Indicators of Policy Requirements



(B) Overall Stringency Score Distribution



(C) Beyond Local Law Score Distribution

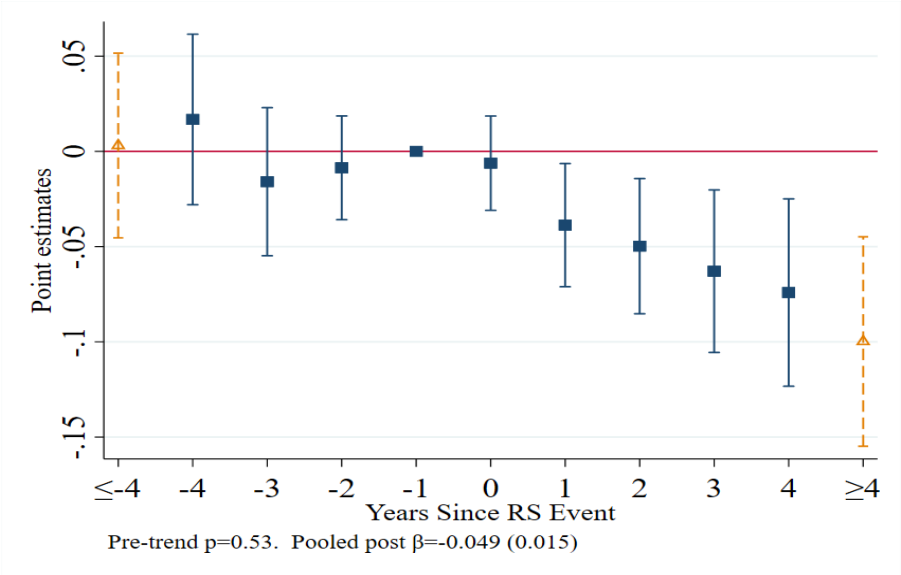


(D) Explicit Penalties for Non-Compliance

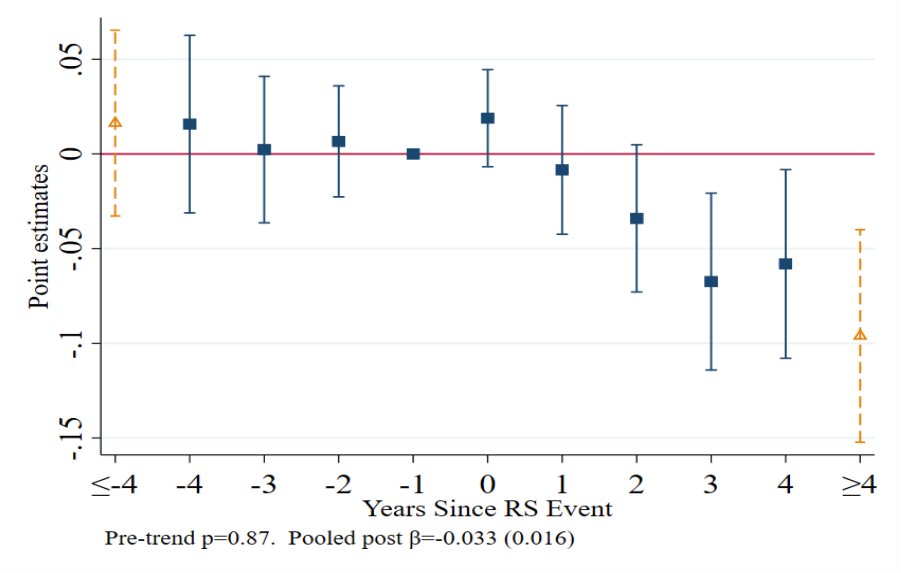
Notes: Figure summarizes a text analysis of RS policy documents conducted with OpenAI's GPT-5 model. Panel 1A reports the share of RS policies that explicitly require each of eight categories of labor provisions: A = Health & Safety, B = No Discrimination, C = No Child/Forced Labor, D = No Harassment, E = Wages & Compensation, F = Freedom of Association, G = Hours Cap, H = Rest Days. Panel 1B reports the distribution of policies by overall stringency, defined as the average of three dimensions each scored from 1 to 10: specificity of requirements, monitoring provisions, and consequences for non-compliance. Panel 1C reports the distribution of policies across three categories: those requiring only compliance with Costa Rican labor law and regulations ("Local Only"), those setting standards that may exceed local law ("May Exceed"), and those explicitly exceeding local minimums ("Explicitly Exceed"). Panel 1D reports the share of policies that specify explicit consequences for failure to adhere to the stated labor practices. Appendix Tables A6–A9 provide further details.

Figure 2: Supplier-Level Effects of RS Rollouts on Annual Sales and Employment

Panel A: Supplier Annual Sales



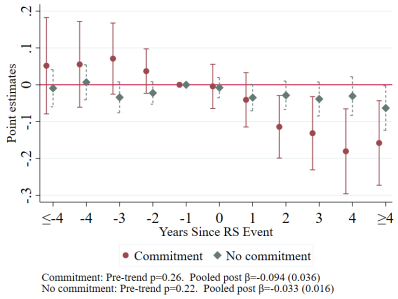
Panel B: Supplier Employment



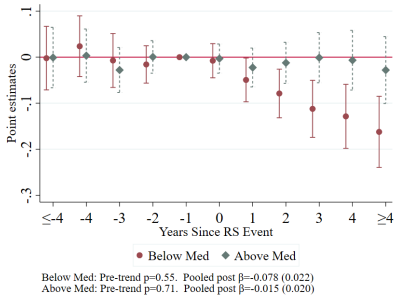
Notes: Panels A and B plot estimates from the supplier-level IV event-study specification in equation (1) and correspond to columns 4 and 8 of Table 1. The outcomes in Panels A and B are log total sales and log number of employees, respectively. Both panels show 95 percent confidence intervals based on standard errors clustered at the firm level. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

Figure 3: Heterogeneity of Supplier-Level Effects of RS Rollouts for Total Sales

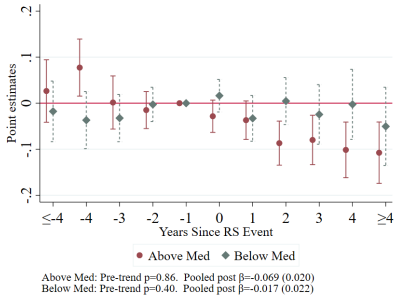
(A) Policy Requires Above Minimum Wage Pay



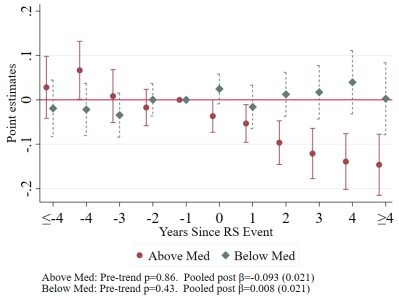
(B) Supplier Sector with Small vs. Large Firms



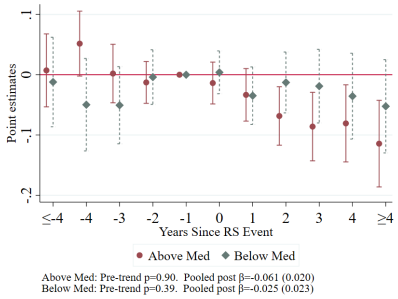
(C) Policy Overall Labor Score



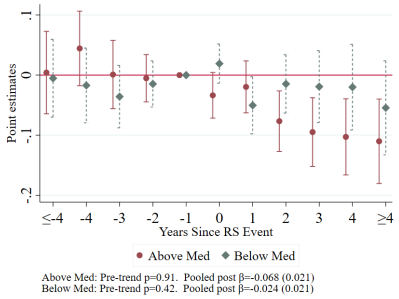
(D) Policy Compensation-Related Word Share



(E) Policy Health & Safety Word Share

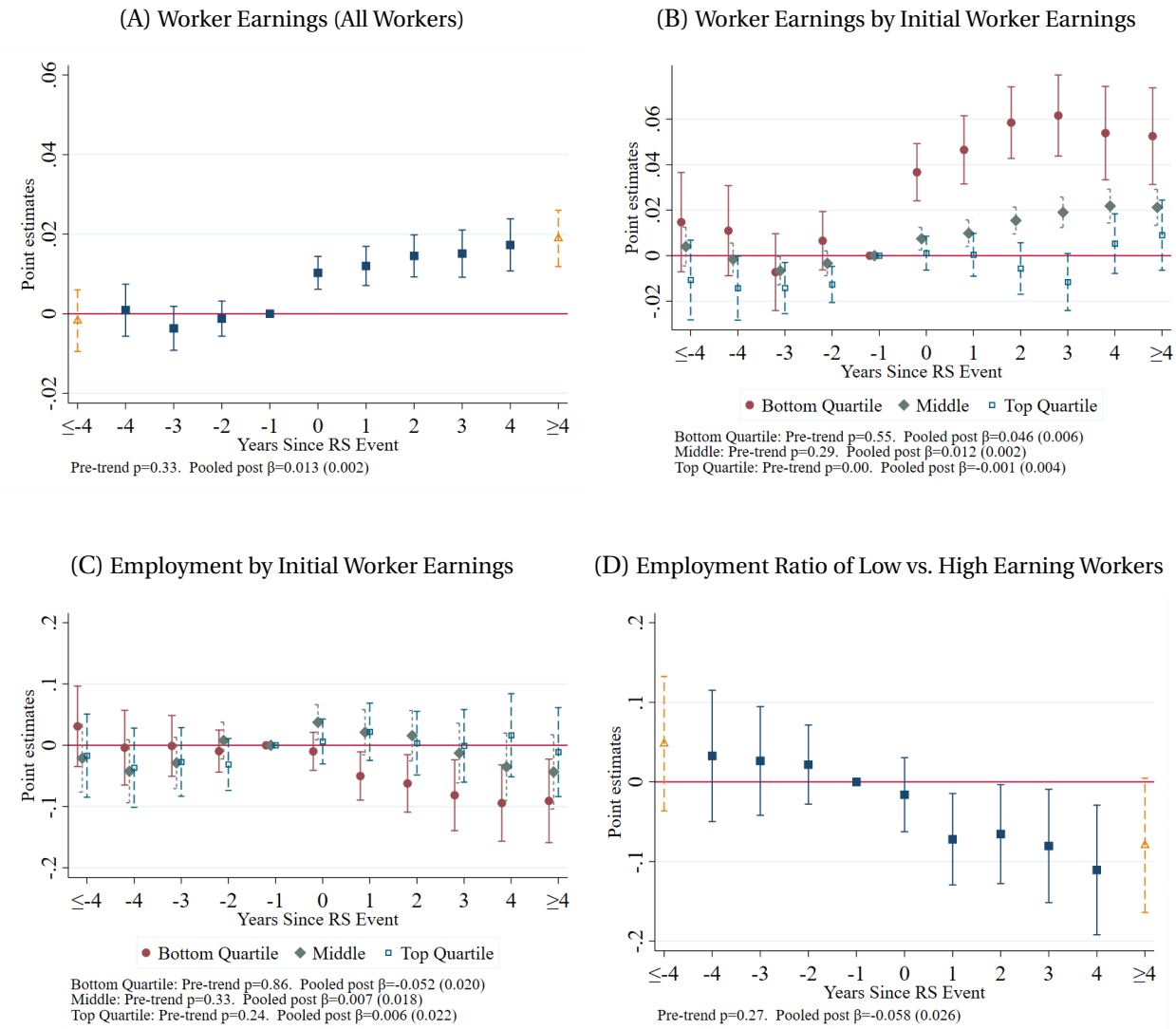


(F) Policy Enforcement & Compliance Word Share



Notes: Figure explores the heterogeneity of the supplier-level average sales effect of RS rollouts from Panel A of Figure 2. Panel 3A splits suppliers by whether the triggering MNE's RS policy requires suppliers to commit to paying wages above the legal minimum, as coded by OpenAI's GPT-5 model based on the policy text. Panel 3B splits suppliers by whether their sector's median sales fall below or above the sample-wide median. Panel 3C splits suppliers at the median of an overall labor score for the RS policy, defined as the average of two standardized components derived from OpenAI's GPT-5 model's reading of the policy: (i) a labor-focus score that rates, on a one-to-ten scale, how much the document is devoted to labor standards (working conditions, wages, workplace safety, and non-discrimination); and (ii) the number of wage and compensation commitments required by the policy (see Appendix Table A6 for details). Panels 3D, 3E, and 3F split suppliers at the median share of RS policy-document words falling in the compensation, health-and-safety, and enforcement-and-compliance keyword categories, respectively. All panels show 95 percent confidence intervals based on standard errors clustered at the firm level. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

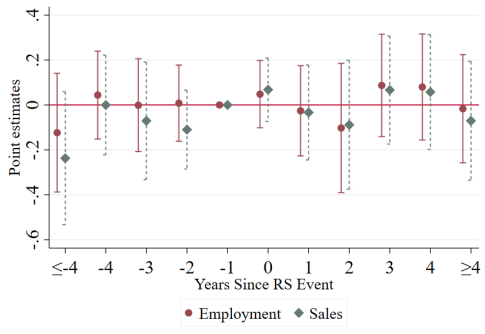
Figure 4: Effects of RS Rollouts on Worker-Level Earnings and Supplier-Level Employment



Notes: Panels 4A and 4B plot estimates from the worker-level IV event-study specification in equation (2). The outcome in both panels is the log of labor earnings. The point estimates are displayed in Appendix Tables C2, C3, C4, and C5. Panels 4C and 4D plot IV estimates from the supplier-level event-study specification in equation (1). The outcome in Panel 4C is log total employment by worker category based on initial monthly earnings. The point estimates are displayed in Appendix Tables C6, C7, and C8. The outcome in Panel 4D is the log employment ratio of the bottom to top quartiles of workers, again based on initial monthly earnings. The point estimates are displayed in Appendix Table C1. Panel 4A includes all workers employed by the suppliers, whereas Panels 4B and 4C display estimates from separate regressions by quartile of initial worker earnings. All panels show 95 percent confidence intervals. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

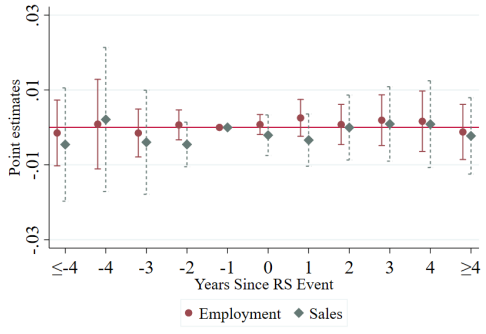
Figure 5: Effects of RS Rollouts on MNE-Level Outcomes and of NGO Campaigns on RS Probability

(A) Sales and Employment of the MNE subsidiary in CR



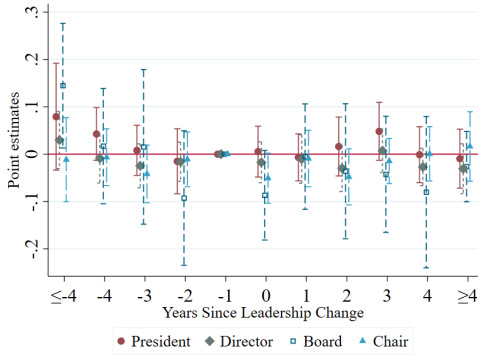
Employment: Pre-trend p=0.60. Pooled post β=-0.025 (0.102)
Sales: Pre-trend p=0.13. Pooled post β=-0.036 (0.109)

(B) Ratio of MNE Sales and Employment in CR vs. Global

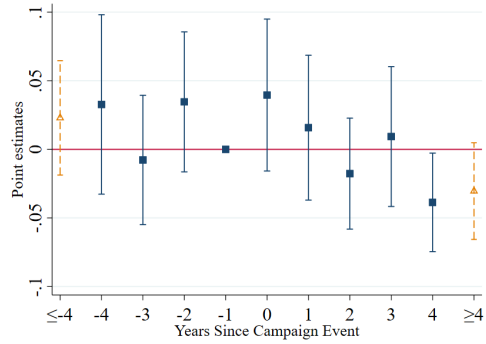


Employment: Pre-trend p=0.90. Pooled post β=-0.000 (0.002)
Sales: Pre-trend p=0.56. Pooled post β=-0.004 (0.003)

(C) MNE Changes of Key MNE Employees



(D) RS Rollout Probability on Negative NGO Campaigns

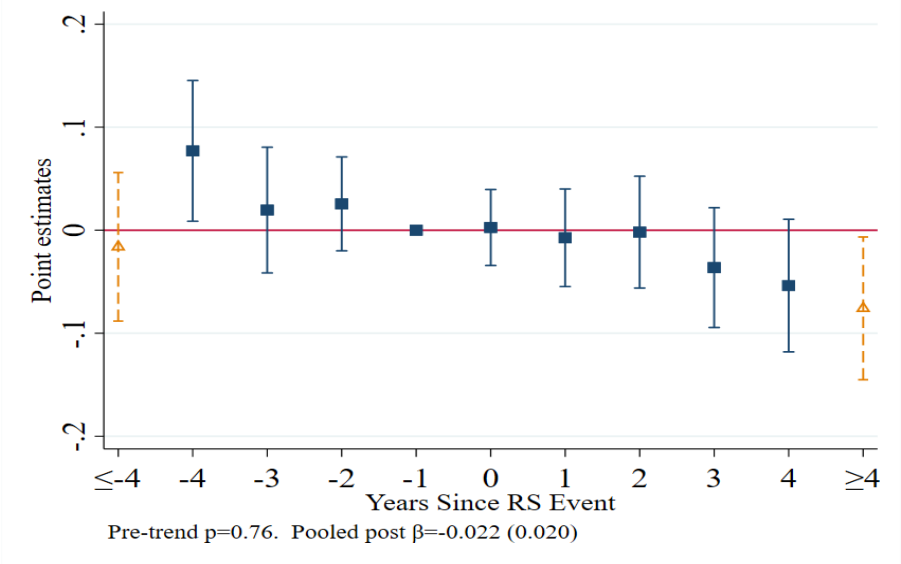


Pre-trend p=0.39. Pooled post β=-0.002 (0.018)

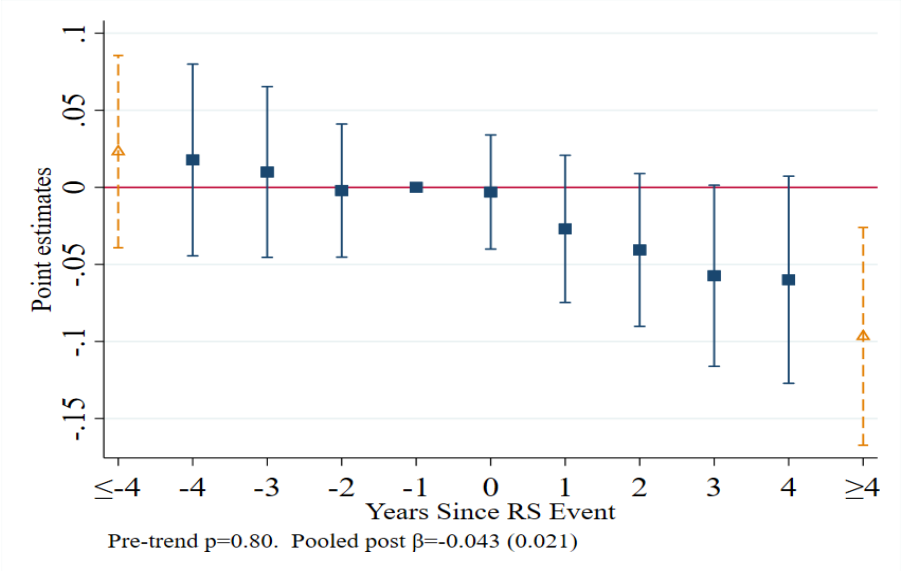
Notes: Panels 5A and 5B display event-study estimates of the effects of RS rollouts on MNE-level outcomes. Panel 5A plots estimates from MNE-level regressions, where the outcomes are the log sales and log employment of the MNE subsidiary in CR. Panel 5B plots estimates from MNE-level regressions, where the outcomes are the ratios of subsidiary sales and employment to the global sales and employment of the MNE. The point estimates for Panels 5A and 5B are displayed in Appendix Tables C9 and C10. Panel 5C plots estimates from MNE-level regressions, where the outcome is a dummy for an RS rollout by the MNE, and event time counts the years since a change in one of four types of leadership positions within the MNE. This panel uses the BoardEx and Orbis data, as described in Section 2. These data provide the start and end dates of employment for individuals in key positions at each MNE. Given the variation in job titles across MNEs, we focus on leadership changes involving positions with titles that include "president", "director" (including "MD" for managing director), "board", or "chair". Panel 5D implements MNE-level event studies with a dummy for RS rollout decisions as an outcome. The event timeline is based on NGO campaigns against the MNE. Panel 5D uses the Sigwatch database described in Section 2, which provides comprehensive coverage of NGO-led campaigns against MNEs. We assess whether negative NGO campaigns about MNE production practices increase the likelihood of the MNE rolling out an RS policy. All panels show 95 percent confidence intervals based on standard errors clustered at the MNE level. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

Figure 6: Effects on Sales to the MNE Buyer and Supplier Sales to Other Buyers

Panel A: Intensive Margin of MNE Transaction Sales among Continuing Suppliers



Panel B: Supplier Sales to Other (Non-RS) Buyers



Notes: Panel A plots estimates from the transaction-level IV event-study specification in equation (2). The outcome is the log annual transaction value of sales made by domestic suppliers to their MNE buyers. The sample includes transactions with RS-MNE buyers for treated suppliers (note that these buyers had not rolled out RS before event time 0) and transactions with any MNE for non-treated suppliers. The point estimates are displayed in Appendix Table C11. Panel B plots estimates from the supplier-level IV event-study specification in equation (1). The outcome is the log total sales to domestic (non-RS) buyers. The point estimates are displayed in Appendix Table C12. Both panels show 95 percent confidence intervals based on standard errors clustered at the firm level. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

Tables

Table 1: Supplier-Level Effects of Exposure to MNE RS Rollouts on Sales and Employment

	Panel A: Log Supplier Sales				Panel B: Log Supplier Employment			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	TWFE	TWFE	SA	IV	TWFE	TWFE	SA	IV
$\eta \leq -4$	-0.216***	0.010	0.042*	0.003	-0.175***	0.013	0.038	0.016
	(0.024)	(0.024)	(0.025)	(0.025)	(0.024)	(0.024)	(0.025)	(0.025)
$\eta = -4$	-0.139***	0.010	0.035	0.017	-0.111***	0.009	0.032	0.016
	(0.023)	(0.023)	(0.023)	(0.023)	(0.024)	(0.023)	(0.023)	(0.024)
$\eta = -3$	-0.121***	-0.021	0.001	-0.016	-0.076***	-0.003	0.017	0.002
	(0.020)	(0.019)	(0.019)	(0.020)	(0.020)	(0.019)	(0.019)	(0.020)
$\eta = -2$	-0.058***	-0.005	0.008	-0.009	-0.029**	0.003	0.014	0.007
	(0.014)	(0.013)	(0.013)	(0.014)	(0.015)	(0.014)	(0.014)	(0.015)
$\eta = 0$	-0.026**	-0.015	-0.019	-0.006	0.019	0.016	0.014	0.019
	(0.012)	(0.012)	(0.012)	(0.013)	(0.013)	(0.013)	(0.013)	(0.013)
$\eta = 1$	-0.068***	-0.035**	-0.042***	-0.039**	-0.010	-0.007	-0.008	-0.008
	(0.016)	(0.015)	(0.015)	(0.016)	(0.017)	(0.017)	(0.017)	(0.017)
$\eta = 2$	-0.106***	-0.056***	-0.062***	-0.050***	-0.048**	-0.037**	-0.036*	-0.034*
	(0.017)	(0.017)	(0.017)	(0.018)	(0.019)	(0.019)	(0.019)	(0.020)
$\eta = 3$	-0.139***	-0.067***	-0.074***	-0.063***	-0.090***	-0.067***	-0.063***	-0.067***
	(0.021)	(0.020)	(0.020)	(0.022)	(0.022)	(0.022)	(0.022)	(0.024)
$\eta = 4$	-0.190***	-0.085***	-0.099***	-0.074***	-0.113***	-0.070***	-0.071***	-0.058**
	(0.024)	(0.024)	(0.024)	(0.025)	(0.024)	(0.024)	(0.025)	(0.025)
$\eta \geq 4$	-0.244***	-0.107***	-0.150***	-0.100***	-0.165***	-0.103***	-0.139***	-0.096***
	(0.026)	(0.026)	(0.029)	(0.028)	(0.027)	(0.027)	(0.031)	(0.029)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year-Sect FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Adjusted R^2	0.80	0.82	0.82	–	0.76	0.77	0.77	–
# Observations	136,590	136,590	136,590	136,590	136,590	136,590	136,590	136,590
# Treated	2,130	2,130	2,130	2,130	2,130	2,130	2,130	2,130
# Never Treated	15,583	15,583	15,583	15,583	15,583	15,583	15,583	15,583
# Sector-Year Bins	2,638	2,638	2,638	2,638	2,638	2,638	2,638	2,638

Notes: Table presents two-way fixed effects (TWFE), Sun and Abraham (2021) (SA) and IV estimates for the supplier-level specification in equation (1). Outcomes are suppliers' log annual sales in Panel A and log annual employment in Panel B. Columns 4 and 8 correspond to Panels A and B of Figure 2. The first-stage F-statistics for the IV columns exceed 50. Columns with "Yes" in the "Controls" row include the full timeline of event-study indicators for having had positive sales to any MNE subsidiary in CR. Standard errors clustered at the firm level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Responsible Sourcing?

Evidence from Costa Rica

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Appendix for online publication

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This Appendix supplements the paper "Responsible Sourcing? Evidence from Costa Rica" with the following material:

- Appendix A provides additional descriptive statistics referenced in Section 2.
- Appendix B provides additional event-study figures referenced in Section 3.
- Appendix C provides event-study tables that correspond to the event-study figures discussed in Section 3.
- Appendix D provides derivations, additional details and extensions for the model in Section 4.
- Appendix E provides additional counterfactual results discussed in Section 5.

Appendix A Additional descriptive statistics

Table A1: MNE Sample Coverage

	Share of Total
Total Sales	78.9%
Number of Workers	78.6%
Wage Bill	79.4%
Exports	91.2%
Imports	81.1%
Value Added	80.4%
Domestic Purchases	72.9%
Total Net Assets	78.2%

Notes: Table presents the coverage for the period 2008 to 2019 (for eight variables, summing across all years) of the total values for the 481 MNEs in our sample, expressed as a percentage of the total values for the full sample of 2,156 firms that are part of a corporate group with partial foreign ownership. See Section 2.1 for discussion.

Table A2: Sample of RS Rollouts Between 2000 and 2019 by MNEs with Subsidiaries in CR

Year	MNE	Year	MNE	Year	MNE	Year	MNE
2000	McDonald's	2007	The Bank of Nova Scotia	2011	G4S*	2015	Cargill
2001	Baxter Americas Services	2008	Abertis Infraestructuras	2011	Huawei Technologies*	2015	Credomatic*
2001	Baxter International	2008	American Airlines	2011	Maersk*	2015	Emerson Electric*
2001	British American Tobacco	2008	DHL	2011	McCann Advertising*	2015	Griffith Foods International*
2002	Enel Green Power	2008	DHL Customer Support	2011	Philip Morris*	2015	Grupo Aval*
2002	Fyffes	2008	Florida Ice and Farm	2011	Puratos Group*	2015	Kuehne + Nagel Logistics*
2002	GlaxoSmithKline	2008	Heinz	2012	American & Efird LLC*	2015	Optica Industrial*
2003	Abbott	2008	Inabensa	2012	Bimbo*	2015	Pipasa*
2003	BASF SE	2008	Millicom	2012	BioMar	2015	Productos Florida*
2003	Hewlett-Packard	2008	Nestle	2012	C&K Components, Inc*	2015	Reca Quimica*
2004	Cisco	2008	Paradise Ingredients	2012	Camino Real Hotels*	2015	VMware*
2004	F. Hoffmann-La Roche AG	2008	Sodexo	2012	Claro Company*	2015	Volcafe*
2004	IBM	2008	Vitec	2012	Edwards Lifesciences	2015	Yamaha*
2004	Intel	2009	AstraZeneca*	2012	Havells Sylvania*	2016	Aluma Systems*
2004	J. Walter Thompson	2009	ERIAL*	2012	Mexichem Orbia*	2016	Baltimore Spice Company*
2004	Microsoft	2009	Eaton Electrical*	2012	Nutresa*	2016	Burger King*
2004	Novartis	2009	Ericsson*	2012	POPS Ice Creams*	2016	Felguera IHI*
2004	Panasonic	2009	Oracle*	2012	Pfizer*	2016	Kuehne + Nagel Service Center
2004	Philips	2009	Praxair*	2012	Samtec Inc*	2016	Productores Monteverde*
2004	Teradyne	2009	Toyota*	2012	The Westin - Marriott International*	2016	Sigma Alimentos*
2005	Granja Avicola Ricura	2009	UPS*	2012	Zoetis	2017	Brightpoint Group*
2005	Holcim	2010	Accenture*	2013	Allergan*	2017	Cemaco*
2005	The Procter & Gamble Company	2010	Bayer Medical*	2013	ArcelorMittal*	2017	Concentrix*
2005	Unilever	2010	Citi Business Services*	2013	Bimbo Servicios	2017	Constructora los Negros*
2005	Walmart	2010	Clorox de Centroamerica*	2013	Dole plc*	2017	FHACASA*
2006	3M	2010	Desatur Corobici*	2013	ICA Construction*	2017	Panduit*
2006	Cemex	2010	El Gallito Industrial*	2013	Medtronic*	2017	Radisson Hotels*
2006	Ecolab	2010	Embotelladora Centroamericana*	2013	Qorvo*	2017	Sonepar Company*
2006	Fujitsu Global	2010	Expeditors International*	2013	Swissport*	2017	Zevex INC*
2006	ICU Medical	2010	Fertilizers Pacific*	2013	Sykes Enterprises*	2017	Zollner Electronic*
2006	Leo Burnett Worldwide, Inc.	2010	Florida bebidas*	2013	Telefonica*	2018	Itinera S.p.A*
2006	Ricoh	2010	Productora la Florida*	2013	Wyndham Hotels & Resorts*	2018	McKinsey & Company*
2007	BTicino S.p.A	2010	Prosix Engineering*	2014	Alpla Group*	2018	Medplast Medical*
2007	Bacardi	2010	Ristic AG*	2014	Avianca*	2018	The Crowley Company*
2007	Citibank	2010	Smurfit Kappa*	2014	BA Continuum*	2018	Vertiv
2007	Coca Cola	2011	Amway*	2014	Boston Scientific*	2019	Amazon*
2007	Coca Cola Service Center	2011	BATO Shared Services	2014	Davivienda*	2019	IGT Global Solutions*
2007	Getinge	2011	Bourns, Inc*	2014	Del Monte Agric*	2019	Microvention*
2007	Jones Lang LaSalle	2011	Bridgestone*	2014	Del Monte Frozen Products*	2019	R.R. Donnelley*
2007	Kimberly Clark	2011	Cognizant	2014	Greif, Inc*	2019	UPL Limited*
2007	PepsiCo	2011	Colgate-Palmolive Company*	2014	Securitas*		
2007	Robert Bosch	2011	Ernst & Young*	2015	Bekaert*		
2007	Siemens	2011	Essity AB*	2015	CWT Company*		

Notes: Table presents the list of first-time labor-related RS rollouts among the 481 MNEs with subsidiaries in CR. We focus on rollouts related to labor standards. This information is based on public records and does not disclose any confidential information. To construct these data, we implemented a double-blind search process conducted by two independent research teams, whose outputs we then cross-checked and combined into one final database. For each MNE, we searched all publicly available company reports, press releases, corporate filings, and online publications, including company websites of the subsidiaries in CR and the MNE groups, for information on corporate social responsibility, responsible sourcing practices and supplier codes of conduct. In addition, for each MNE, we conducted online searches in both local CR and international media outlets. The star "*" next to certain MNE names indicates that these MNEs triggered a first-time RS exposure event for suppliers in our estimation sample. See Section 2.3 for discussion.

Table A3: Estimation Sample Construction

	Number of Firms
Panel A: MNEs	
MNEs with subsidiaries in CR and $\geq$ 1 mill USD purchases	481
No labor RS policy found	312
Labor RS policy found	169
RS rollout in or before 2008	56
No first-time suppliers available	9
Triggering first-time exposure (2009–2019)	*104*
of which: global RS policy	93
Panel B: Suppliers	
Never-treated suppliers	16,627
Treated suppliers	2,139
Total	*18,766*
Excluded from estimation	11,591
Never-treated suppliers (in estimation sample)	15,583
Treated suppliers (in estimation sample)	2,130
Total (in estimation sample)	*17,713*
Other domestic firms	82,480

Notes: Table summarizes the construction of the estimation sample. Panel A reports the number of MNEs at each step. We begin with 481 MNEs with subsidiaries in Costa Rica that have average annual domestic input purchases exceeding 1 million U.S. dollars and average annual employment of more than 50 workers. Of these, 169 MNEs have a publicly documented RS policy with labor-related provisions. Among the 169 MNEs with labor RS policies, 56 implemented their first RS rollout in or before 2008 (2009 is the first year in which our firm-to-firm transaction data allow us to compute supplier exposure), and 9 had no suppliers experiencing first-time RS exposure (all their suppliers had prior RS exposure from other MNEs). Therefore, 104 trigger a first-time RS exposure event for at least one domestic supplier during our estimation period. Of these 104 triggering MNEs, 93 (89%) have a global RS policy originating from the parent company that applies to all suppliers worldwide. Panel B reports the number of domestic suppliers. Treated suppliers are those selling to an MNE in the year before that MNE's first RS rollout and who had never previously sold to an RS-active MNE. Never-treated suppliers are firms that supply to MNEs at some point but never sell to RS-active MNEs, restricted to the 4-digit sectors of the treated suppliers. Domestic suppliers of RS-active MNEs that are neither treated nor never-treated are reported as "Excluded from estimation"; they include those with prior exposure to an RS-active MNE, and those entering the RS relationship at or after the event. For completeness, the row labeled "Other domestic firms" reports domestic firms that appear as sellers in the transaction data but never supplied an MNE; these are shown for reference and are not part of any estimation sample. The estimation sample of 17,713 suppliers is a subset of the sample of 18,766 suppliers, excluding singleton observations in the regression (due to fixed effects). See Section 2.3 for further details on data construction and Section 3.1 for the empirical strategy.

Table A4: Descriptive Statistics for the Sample of MNEs

	No RS policy	RS policy, no events	RS policy, triggering events
	(1)	(2)	(3)
Total Sales	51,913.1	96,532.0	74,015.7
	(153,624.7)	(149,017.5)	(91,176.0)
Employment	451.8	626.0	724.9
	(978.5)	(943.2)	(1,673.0)
Wage Bill	5,252.6	13,491.0	10,724.9
	(10,178.1)	(21,154.4)	(16,916.3)
Exports	11,027.0	26,685.4	22,245.9
	(31,716.3)	(69,744.2)	(59,660.3)
Imports	11,735.0	18,892.8	20,548.9
	(26,268.2)	(34,474.8)	(33,986.6)
Value Added	10,659.4	34,017.7	28,701.1
	(26,367.3)	(48,600.6)	(52,505.0)
Domestic Purchases	6,844.8	7,266.6	8,344.3
	(11,829.5)	(6,988.5)	(11,089.5)
Total Net Assets	58,350.6	146,429.6	142,314.1
	(119,388.3)	(419,194.5)	(477,344.8)
Share in Manufacturing	32.7	21.5	42.3
	(47.0)	(41.4)	(49.6)
Share in Agriculture and Mining	7.1	1.5	1.0
	(25.6)	(12.4)	(9.8)
Share in Retail and Wholesale	16.7	20.0	12.5
	(37.3)	(40.3)	(33.2)
Share in Services	43.6	56.9	44.2
	(49.7)	(49.9)	(49.9)
Firms with HQ in USA	28.8	46.2	51.9
	(45.4)	(50.2)	(50.2)
Firms with HQ in Europe	17.3	38.5	27.9
	(37.9)	(49.0)	(45.1)
Number of Firms	312	65	104

Notes: Table presents descriptive statistics for three MNE groups: (1) MNEs that did not implement an RS policy; (2) MNEs that implemented an RS policy but did not trigger first-time RS exposure events in our analysis; and (3) MNEs implementing an RS policy and triggering first-time RS exposure events in our analysis. See Section 2.3 for more details. Standard deviations are reported in parentheses. Nominal variables are in thousands of CPI-deflated 2013 U.S. dollars. These statistics represent averages from 2008 to 2019. The four sector rows report the share of each group's firms in four mutually exclusive categories that span all sectors: Manufacturing; Agriculture and Mining; Retail and Wholesale; and Services. The latter comprises electricity and gas; water supply, sewerage and waste management; construction; transportation and storage; accommodation and food services; information and communication; real estate; professional, scientific and technical activities; administrative and support services; education; human health and social work; arts, entertainment and recreation; financial activities; and other services.

Table A5: Descriptive Statistics for Exposed and Non-Exposed Domestic Suppliers to MNEs

	Non-exposed suppliers	Exposed suppliers	Difference
	(1)	(2)	(3)
Time-Invariant Characteristics			
Share in Manufacturing	9.5	12.9	-3.4
	(29.3)	(33.5)	(0.7)
Share in Agriculture and Mining	12.5	11.2	1.3
	(33.1)	(31.5)	(0.8)
Share in Retail and Wholesale	25.4	27.7	-2.3
	(43.5)	(44.7)	(1.0)
Share in Services	52.7	48.3	4.3
	(49.9)	(50.0)	(1.2)
Time-Varying Characteristics			
Total Sales (thous. U.S. dollars)	797.1	1,572.1	-775.0
	(3,261.3)	(5,080.5)	(72.2)
Number of Workers	10.7	18.3	-7.6
	(35.3)	(54.2)	(0.8)
Total Sales (thous. U.S. dollars) / Worker	124.5	124.9	-0.4
	(574.8)	(252.3)	(12.5)
Wage Bill per Worker	6.4	6.9	-0.5
	(6.9)	(5.8)	(0.1)
Share of Importers	19.3	29.2	-9.8
	(39.5)	(45.5)	(0.9)
Share of Exporters	4.8	9.4	-4.6
	(21.4)	(29.2)	(0.5)
Number of Firms	15,583	2,130	

Notes: Table presents descriptive statistics for the sample of suppliers that experience a first-time MNE RS-policy exposure event during our sample period (2009–2019) (column 2), other suppliers to MNEs that are never exposed to RS (column 1), and the difference between non-RS-exposed and RS-exposed suppliers (column 3). All time-varying variables are calculated as averages over time for each supplier. For column 2, we compute the averages using only the year prior to the first RS exposure event. Standard deviations are reported in parentheses in columns 1 and 2, and the standard error of the difference in column 3. Nominal variables are in thousands of CPI-deflated 2013 U.S. dollars. See Section 2.3 for discussion. The four sector rows report the share of suppliers in four mutually exclusive categories that span all sectors: Manufacturing; Agriculture and Mining; Retail and Wholesale; and Services. The latter comprises electricity and gas; water supply, sewerage and waste management; construction; transportation and storage; accommodation and food services; information and communication; real estate; professional, scientific and technical activities; administrative and support services; education; human health and social work; arts, entertainment and recreation; financial activities; and other services.

Table A6: Summary Statistics of RS Policy Text Measures

Variable	Mean	Median	Std. Dev.	Min.	Max.
Number of words	3491.56	2444.50	3089.05	203.00	17045.00
Labor-focus score	6.04	7.00	1.97	2.00	10.00
Number of commitments	1.67	1.00	1.61	0.00	6.00
Overall labor score	0.00	0.03	0.81	-1.54	2.35

Notes: Table reports summary statistics across the 104 RS policy documents in the analysis sample. "Number of words" is the word count of the source PDF. "Labor-focus score" is OpenAI's GPT-5 model's 1–10 assessment of how much of each document is devoted to labor standards (1 = no reference to labor standards, 10 = exclusive focus on them). "Number of commitments" counts, for each document, how many of six binary indicators related to wages and compensation are explicitly required: minimum wage compliance, above-minimum wage, overtime pay, no illegal deductions, timely payment, and an hours cap. See Appendix Table A7 for additional summary statistics on these binary indicators. "Overall labor score" is the average of the standardized "Number of commitments" and the standardized "Labor-focus score," each standardized over the 104 documents; by construction it has mean zero.

Table A7: Frequency of Specific RS Policies Based on Binary Indicators

Indicator	Yes	No	Unclear
Labor Provisions			
Wages and Compensation	79	22	3
Min. Wage Compliance	74	27	3
Above Min. Wage	11	80	13
Overtime Pay	26	59	19
No Illegal Deductions	19	72	13
Timely Payment	14	67	23
Leave Entitlements	11	78	15
Hours Cap	30	55	19
Rest Days	16	73	15
No Child or Forced Labor	93	8	3
No Child Labor	93	8	3
No Forced Labor	82	15	7
Freedom of Association	67	27	10
Health & Safety	99	2	3
No Discrimination	95	6	3
No Harassment	82	11	11
Other Non-Labor Provisions			
Environmental	99	5	0
Anti-Corruption	82	12	10

Notes: Table shows total "Yes"/"No"/"Unclear" counts for a series of binary indicators, capturing whether OpenAI's GPT-5 model classified specific labor practices as explicitly required, explicitly prohibited, or neither, across the 104 RS policy documents in the analysis sample. Responses are classified as "Yes" if the document explicitly requires or prohibits the practice, "No" if the document does not mention it, or "Unclear" if the practice is mentioned but it is ambiguous whether it is required or prohibited. Some individual indicators are grouped into broader categories, with sub-indicators shown indented below their group. For example, the "wages and compensation" category encompasses six sub-indicators: "minimum wage compliance" (supplier must pay at least the statutory minimum wage), "above minimum wage" (wages above the statutory minimum, e.g., living wage or industry standards), "overtime pay" (premium compensation for hours beyond standard limits), "no illegal deductions" (prohibition of unauthorized deductions from wages), "timely payment" (wages paid on schedule), and "leave entitlements" (legally mandated leave provided). A grouped category is classified as "Yes" if at least one constituent indicator is "Yes", "No" if all indicators are "No", and "Unclear" otherwise.

Table A8: Summary of RS Labor Policy Stringency Dimensions

Dimension	**Mean**	**Std. Dev.**	**Min.**	**Max.**
Specificity	5.11	1.38	2.00	8.00
Monitoring	4.59	1.60	1.00	9.00
Consequences	4.58	2.41	1.00	9.00
Overall	4.76	1.46	1.67	8.67

Notes: Table shows descriptive statistics for three stringency dimensions. For each RS policy document and dimension of stringency, OpenAI's GPT-5 model provides a score on a 1–10 scale. The "Specificity" dimension measures whether requirements are concrete and measurable or vague and aspirational (1–3: vague principles only, e.g., "fair wages," "safe workplace"; 4–6: some specific requirements but many left vague; 7–10: concrete, measurable standards, e.g., specific hours, procedures, or requirements). The "Monitoring" dimension assesses whether the document includes explicit audit, inspection, or verification provisions (1–3: no monitoring mentioned; 4–6: general mention of audits or inspections; 7–10: detailed audit procedures, frequency, or third-party verification). The "Consequences" dimension assesses whether there are explicit penalties for non-compliance (1–3: no consequences mentioned; 4–6: vague consequences, e.g., "may affect relationship"; 7–10: explicit consequences such as termination, corrective action requirements, or timelines). The scoring guide does not specify how to choose a score within each range; this choice is left to the model. The overall score is the simple average of the three dimension scores.

Table A9: Beyond-Law Scores and Common Beyond-Law RS Provisions

Panel A: Beyond-Law Score Distribution

Score	Frequency	Percentage
1.0	27	26.0
2.0	46	44.2
3.0	31	29.8

Panel B: Common Beyond-Law Provisions

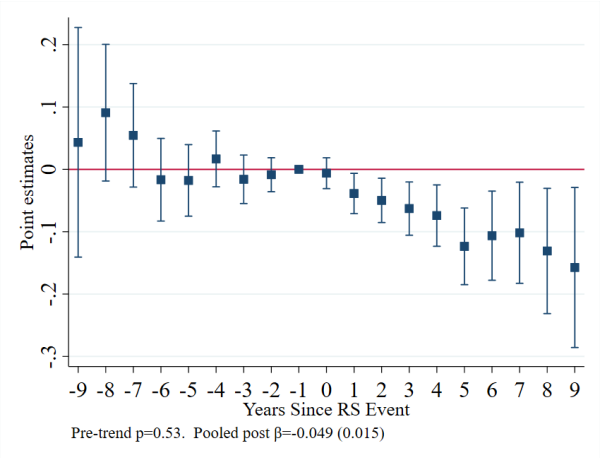
Provision Category	Frequency	Percentage
Labor Provisions		
Health & Safety Standards	43	41.3
Wages & Compensation	39	37.5
No Child/Forced Labor & Trafficking	32	30.8
Non-Discrimination & Inclusion	29	27.9
Working Hours & Leave Entitlements	21	20.2
International Standards Alignment	20	19.2
Grievance & Whistleblower Mechanisms	14	13.5
Other Non-Labor Provisions		
Supply Chain Compliance & Audit Rights	63	60.6
Environmental Responsibility	39	37.5
Anti-Corruption & Business Ethics	15	14.4

Notes: For each document, OpenAI's GPT-5 model compared the RS policy against Costa Rica's key labor regulations (see the description of Panel B) and assigned a score: 1 (only requires compliance with local/national laws), 2 (requires some standards that may exceed local law, e.g., "fair wages," "reasonable hours"), or 3 (explicitly requires standards above local minimums). Panel A shows beyond-law score frequency counts and percentages. Panel B shows the ten most common thematic categories among all provisions identified as going beyond Costa Rican legal minimums, grouped into labor and non-labor provisions and ranked by frequency within each group. For each RS policy document, the LLM compared the RS policy against Costa Rica's key labor regulations—including minimum wage (approx. $650–830 USD/month), a 48-hour standard workweek, overtime paid at 150%, a minimum working age of 15 (18 for hazardous work), four months of maternity leave, mandatory social security contributions, constitutionally protected freedom of association, a minimum of two weeks of annual leave, and severance for unjustified dismissal. The LLM was then asked to identify specific provisions that explicitly exceed these local regulatory standards. All extracted provisions were then categorized by the research team into one of ten thematic categories. Frequency counts reflect the number of RS policy documents, out of 104, containing at least one provision mapped to each category; percentages are relative to the total number of documents.

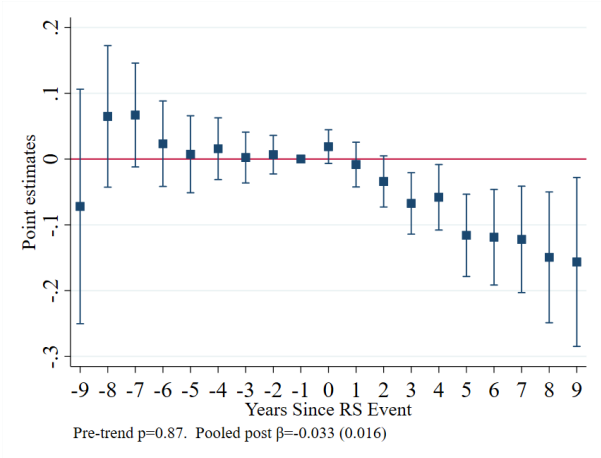
Appendix B Additional figures

Figure B1: Supplier-Level Effects of RS Rollouts on Total Sales and Employment for Extended Event-Study Periods

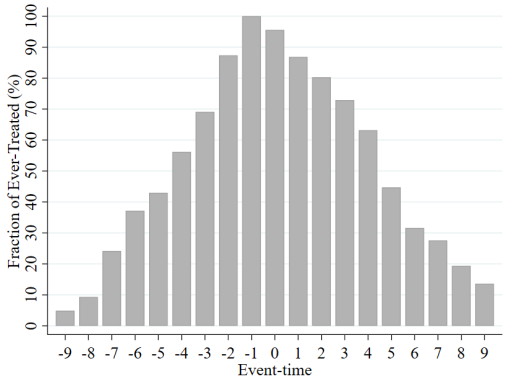
(A) Supplier Total Annual Sales



(B) Supplier Total Employment

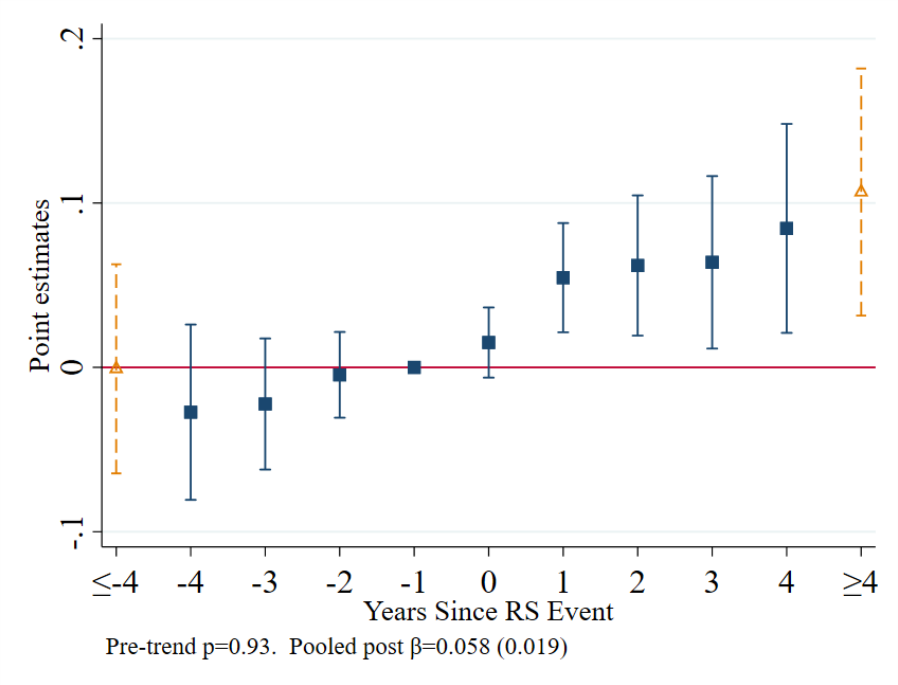


(C) Percentage of Ever-Treated Suppliers by Event Time



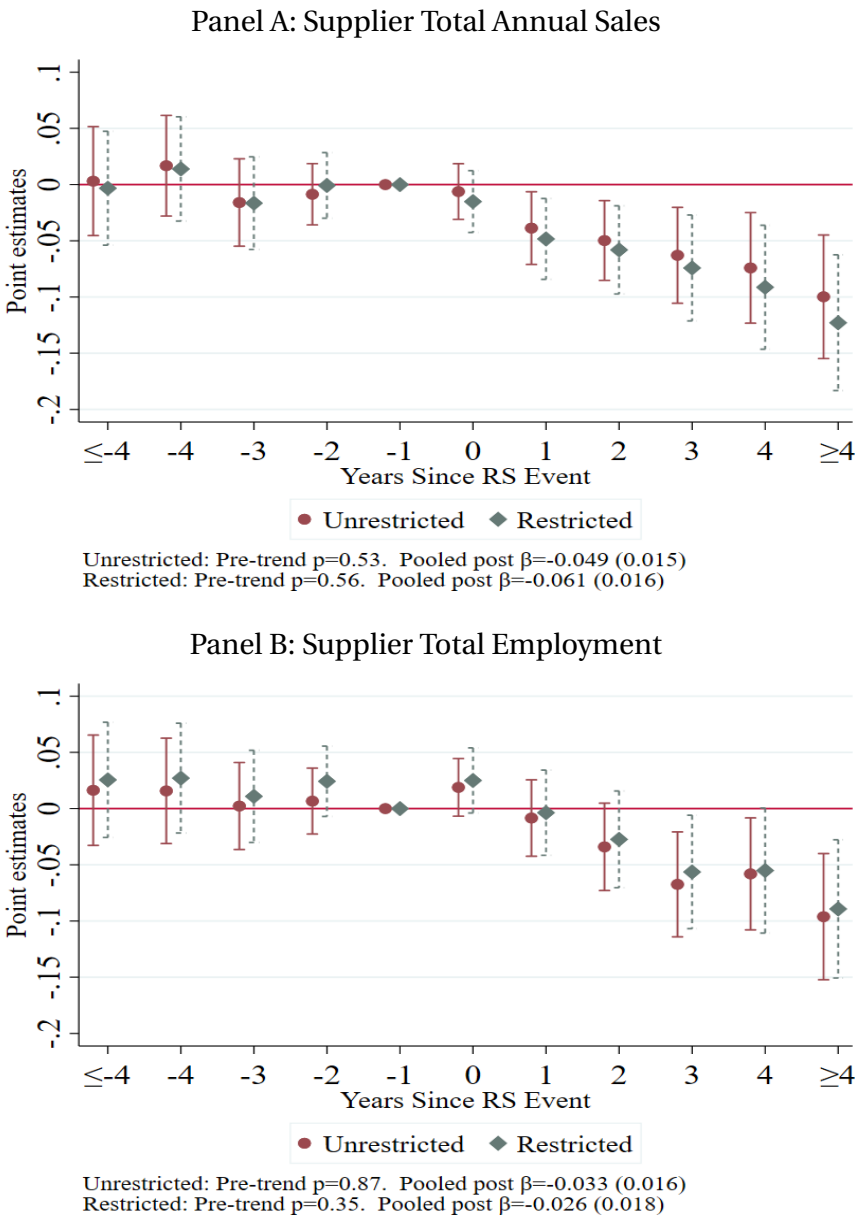
Notes: Panels A and B plot estimates from the supplier-level IV event-study specification in equation (1) for event-study lags and leads between -9 and 9. The outcomes in Panels A and B are log total sales and log number of employees, respectively. These results correspond to columns 4 and 8 of Table 1 for event-study lags and leads between -4 and 4. Both panels show 95 percent confidence intervals based on standard errors clustered at the firm level. Panel C shows the percentage of ever-treated suppliers that report positive sales and employment in the data for each event-study lag and lead between -9 and 9. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

Figure B2: Worker Separation Rate at Exposed Suppliers



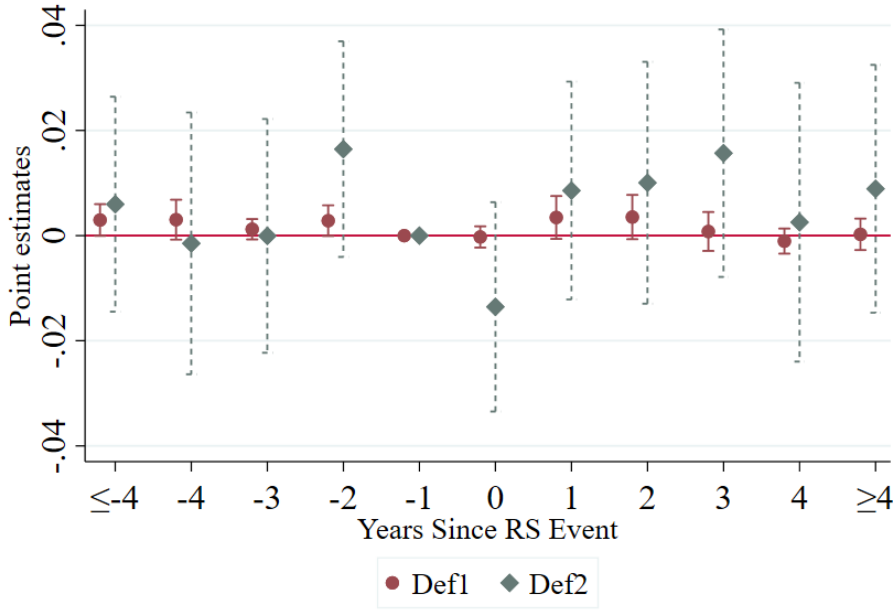
Notes: The figure reports IV event-study estimates of the effect of first-time RS exposure on the supplier-level cumulative worker separation rate, using the supplier-level event-study specification (1). The outcome is constructed as follows. In each firm-year we count separations (workers in the last year of a worker–firm employment spell, dated to the following year) and divide the count by the supplier's average employment over the sample period to obtain a per-period separation rate. The plotted outcome is the within-supplier cumulative sum of this rate: at supplier i and year t, it is the sum of that supplier's separation rate from its first observed year through year t, so the coefficient at event time η measures the accumulated separation flow η years from the event, relative to event time $\eta = -1$. The first and last panel years are dropped to avoid censoring of spells. 95 percent confidence intervals are based on standard errors clustered at the firm level. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

Figure B3: Effects of RS Rollouts on Supplier-Level Sales and Employment: Baseline Sample vs. Sample Restricted to Suppliers with a Single RS Exposure



Notes: Both panels plot estimates from the supplier-level IV event-study specification in equation (1). Each panel shows the baseline "unrestricted" results from Figure 2 together with results from a subsample where the treatment group is restricted to suppliers exposed to exactly one RS rollout. Everything else in the "restricted" subsample and regression is kept the same as in the baseline analysis. Panel A presents results for log total sales as an outcome. Panel B presents results for log number of employees as an outcome. 95 percent confidence intervals are based on standard errors clustered at the firm level. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

Figure B4: No Evidence of Supplier Splitting After Exposure to MNE RS Rollouts

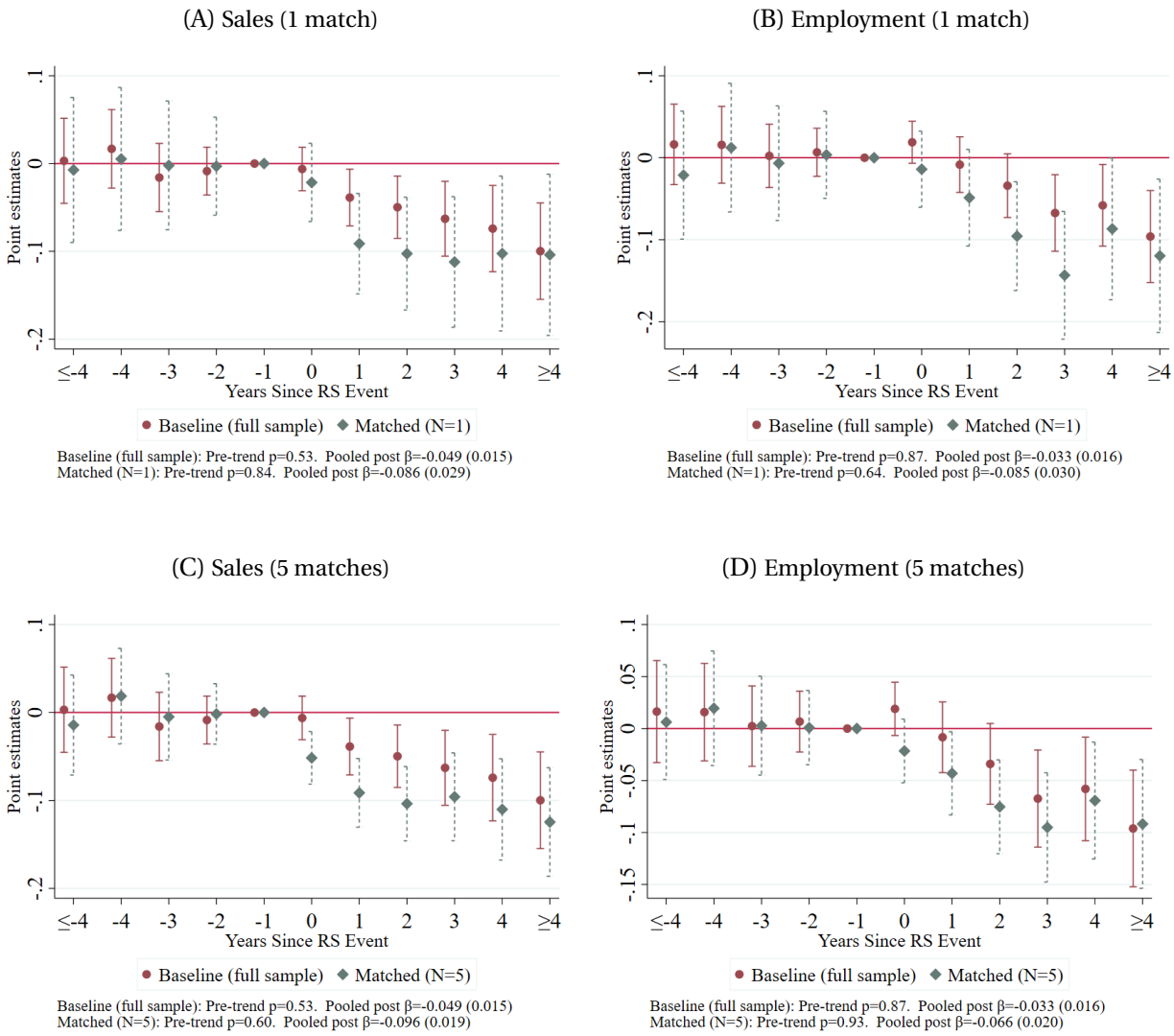


Notes: Figure explores whether some of the negative effects of a first RS rollout exposure on supplier sales and employment could be explained by suppliers "splitting up" into several entities (tax IDs). In this scenario, either the old or new entity could specialize in supplying to the RS-MNE, thereby avoiding costly compliance requirements for the rest of their production. To assess this possibility, we use the matched employer-employee data to investigate whether the propensity for linked worker movements to new tax IDs increases due to first-time exposure to an RS code of conduct. We estimate the same supplier-level IV event-study specification (1). 95 percent confidence intervals are based on standard errors clustered at the firm level. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

We use two alternative measures of "firm splitting." The first (Def1) is an outsourcing measure similar to that used in Goldschmidt and Schmieder (2017). The outcome is an indicator equal to 1 if there is a flow of workers from one tax ID (an MNE supplier) to another tax ID satisfying four conditions: (i) the linked movement of workers represents more than 30% of the initial firm's employment in year $(t-1)$; (ii) either the new employer is a new firm (new tax ID), or the inflow represents at least 66% of the receiving firm's employment in year t; (iii) the number of workers at the initial firm in year $(t-1)$ is at least 5; and (iv) the receiving firm operates in the same sector as the originating firm.

The outcome variable in the second definition (Def2) is: $X_{i,t} = \max_{j \neq i} \frac{w_{i,t}^{j,t+1}}{\sum_{k \neq i} w_{i,t}^{k,t+1}}$, where $w_{i,t}^{j,t+1}$ is the number of workers who move from firm i in year t to firm j in year $(t+1)$. We only consider workers who move to another firm (excluding those who continue in the same firm or are unemployed) in year $(t+1)$. Hence, this variable measures the largest relative importance of a receiving firm j in terms of attracting new workers from i across all firms receiving workers from i over the same t to $(t+1)$ period, with a value of 1 at the maximum.

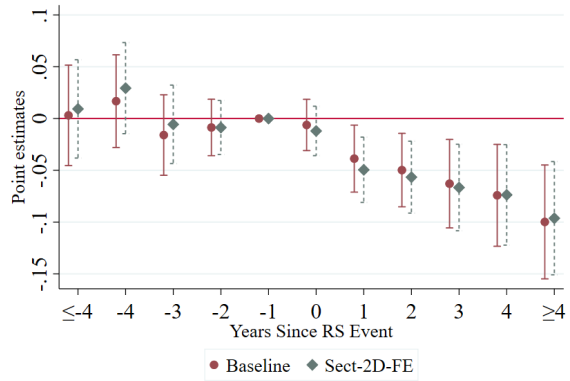
Figure B5: Baseline vs. Matched-DiD Event Studies for Supplier-Level Sales and Employment



Notes: The panels plot estimates from the supplier-level IV event-study specification in equation (1), overlaying our baseline estimates with the matching estimation. The outcomes are log total sales in Panels B5A and B5C and log number of employees in Panels B5B and B5D. The matching estimates compare changes in outcomes among treated suppliers to those of matched untreated suppliers. N indicates the number of matched suppliers (nearest or nearest 5). All panels show 95 percent confidence intervals; standard errors are clustered at the firm level. The pre-trend p-value and the pooled post β (SE) are reported for each series and come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

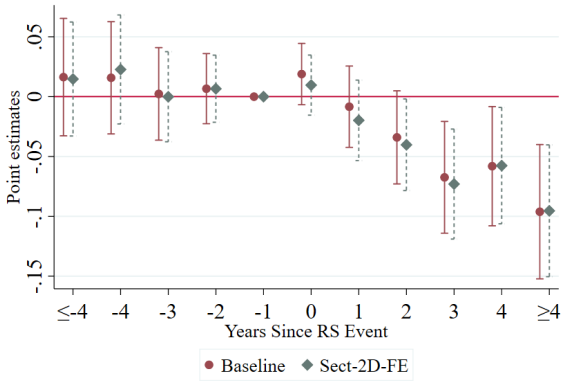
Figure B6: Spillover Robustness of Supplier-Level Effects of RS Rollouts

(A) 2-Digit Sector-by-Year FE—Log Sales



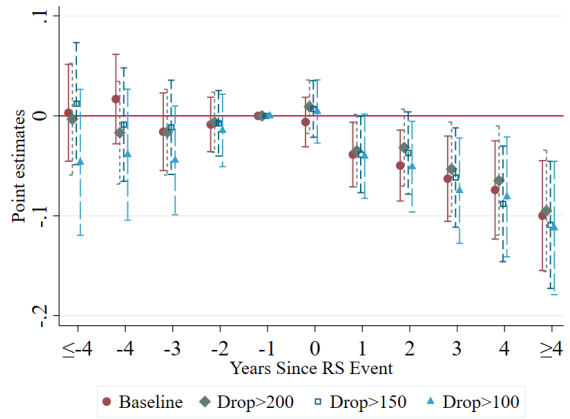
Baseline: Pre-trend p=0.53. Pooled post β=-0.049 (0.015)
Sect-2D-FE: Pre-trend p=0.79. Pooled post β=-0.054 (0.015)

(B) 2-Digit Sector-by-Year FE—Log Empl.

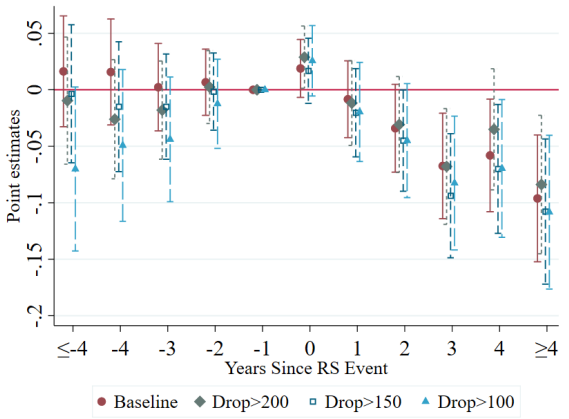


Baseline: Pre-trend p=0.87. Pooled post β=-0.033 (0.016)
Sect-2D-FE: Pre-trend p=0.89. Pooled post β=-0.039 (0.016)

(C) Dropping Largest Events—Log Sales

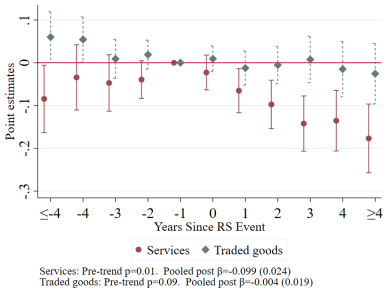


(D) Dropping Largest Events—Log Empl.

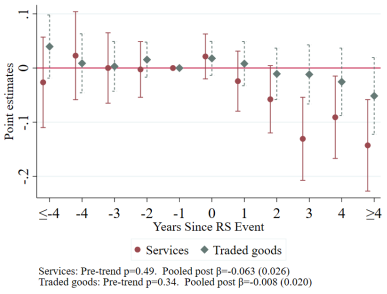


Notes: The figure plots supplier-level IV event-study estimates from equation (1) for sales (Panels B6A and B6C) and employment (Panels B6B and B6D). In each panel, the "Baseline" series reproduces the corresponding results from Figure 2 and is overlaid with estimates from alternative specifications. Panels B6A and B6B present the "Sect-2D-FE" specification, which replaces the baseline 4-digit sector-by-year fixed effects with 2-digit sector-by-year fixed effects (spanning 73 distinct 2-digit sectors). Panels B6C and B6D re-estimate the baseline after progressively excluding the treated suppliers belonging to RS rollout events that exposed more than 200, 150, or 100 suppliers from the estimation sample. All panels report IV estimates with 95 percent confidence intervals based on standard errors clustered at the firm level. In Panels B6A and B6B, the pre-trend p-value and the pooled post β (SE) are reported for each series and come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

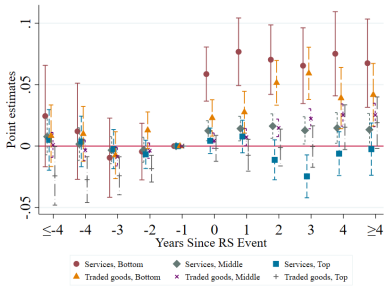
Figure B7: Heterogeneity of Supplier Sales, Employment, and Worker Earnings Effects of MNE RS Rollouts, by Supplier Sector and MNE Headquarters Management Score



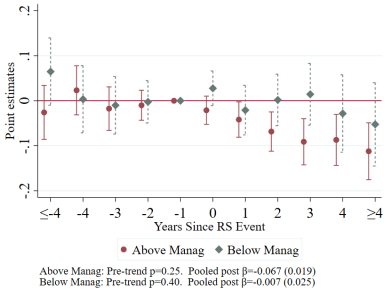
(A) Supplier Sales, by Sector



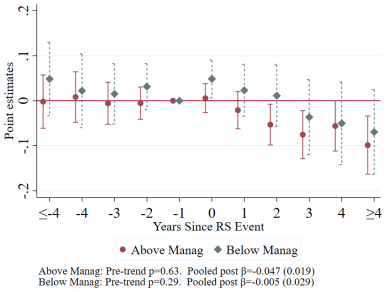
(B) Supplier Employment, by Sector



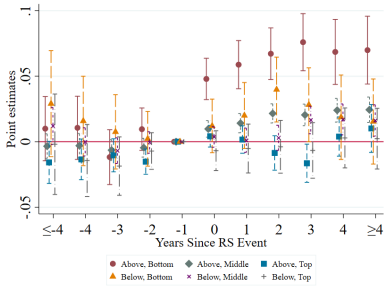
(C) Worker Earnings, by Sector



(D) Supplier Sales, by MNE Management



(E) Supplier Employment, by MNE Management

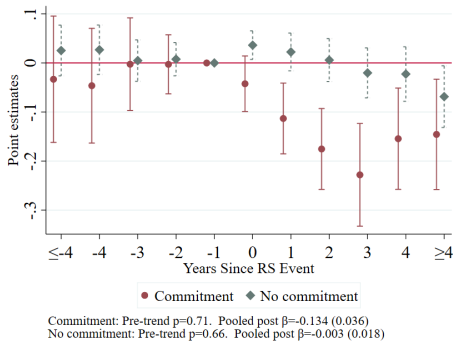


(F) Worker Earnings, by MNE Management

Notes: Figure presents the heterogeneity of the supplier-level sales and employment effects and the worker-level earnings effects of RS rollouts along two supplier groupings. The top row (Panels A–C) separates suppliers into services and traded-goods sectors. Services comprises all sectors other than manufacturing, agriculture and mining, and retail and wholesale trade. See note in Appendix Table A4 for more details. The bottom row (Panels D–F) separates suppliers based on whether the MNE buyer that triggered their event is headquartered in a country whose average firm management score is above or below the median of the set of MNE HQ countries; the country-level management score is taken from Bloom et al. (2021). Panels A (B) and D (E) plot the supplier-level total sales (employment) effect from the IV event-study specification equation (1). Panels C and F plot the worker-level log-earnings effect from the worker-level IV event-study specification in equation (2), with estimates reported separately by workers' initial wage group. All panels show 95 percent confidence intervals. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference. See Section 3.2 for discussion.

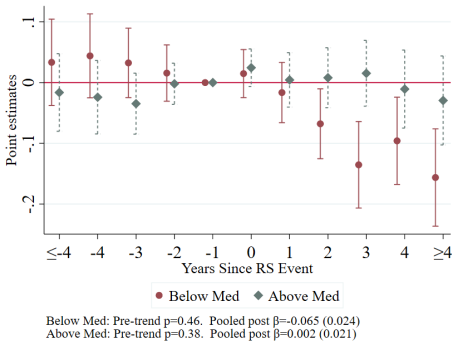
Figure B8: Heterogeneity of Supplier-Level Effects of RS Rollouts for Employment

(A) Policy Requires Above Minimum Wage Pay



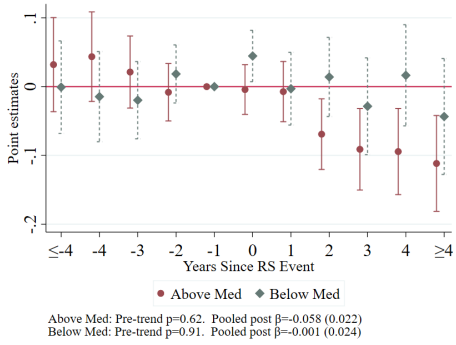
Commitment: Pre-trend p=0.71. Pooled post β=-0.134 (0.036)
No commitment: Pre-trend p=0.66. Pooled post β=-0.003 (0.018)

(B) Supplier Sector with Small vs. Large Firms



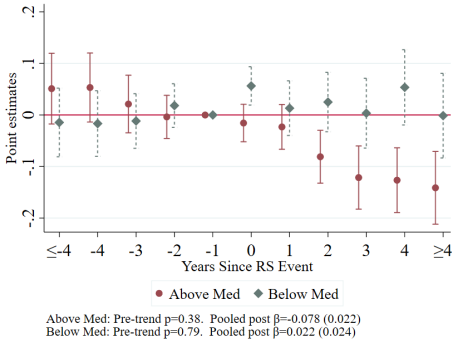
Below Med: Pre-trend p=0.46. Pooled post β=-0.065 (0.024)
Above Med: Pre-trend p=0.38. Pooled post β=0.002 (0.021)

(C) Policy Overall Labor Score



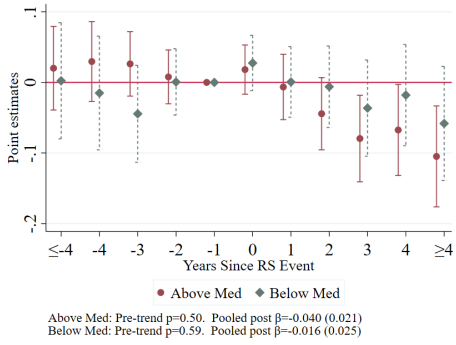
Above Med: Pre-trend p=0.62. Pooled post β=-0.058 (0.022)
Below Med: Pre-trend p=0.91. Pooled post β=-0.001 (0.024)

(D) Policy Compensation-Related Word Share



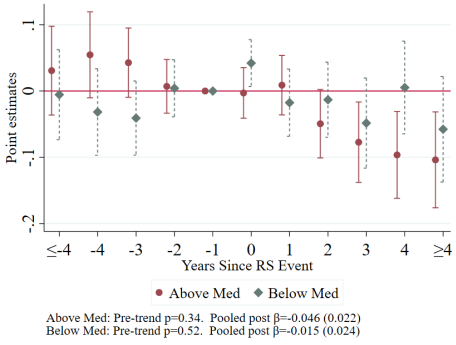
Above Med: Pre-trend p=0.38. Pooled post β=-0.078 (0.022)
Below Med: Pre-trend p=0.79. Pooled post β=0.022 (0.024)

(E) Policy Health & Safety Word Share



Above Med: Pre-trend p=0.50. Pooled post β=-0.040 (0.021)
Below Med: Pre-trend p=0.59. Pooled post β=-0.016 (0.025)

(F) Policy Enforcement & Compliance Word Share

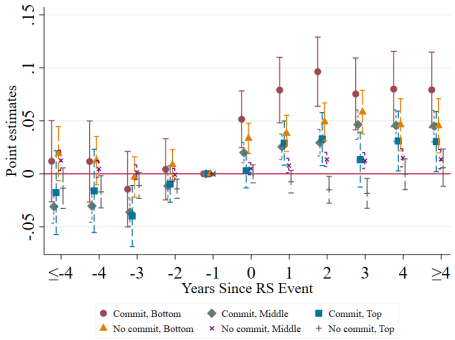


Above Med: Pre-trend p=0.34. Pooled post β=-0.046 (0.022)
Below Med: Pre-trend p=0.52. Pooled post β=-0.015 (0.024)

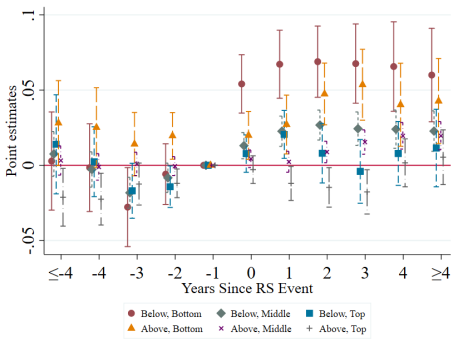
Notes: Figure replicates the heterogeneity analysis of Figure 3 for log total employment, estimated from the supplier-level IV event-study specification in equation (1). The supplier groups and panels are defined exactly as in Figure 3; see the note in that figure for the detailed definition of each split. All panels show 95 percent confidence intervals based on standard errors clustered at the firm level. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

Figure B9: Heterogeneity of Worker-Level Earnings Effects of RS Rollouts, by Worker Wage Group

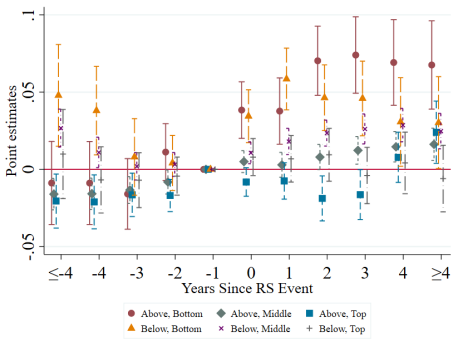
(A) Policy Requires Above Minimum Wage Pay



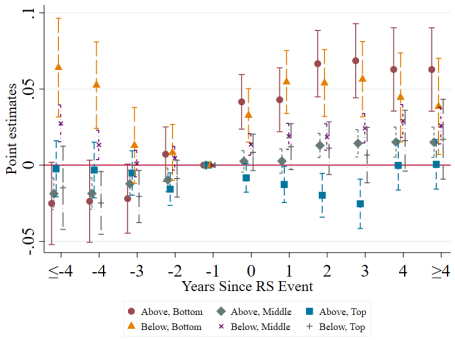
(B) Supplier Sector with Small vs. Large Firms



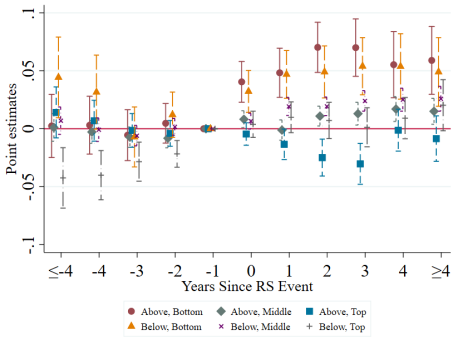
(C) Policy Overall Labor Score



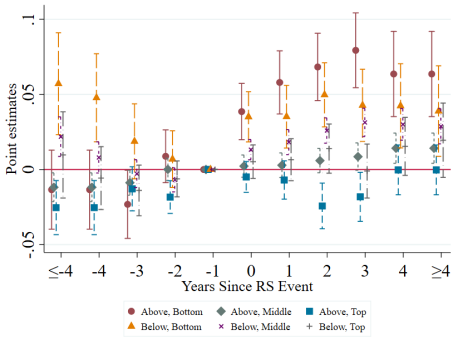
(D) Policy Compensation-Related Word Share



(E) Policy Health & Safety Word Share

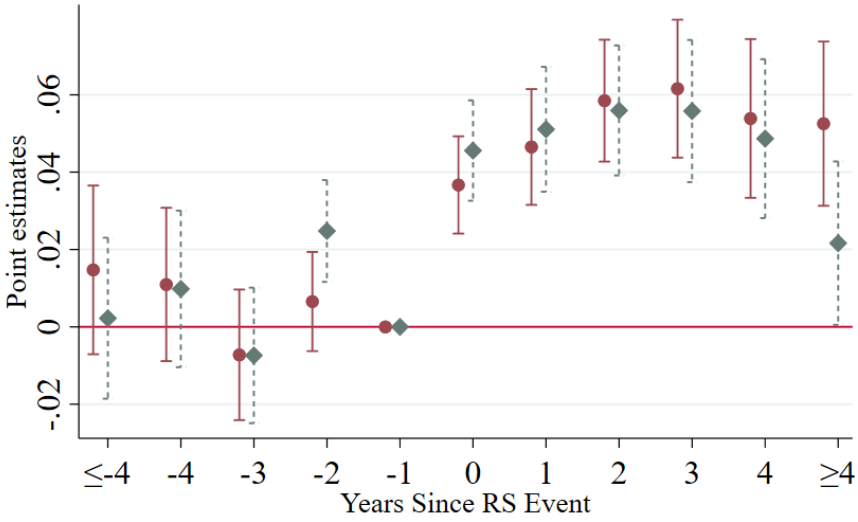


(F) Policy Enforcement & Compliance Word Share



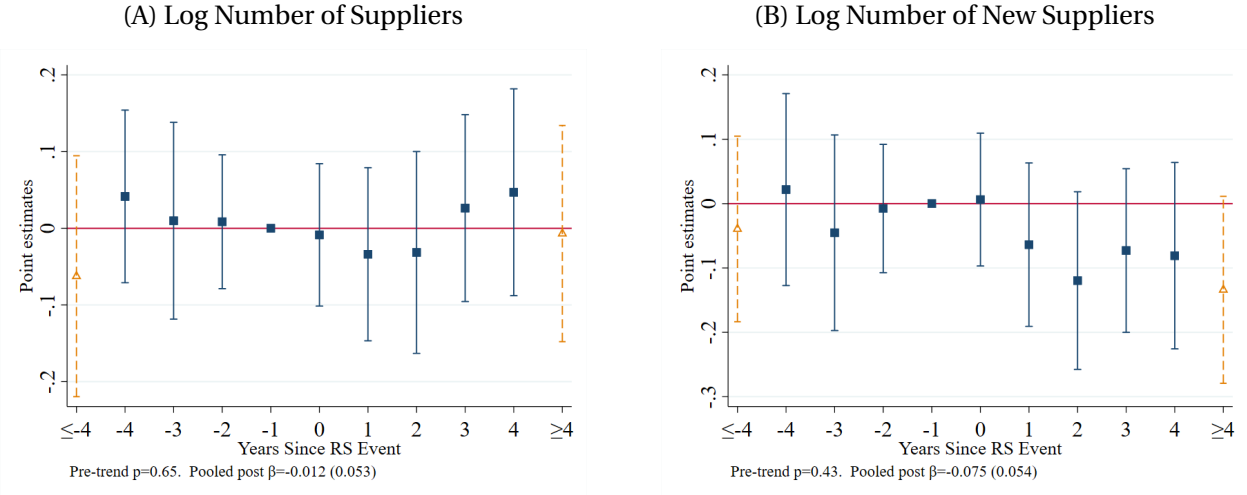
Notes: Figure replicates the heterogeneity analysis of Figure 3 for log worker earnings, estimated from the worker-level IV event-study specification in equation (2). Within each panel and supplier group, estimates are reported separately by worker wage group (the bottom, middle, and top quartiles of initial worker earnings). The supplier groups and panels are defined exactly as in Figure 3; see the note in that figure for the detailed definition of each split. All panels show 95 percent confidence intervals based on standard errors clustered at the worker level.

Figure B10: Earnings Effects on Exposed Bottom-Quartile Workers Regardless of Employer



Notes: The figure reports IV event-study estimates from two specifications. The series labeled "Baseline" comes from our baseline worker-level specification (2), in which exposure is defined at the level of the supplier. The series labeled "Worker timeline" comes from a worker-level specification, in which exposure is defined at the level of the worker (being employed at an exposed supplier at $\eta = -1$) rather than at the level of the supplier. That specification includes worker fixed effects and sector-by-year fixed effects but not worker-by-firm fixed effects. The pre-trend p-value and the pooled post β (SE) are reported for each series and come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

Figure B11: Effects of RS Rollouts on the Number of CR Suppliers to the MNE



Notes: The figure reports MNE-level event-study estimates of the effect of RS rollouts on the number of Costa Rican suppliers selling to the MNE. Panel B11A uses the log of the total number of suppliers transacting with the MNE in each year. Panel B11B uses the log of one plus the number of new suppliers, where a supplier is classified as new in year t if t is the first year in which that supplier–MNE pair appears in the transaction data. Estimates include MNE fixed effects and sector-by-year fixed effects. 95 percent confidence intervals are based on standard errors clustered at the MNE level. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

Appendix B.1 Workplace amenities, non-discrimination and environmental practices

Panel A of Appendix Figure B12 presents event-study results suggesting that RS exposure reduces workers' time on accident leave, which could be due to fewer workplace accidents and less severe ones. Panel B shows a positive, although insignificant, effect of RS on the average time of maternity leave.[48] Panel C of Appendix Figure B12 indicates positive effects on suppliers' purchases of health services, though only the pooled post-rollout estimate is significant at conventional levels. Finally, Panel D presents suggestive evidence of positive effects of RS rollouts on the share of female workers, which could reflect commonly included RS clauses on non-discrimination in hiring practices. Together, these findings suggest that RS policies not only increase worker earnings but may also improve some dimensions of workplace amenities. Both our theory and quantification speak to this possibility (Sections 4.5 and 5).

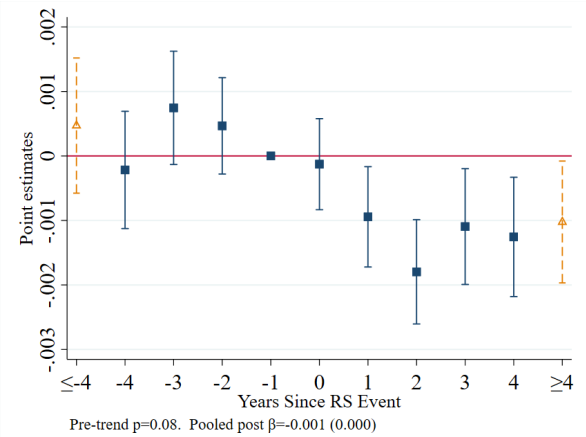
While the RS policies in our sample primarily focus on labor standards, they also include environmental provisions such as waste reduction and recycling, pollution control, and climate change mitigation. A natural question is to what extent such non-labor-related provisions may be driving the effects of RS on suppliers that we estimate in Section 3. To assess this, we can first note from the results in Section 3.2 that the adverse effect of RS on supplier sales is concentrated in

[48] Appendix Figure B13 provides suggestive evidence that both effects are concentrated among low-earning workers.

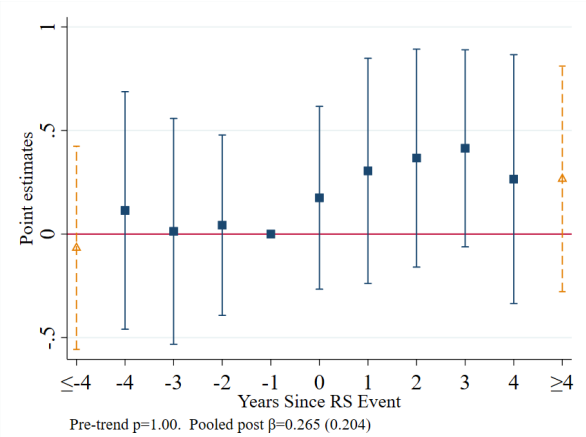
the service sector. Given services' generally low environmental impact (relative to manufacturing, agriculture or mineral extraction), this suggests that environmental standards are unlikely to be the primary constraints imposed by RS policies in our context. Second, according to the International Energy Agency, CR has one of the highest shares of energy supplied from renewable sources worldwide (e.g., geothermal, hydro, and biofuels). As a result, more than 99% of total CO_2 emissions in CR stem from fuel combustion, with 75% coming from the transportation sector. In this context, we can use our sales transaction data to test whether RS rollouts affect the suppliers' expenditures on transportation services, sewage and waste management services, electricity, or gas. We find no evidence for such effects, with point estimates close to zero (Appendix Figure B14). Finally, we can examine whether supplier exposure to "placebo" RS rollouts—those without labor-related provisions that we exclude in our estimation, focused on other areas such as environmental practices—impacts supplier outcomes in CR in similar ways as in our estimation sample. Implementing the same supplier-level event-study design as in Section 3, we find no significant treatment effects of RS codes without labor standards, with point estimates close to zero (see Appendix Figure B15). Together, these findings suggest that the effects of RS rollouts in our context are primarily driven by the labor-related provisions that are the main focus of RS codes in our estimation sample. We also discuss these questions as part of both the model and counterfactual analysis, where we consider alternative assumptions about the nature of the RS-induced increase in supplier costs.

Figure B12: Effects of RS Rollouts on Workplace Amenities and Non-Discrimination

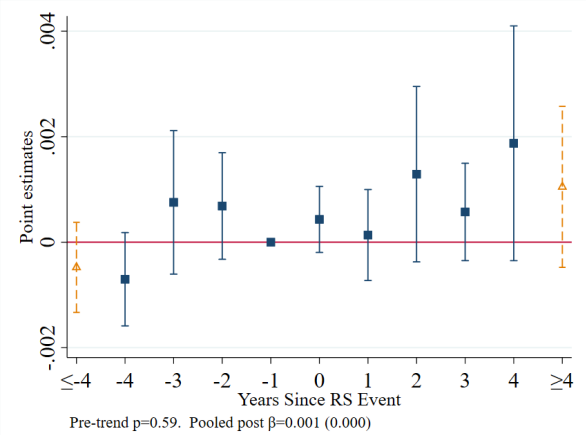
(A) Accident Leave Among All Employees



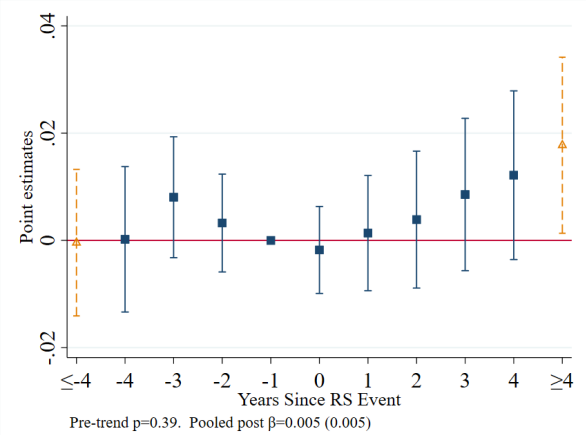
(B) Maternity Leave Duration Among Women



(C) Supplier Purchases of Health Services



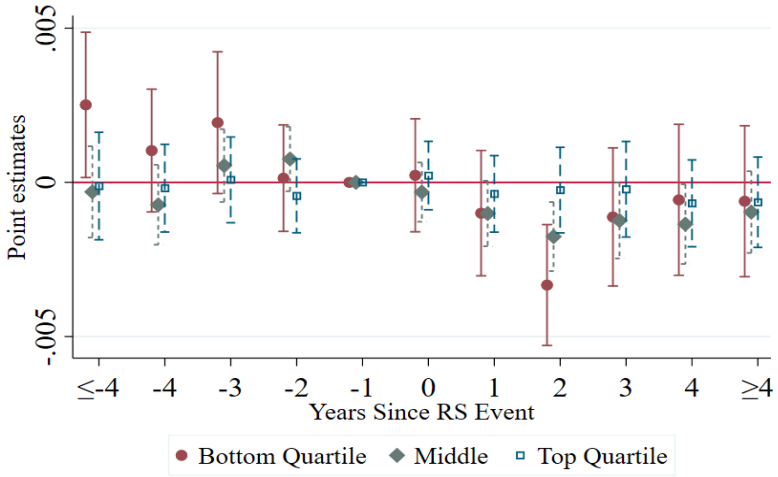
(D) Female Share Among All Employees



Notes: Panel B12A shows worker-level event-study estimates for the share of months a worker was on leave due to a work-related accident. All other panels show estimates from supplier-level regressions. In Panel B12B, the outcome is the average number of months of paid maternity leave among women who take such leave. In Panel B12C, the outcome is the share of total input purchases from suppliers in ISIC Rev. 4 sector 8690 (Other human health activities), which encompasses services such as physiotherapy, vision testing (optometry), medical therapeutic massage, occupational therapy, chiropractic, homeopathy, spinal massage, and acupuncture. In Panel B12D, the outcome is the share of female workers among all employees. All panels show IV estimates and their 95 percent confidence intervals. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

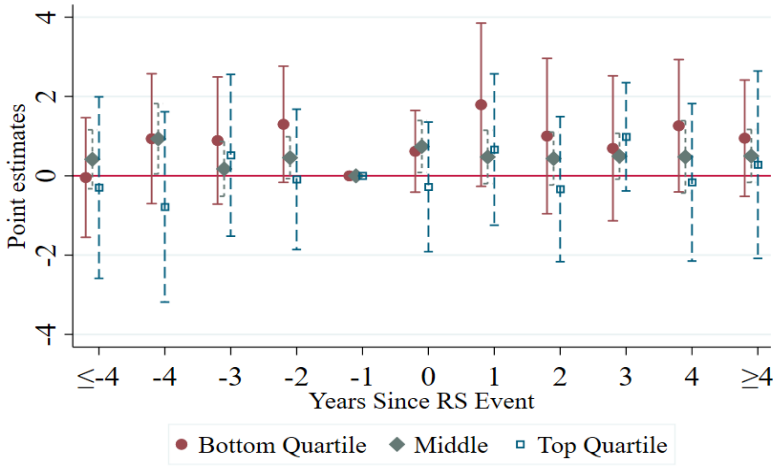
Figure B13: Heterogeneous Effects of Exposure to MNE RS Rollouts on Workplace Accidents and Maternity Leave by Workers' Initial Earnings Quartile

Panel A: Accident Leave: By Initial Worker Earnings



Bottom Quartile: Pre-trend p=0.12. Pooled post β=-0.001 (0.001)
Middle: Pre-trend p=0.27. Pooled post β=-0.001 (0.000)
Top Quartile: Pre-trend p=0.67. Pooled post β=-0.000 (0.000)

Panel B: Maternity Leave Duration: By Initial Worker Earnings

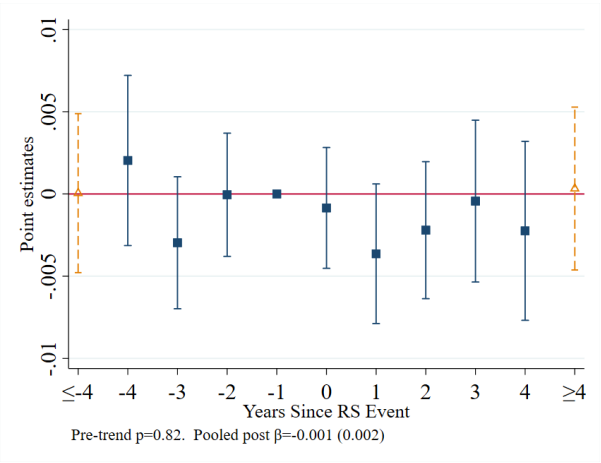


Bottom Quartile: Pre-trend p=0.49. Pooled post β=0.761 (0.568)
Middle: Pre-trend p=0.14. Pooled post β=0.579 (0.254)
Top Quartile: Pre-trend p=0.98. Pooled post β=0.023 (0.709)

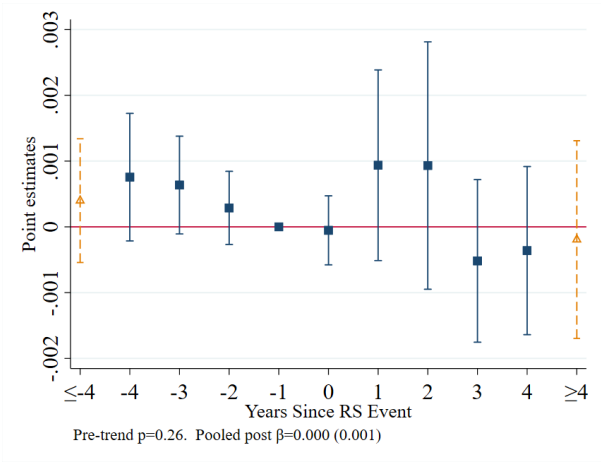
Notes: Panel A shows worker-level IV event-study estimates for the share of months a worker was on leave due to a work-related accident. Panel B shows supplier-level IV event-study estimates for the average number of months of paid maternity leave for female workers, conditional on taking maternity leave. These panels present a heterogeneity decomposition of the findings established across all workers in Panels A and B of Appendix Figure B12, with workers separated by their initial earnings quartile. All panels show 95 percent confidence intervals. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

Figure B14: Effects of RS Rollouts on Supplier Environmental Practices

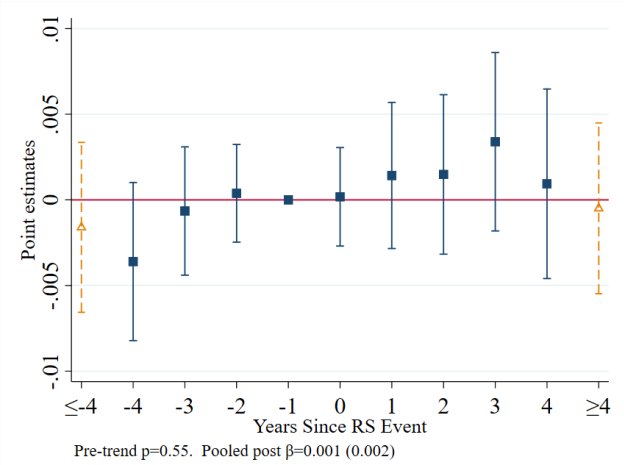
(A) Purchases of Transportation & Travel Services



(B) Purchases of Sewage & Waste Mngmt Services



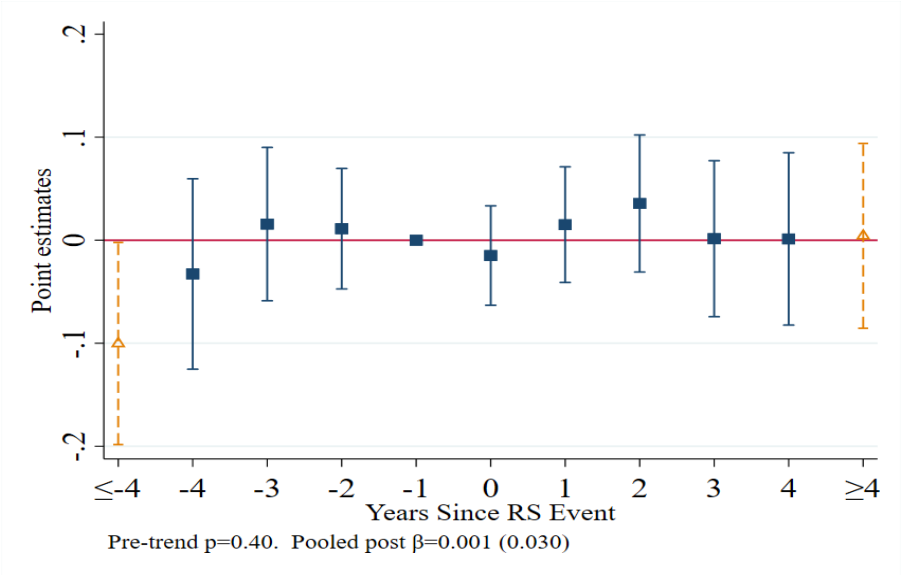
(C) Purchases of Electricity and Gas



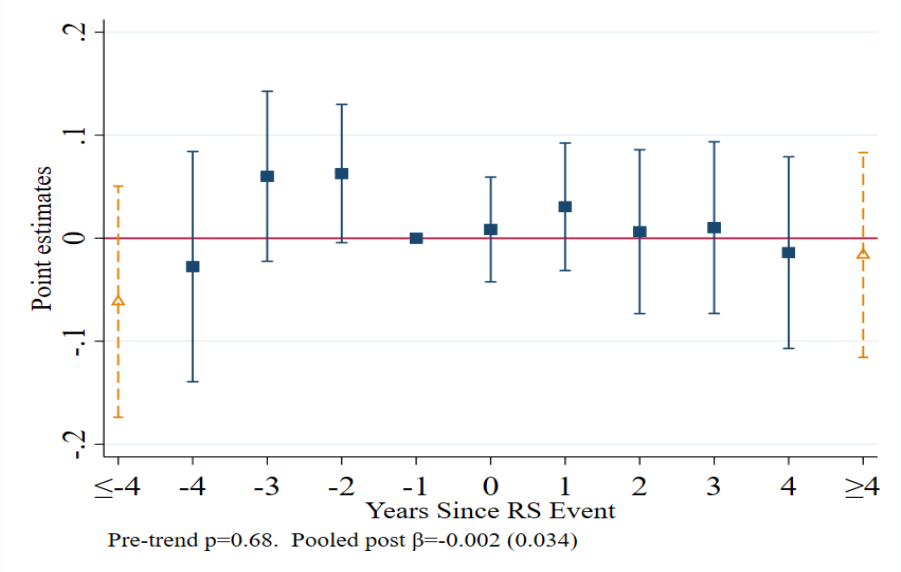
Notes: All panels plot estimates from the supplier-level IV event-study specification in equation (1). The outcomes in all panels represent the value share of arm's-length input purchases by suppliers from firms operating in specific ISIC Rev. 4 sectors. Panel A focuses on transportation (land, rail, water, and air) and travel (sectors 49–51 and 79). Panel B focuses on sewerage and waste collection (sectors 36*–39*). Panel C focuses on electricity and gas (sectors 3510 and 3520). 95 percent confidence intervals are based on standard errors clustered at the firm level. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

Figure B15: Supplier-Level Effects of Placebo Rollouts on Annual Sales and Employment

Panel A: Supplier Annual Sales



Panel B: Supplier Employment



Notes: Panels A and B plot estimates from the supplier-level IV event-study specification in equation (1), using placebo events. These placebo events are triggered by MNEs whose first RS rollout does not include directives related to labor standards but instead focuses on other areas, such as environmental or governance practices. These rollouts account for 19% of all first MNE RS rollouts in our search protocol. The placebo analysis sample excludes suppliers that have ever sold to an RS-labor-active MNE, regardless of whether they belong to the treated or never-treated placebo groups. Apart from this exclusion, the placebo analysis sample is constructed identically to the baseline analysis sample. The outcomes in Panels A and B are log total sales and log number of employees, respectively. Both panels show 95 percent confidence intervals based on standard errors clustered at the firm level. The pre-trend p-value and the pooled post β (SE) come from one regression that replaces the event dummies with a single pre-rollout indicator ($\eta \leq -2$) and a single post-rollout indicator ($\eta \geq 0$), keeping $\eta = -1$ as the omitted reference.

Appendix C Additional event-study tables

Table C1: Effects of RS Rollouts on Suppliers' Relative Employment of Initially Low vs. High Earning Workers

	(1)	(2)	(3)	(4)	(5)
	TWFE	TWFE	SA	IV	PPML
$\eta \leq -4$	0.034	0.019	-0.008	0.048	0.242***
	(0.039)	(0.040)	(0.041)	(0.043)	(0.092)
$\eta = -4$	0.025	0.013	-0.010	0.033	0.180**
	(0.039)	(0.039)	(0.040)	(0.042)	(0.092)
$\eta = -3$	0.026	0.014	0.001	0.026	0.118
	(0.033)	(0.033)	(0.033)	(0.035)	(0.080)
$\eta = -2$	0.030	0.019	0.022	0.022	0.179
	(0.024)	(0.025)	(0.025)	(0.025)	(0.118)
$\eta = 0$	-0.021	-0.022	-0.021	-0.016	-0.001
	(0.021)	(0.022)	(0.022)	(0.024)	(0.052)
$\eta = 1$	-0.057**	-0.066**	-0.067**	-0.072**	-0.054
	(0.026)	(0.027)	(0.027)	(0.029)	(0.056)
$\eta = 2$	-0.050*	-0.063**	-0.066**	-0.066**	-0.087
	(0.028)	(0.029)	(0.029)	(0.032)	(0.063)
$\eta = 3$	-0.065**	-0.085**	-0.088***	-0.080**	-0.090
	(0.032)	(0.033)	(0.034)	(0.036)	(0.067)
$\eta = 4$	-0.067*	-0.100***	-0.106***	-0.111***	-0.169**
	(0.035)	(0.038)	(0.039)	(0.042)	(0.073)
$\eta \geq 4$	-0.021	-0.070*	-0.050	-0.080*	-0.141**
	(0.036)	(0.039)	(0.044)	(0.043)	(0.069)
Firm FE	Yes	Yes	Yes	Yes	Yes
Year-Sect FE	Yes	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes	Yes
Adjusted R^2	0.67	0.68	0.68	–	–
# Observations	60,391	60,391	60,391	60,391	103,881
# Treated	1,456	1,456	1,456	1,456	1,664
# Never Treated	9,128	9,128	9,128	9,128	11,362
# Sector-Year Bins	2,400	2,400	2,400	2,400	2,526

Notes: Table presents two-way fixed effects (TWFE), Sun and Abraham (2021) (SA), IV and PPML estimates for the supplier-level specification in equation (1). The outcome variable in columns 1–4 is the log ratio of initially low-earning to initially high-earning workers. In the PPML specification (column 5), the outcome is the same ratio but without the log transformation. The first-stage F-statistic for the IV column exceeds 50. Columns with "Yes" in the "Controls" row include the full timeline of event-study indicators for having had positive sales to any MNE subsidiary in CR. Standard errors are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. See Section 3.2 for discussion.

Table C2: Worker-Level Effects of Exposure to MNE RS Rollouts on Labor Earnings: All Workers

	(1)	(2)	(3)	(4)
	TWFE	TWFE	SA	IV
$\eta \leq -4$	-0.001	0.005	0.010***	-0.002
	(0.003)	(0.003)	(0.003)	(0.004)
$\eta = -4$	0.002	0.005	0.007**	0.001
	(0.003)	(0.003)	(0.003)	(0.003)
$\eta = -3$	-0.002	-0.000	0.002	-0.004
	(0.003)	(0.003)	(0.003)	(0.003)
$\eta = -2$	-0.002	-0.001	-0.001	-0.001
	(0.002)	(0.002)	(0.002)	(0.002)
$\eta = 0$	0.007***	0.006***	0.006***	0.010***
	(0.002)	(0.002)	(0.002)	(0.002)
$\eta = 1$	0.007***	0.007***	0.007***	0.012***
	(0.002)	(0.002)	(0.002)	(0.003)
$\eta = 2$	0.008***	0.009***	0.009***	0.015***
	(0.002)	(0.002)	(0.002)	(0.003)
$\eta = 3$	0.001	0.003	0.002	0.015***
	(0.003)	(0.003)	(0.003)	(0.003)
$\eta = 4$	-0.001	0.003	0.004	0.017***
	(0.003)	(0.003)	(0.003)	(0.003)
$\eta \geq 4$	0.002	0.003	0.004	0.019***
	(0.003)	(0.003)	(0.004)	(0.004)
Year-Sect FE	Yes	Yes	Yes	Yes
Worker-Firm FE	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes
Adjusted R^2	0.81	0.81	0.81	–
# Observations	2,847,619	2,847,619	2,847,619	2,847,619
# Firms	57,487	57,487	57,487	57,487
# Workers	435,450	435,450	435,450	435,450

Notes: Table presents two-way fixed effects (TWFE), Sun and Abraham (2021) (SA) and IV estimates from the worker-level version of the specification in equation (2). The reported estimates correspond to all workers. The outcome is the log of annual worker earnings divided by the number of months of employment, restricted to worker-months in full-time employment. The first-stage F-statistic for the IV column exceeds 50. Columns with "Yes" in the "Controls" row include the full timeline of event-study indicators for having had positive sales to any MNE subsidiary in CR. Standard errors clustered at the worker level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. See Section 3.2 for discussion.

Table C3: Worker-Level Effects of RS Rollouts on Labor Earnings: Bottom Quartile of Initial Earnings

	(1) TWFE	(2) TWFE	(3) SA	(4) IV
$\eta \leq -4$	-0.007	0.005	-0.006	0.015
	(0.009)	(0.009)	(0.010)	(0.011)
$\eta = -4$	0.000	0.005	0.005	0.011
	(0.009)	(0.009)	(0.010)	(0.010)
$\eta = -3$	-0.019**	-0.016**	-0.018**	-0.007
	(0.008)	(0.008)	(0.008)	(0.009)
$\eta = -2$	-0.004	-0.003	-0.003	0.007
	(0.006)	(0.006)	(0.006)	(0.007)
$\eta = 0$	0.029***	0.028***	0.029***	0.037***
	(0.006)	(0.006)	(0.006)	(0.006)
$\eta = 1$	0.041***	0.040***	0.043***	0.047***
	(0.007)	(0.007)	(0.007)	(0.008)
$\eta = 2$	0.046***	0.045***	0.047***	0.058***
	(0.007)	(0.007)	(0.008)	(0.008)
$\eta = 3$	0.045***	0.045***	0.047***	0.062***
	(0.008)	(0.008)	(0.009)	(0.009)
$\eta = 4$	0.036***	0.038***	0.042***	0.054***
	(0.009)	(0.010)	(0.010)	(0.010)
$\eta \geq 4$	0.043***	0.038***	0.048***	0.053***
	(0.009)	(0.010)	(0.012)	(0.011)
Year-Sect FE	Yes	Yes	Yes	Yes
Worker-Firm FE	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes
Adjusted R^2	0.71	0.71	0.71	–
# Observations	669,664	669,664	669,664	669,664
# Firms	34,261	34,261	34,261	34,261
# Workers	119,369	119,369	119,369	119,369

Notes: Table presents two-way fixed effects (TWFE), Sun and Abraham (2021) (SA) and IV estimates from the worker-level version of the specification in equation (2). We group workers based on their quartile in the distribution of (inflation-adjusted) monthly earnings in the first year we observe each worker, starting from 2006. The reported estimates correspond to workers in the bottom quartile of initial earnings. The outcome is the log of annual worker earnings divided by the number of months of employment, restricted to worker-months in full-time employment. The first-stage F-statistic for the IV column exceeds 50. Columns with "Yes" in the "Controls" row include the full timeline of event-study indicators for having had positive sales to any MNE subsidiary in CR. Standard errors clustered at the worker level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. See Section 3.2 for discussion.

Table C4: Worker-Level Effects of RS Rollouts on Labor Earnings: Middle Quartiles of Initial Earnings

	(1)	(2)	(3)	(4)
	TWFE	TWFE	SA	IV
$\eta \leq -4$	-0.007**	0.001	0.007*	0.004
	(0.003)	(0.003)	(0.004)	(0.004)
$\eta = -4$	-0.001	0.001	0.002	-0.002
	(0.003)	(0.003)	(0.004)	(0.004)
$\eta = -3$	-0.002	-0.000	-0.001	-0.007**
	(0.003)	(0.003)	(0.003)	(0.003)
$\eta = -2$	-0.004	-0.003	-0.003	-0.003
	(0.002)	(0.002)	(0.002)	(0.003)
$\eta = 0$	0.005**	0.004*	0.005**	0.007***
	(0.002)	(0.002)	(0.002)	(0.003)
$\eta = 1$	0.007***	0.006**	0.007***	0.010***
	(0.003)	(0.003)	(0.003)	(0.003)
$\eta = 2$	0.013***	0.013***	0.014***	0.015***
	(0.003)	(0.003)	(0.003)	(0.003)
$\eta = 3$	0.006**	0.007**	0.007**	0.019***
	(0.003)	(0.003)	(0.003)	(0.003)
$\eta = 4$	0.007**	0.010***	0.009***	0.022***
	(0.003)	(0.003)	(0.004)	(0.004)
$\eta \geq 4$	0.012***	0.010***	0.015***	0.021***
	(0.003)	(0.003)	(0.004)	(0.004)
Year-Sect FE	Yes	Yes	Yes	Yes
Worker-Firm FE	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes
Adjusted R^2	0.70	0.70	0.70	–
# Observations	1,655,021	1,655,021	1,655,021	1,655,021
# Firms	43,920	43,920	43,920	43,920
# Workers	248,429	248,429	248,429	248,429

Notes: Table presents two-way fixed effects (TWFE), Sun and Abraham (2021) (SA) and IV estimates from the worker-level version of the specification in equation (2). We group workers based on their quartile in the distribution of (inflation-adjusted) monthly earnings in the first year we observe each worker, starting from 2006. The reported estimates correspond to workers in the two middle quartiles of initial earnings. The outcome is the log of annual worker earnings divided by the number of months of employment, restricted to worker-months in full-time employment. The first-stage F-statistic for the IV column exceeds 50. Columns with "Yes" in the "Controls" row include the full timeline of event-study indicators for having had positive sales to any MNE subsidiary in CR. Standard errors clustered at the worker level are reported in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. See Section 3.2 for discussion.

Table C5: Worker-Level Effects of RS Rollouts on Labor Earnings: Top Quartile of Initial Earnings

	(1)	(2)	(3)	(4)
	TWFE	TWFE	SA	IV
$\eta \leq -4$	-0.016**	-0.011*	0.012	-0.011
	(0.006)	(0.006)	(0.008)	(0.009)
$\eta = -4$	-0.012**	-0.011*	-0.002	-0.014**
	(0.006)	(0.006)	(0.007)	(0.007)
$\eta = -3$	-0.011**	-0.012**	-0.008	-0.014**
	(0.005)	(0.005)	(0.005)	(0.006)
$\eta = -2$	-0.012***	-0.013***	-0.013***	-0.013***
	(0.004)	(0.004)	(0.004)	(0.004)
$\eta = 0$	0.002	0.001	0.001	0.001
	(0.003)	(0.003)	(0.003)	(0.004)
$\eta = 1$	0.000	0.001	-0.000	0.000
	(0.004)	(0.004)	(0.004)	(0.005)
$\eta = 2$	-0.007	-0.005	-0.006	-0.006
	(0.005)	(0.005)	(0.005)	(0.006)
$\eta = 3$	-0.020***	-0.016***	-0.020***	-0.012*
	(0.005)	(0.006)	(0.006)	(0.006)
$\eta = 4$	-0.010*	-0.003	-0.005	0.005
	(0.006)	(0.006)	(0.006)	(0.007)
$\eta \geq 4$	-0.011*	-0.003	-0.007	0.009
	(0.006)	(0.006)	(0.007)	(0.008)
Year-Sect FE	Yes	Yes	Yes	Yes
Worker-Firm FE	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes
Adjusted R^2	0.87	0.87	0.87	–
# Observations	522,356	522,356	522,356	522,356
# Firms	18,798	18,798	18,798	18,798
# Workers	67,633	67,633	67,633	67,633

Notes: Table presents two-way fixed effects (TWFE), Sun and Abraham (2021) (SA) and IV estimates from the worker-level version of the specification in equation (2). We group workers based on their quartile in the distribution of (inflation-adjusted) monthly earnings in the first year we observe each worker, starting from 2006. The reported estimates correspond to workers in the top quartile of initial earnings. The outcome is the log of annual worker earnings divided by the number of months of employment, restricted to worker-months in full-time employment. The first-stage F-statistic for the IV column exceeds 50. Columns with "Yes" in the "Controls" row include the full timeline of event-study indicators for having had positive sales to any MNE subsidiary in CR. Standard errors are clustered at the worker level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. See Section 3.2 for discussion.

Table C6: Effects of RS Rollouts on Suppliers' Employment of Workers in the Initial Bottom Earnings Quartile

	(1)	(2)	(3)	(4)
	TWFE	TWFE	SA	IV
$\eta \leq -4$	-0.047	0.040	0.054	0.031
	(0.032)	(0.033)	(0.033)	(0.033)
$\eta = -4$	-0.055*	0.003	0.021	-0.004
	(0.031)	(0.030)	(0.030)	(0.031)
$\eta = -3$	-0.036	-0.003	0.018	-0.001
	(0.025)	(0.025)	(0.025)	(0.025)
$\eta = -2$	-0.028*	-0.012	0.004	-0.010
	(0.016)	(0.016)	(0.016)	(0.018)
$\eta = 0$	0.003	-0.006	-0.009	-0.010
	(0.016)	(0.016)	(0.016)	(0.016)
$\eta = 1$	-0.025	-0.036*	-0.038*	-0.050**
	(0.019)	(0.020)	(0.020)	(0.020)
$\eta = 2$	-0.041*	-0.057**	-0.059**	-0.062***
	(0.022)	(0.023)	(0.023)	(0.024)
$\eta = 3$	-0.075***	-0.087***	-0.085***	-0.081***
	(0.027)	(0.029)	(0.029)	(0.029)
$\eta = 4$	-0.082***	-0.090***	-0.088***	-0.094***
	(0.029)	(0.031)	(0.032)	(0.032)
$\eta \geq 4$	-0.083***	-0.088**	-0.114***	-0.091***
	(0.032)	(0.035)	(0.040)	(0.035)
Firm FE	Yes	Yes	Yes	Yes
Year-Sect FE	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes
Adjusted R^2	0.81	0.81	0.81	–
# Observations	60,391	60,391	60,391	60,391
# Treated	1,456	1,456	1,456	1,456
# Never Treated	9,128	9,128	9,128	9,128
# Sector-Year Bins	2,400	2,400	2,400	2,400

Notes: Table presents two-way fixed effects (TWFE), Sun and Abraham (2021) (SA) and IV estimates for the supplier-level specification in equation (1). We group workers based on their quartile in the distribution of (inflation-adjusted) monthly earnings in the first year we observe each worker, starting from 2006. The reported estimates correspond to the outcome variable—the logarithm of the number of employees—and focus on workers in the bottom quartile of initial earnings. The first-stage F-statistic for the IV column exceeds 50. Columns with "Yes" in the "Controls" row include the full timeline of event-study indicators for having had positive sales to any MNE subsidiary in CR. Standard errors are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. See Section 3.2 for discussion.

Table C7: Effects of RS Rollouts on Suppliers' Employment of Workers in the Initial Middle Earnings Quartiles

	(1)	(2)	(3)	(4)
	TWFE	TWFE	SA	IV
$\eta \leq -4$	-0.139***	-0.017	0.007	-0.021
	(0.027)	(0.027)	(0.029)	(0.028)
$\eta = -4$	-0.125***	-0.048*	-0.026	-0.042
	(0.026)	(0.025)	(0.026)	(0.026)
$\eta = -3$	-0.063***	-0.019	0.002	-0.029
	(0.021)	(0.021)	(0.021)	(0.021)
$\eta = -2$	-0.015	0.003	0.016	0.008
	(0.015)	(0.015)	(0.015)	(0.015)
$\eta = 0$	0.052***	0.045***	0.043***	0.038**
	(0.014)	(0.014)	(0.014)	(0.015)
$\eta = 1$	0.028	0.024	0.024	0.021
	(0.018)	(0.018)	(0.018)	(0.019)
$\eta = 2$	0.010	0.014	0.015	0.016
	(0.020)	(0.020)	(0.020)	(0.021)
$\eta = 3$	-0.034	-0.023	-0.024	-0.013
	(0.023)	(0.023)	(0.024)	(0.025)
$\eta = 4$	-0.078***	-0.048*	-0.051*	-0.035
	(0.026)	(0.027)	(0.027)	(0.028)
$\eta \geq 4$	-0.100***	-0.050*	-0.079**	-0.043
	(0.028)	(0.029)	(0.032)	(0.031)
Firm FE	Yes	Yes	Yes	Yes
Year-Sect FE	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes
Adjusted R^2	0.79	0.79	0.79	–
# Observations	96,547	96,547	96,547	96,547
# Treated	1,872	1,872	1,872	1,872
# Never Treated	12,295	12,295	12,295	12,295
# Sector-Year Bins	2,545	2,545	2,545	2,545

Notes: Table presents two-way fixed effects (TWFE), Sun and Abraham (2021) (SA) and IV estimates for the supplier-level specification in equation (1). We group workers based on their quartile in the distribution of (inflation-adjusted) monthly earnings in the first year we observe each worker, starting from 2006. The reported estimates correspond to the outcome variable—the logarithm of the number of employees—and focus on workers in the two middle quartiles of initial earnings. The first-stage F-statistic for the IV column exceeds 50. Columns with "Yes" in the "Controls" row include the full timeline of event-study indicators for having had positive sales to any MNE subsidiary in CR. Standard errors are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. See Section 3.2 for discussion.

Table C8: Effects of RS Rollouts on Suppliers' Employment of Workers in the Initial Top Earnings Quartile

	(1)	(2)	(3)	(4)
	TWFE	TWFE	SA	IV
$\eta \leq -4$	-0.081**	0.021	0.061*	-0.017
	(0.033)	(0.033)	(0.034)	(0.035)
$\eta = -4$	-0.080**	-0.010	0.031	-0.037
	(0.031)	(0.031)	(0.031)	(0.033)
$\eta = -3$	-0.062**	-0.016	0.017	-0.027
	(0.027)	(0.027)	(0.027)	(0.029)
$\eta = -2$	-0.058***	-0.031	-0.018	-0.031
	(0.021)	(0.021)	(0.021)	(0.022)
$\eta = 0$	0.024	0.017	0.012	0.006
	(0.018)	(0.018)	(0.018)	(0.019)
$\eta = 1$	0.032	0.031	0.029	0.022
	(0.023)	(0.023)	(0.024)	(0.024)
$\eta = 2$	0.009	0.007	0.007	0.003
	(0.024)	(0.025)	(0.026)	(0.026)
$\eta = 3$	-0.009	-0.001	0.004	-0.001
	(0.027)	(0.028)	(0.029)	(0.030)
$\eta = 4$	-0.015	0.009	0.019	0.016
	(0.030)	(0.032)	(0.033)	(0.035)
$\eta \geq 4$	-0.061*	-0.018	-0.064*	-0.011
	(0.032)	(0.034)	(0.037)	(0.037)
Firm FE	Yes	Yes	Yes	Yes
Year-Sect FE	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes
Adjusted R^2	0.75	0.75	0.75	–
# Observations	60,391	60,391	60,391	60,391
# Treated	1,456	1,456	1,456	1,456
# Never Treated	9,128	9,128	9,128	9,128
# Sector-Year Bins	2,400	2,400	2,400	2,400

Notes: Table presents two-way fixed effects (TWFE), Sun and Abraham (2021) (SA) and IV estimates for the supplier-level specification in equation (1). We group workers based on their quartile in the distribution of (inflation-adjusted) monthly earnings in the first year we observe each worker, starting from 2006. The reported estimates correspond to the outcome variable—the logarithm of the number of employees—and focus on workers in the top quartile of initial earnings. The first-stage F-statistic for the IV column exceeds 50. Columns with "Yes" in the "Controls" row include the full timeline of event-study indicators for having had positive sales to any MNE subsidiary in CR. Standard errors are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. See Section 3.2 for discussion.

Table C9: Effects of RS Rollouts on the MNE Subsidiary Employment and Relative Employment in the MNE Group

	Panel A: Log CR Employment			Panel B: CR Share of MNE Employment		
	(1)	(2)	(3)	(4)	(5)	(6)
	TWFE	SA	IV	TWFE	SA	IV
$\eta \leq -4$	-0.099	-0.149	-0.124	-0.001	-0.004	-0.001
	(0.121)	(0.142)	(0.135)	(0.004)	(0.004)	(0.004)
$\eta = -4$	0.042	0.057	0.044	0.001	-0.000	0.001
	(0.091)	(0.102)	(0.100)	(0.006)	(0.006)	(0.006)
$\eta = -3$	-0.007	-0.006	-0.001	-0.001	-0.002	-0.001
	(0.096)	(0.098)	(0.105)	(0.003)	(0.003)	(0.003)
$\eta = -2$	0.000	-0.017	0.008	0.001	0.000	0.001
	(0.078)	(0.072)	(0.086)	(0.002)	(0.002)	(0.002)
$\eta = 0$	0.040	0.040	0.048	0.001	0.000	0.001
	(0.069)	(0.063)	(0.076)	(0.001)	(0.002)	(0.001)
$\eta = 1$	-0.052	-0.059	-0.026	0.003	0.002	0.003
	(0.093)	(0.086)	(0.102)	(0.002)	(0.003)	(0.003)
$\eta = 2$	-0.118	-0.109	-0.103	0.001	-0.001	0.001
	(0.130)	(0.116)	(0.147)	(0.003)	(0.002)	(0.003)
$\eta = 3$	0.057	0.070	0.087	0.002	-0.000	0.002
	(0.103)	(0.088)	(0.116)	(0.003)	(0.002)	(0.003)
$\eta = 4$	0.051	0.026	0.080	0.002	-0.001	0.002
	(0.107)	(0.090)	(0.120)	(0.004)	(0.002)	(0.004)
$\eta \geq 4$	-0.042	-0.022	-0.017	-0.001	0.000	-0.001
	(0.111)	(0.037)	(0.123)	(0.004)	(0.001)	(0.004)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-Sect FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.74	0.74	–	0.94	0.94	–
# Observations	5,229	5,229	5,229	1,612	1,612	1,612
# Treated MNEs	169	169	169	104	104	104
# Non-Treated MNEs	310	310	310	57	57	57
# Sector-Year Bins	1,748	1,748	1,748	860	860	860

Notes: Table presents two-way fixed effects (TWFE), Sun and Abraham (2021) (SA) and IV estimates for the MNE-level version of the specification in equation (1). In columns 1–3, the outcome is the log employment of the MNE subsidiary in CR. In columns 4–6, the outcome is the fraction of employment of the CR subsidiary in total global employment of the MNE. Columns 4–6 are restricted to the MNEs whose global sales and employment are observed in Orbis, which is why the number of observations and treated MNEs falls relative to columns 1–3. The first-stage F-statistics for the IV columns exceed 50. Standard errors are clustered at the MNE level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. See Section 3.2 for discussion.

Table C10: Effects of RS Rollouts on the MNE Subsidiary Sales and Relative Sales in the MNE Group

	Panel A: Log CR Sales			Panel B: CR Share of MNE Sales		
	(1)	(2)	(3)	(4)	(5)	(6)
	TWFE	SA	IV	TWFE	SA	IV
$\eta \leq -4$	-0.192	-0.261	-0.237	-0.001	-0.009	-0.005
	(0.137)	(0.165)	(0.151)	(0.007)	(0.006)	(0.008)
$\eta = -4$	0.024	0.070	-0.000	0.006	0.003	0.002
	(0.105)	(0.121)	(0.113)	(0.009)	(0.009)	(0.010)
$\eta = -3$	-0.062	-0.049	-0.071	-0.001	-0.003	-0.004
	(0.121)	(0.125)	(0.133)	(0.006)	(0.006)	(0.007)
$\eta = -2$	-0.092	-0.108	-0.110	0.002	0.001	-0.005
	(0.081)	(0.076)	(0.090)	(0.004)	(0.004)	(0.003)
$\eta = 0$	0.069	0.077	0.068	-0.000	-0.002	-0.002
	(0.065)	(0.059)	(0.072)	(0.003)	(0.003)	(0.003)
$\eta = 1$	-0.051	-0.039	-0.033	-0.003	-0.005	-0.003
	(0.098)	(0.090)	(0.108)	(0.003)	(0.003)	(0.004)
$\eta = 2$	-0.101	-0.064	-0.089	-0.002	-0.004	-0.000
	(0.130)	(0.114)	(0.147)	(0.004)	(0.003)	(0.004)
$\eta = 3$	0.034	0.083	0.067	-0.004	-0.003	0.001
	(0.110)	(0.089)	(0.123)	(0.004)	(0.003)	(0.005)
$\eta = 4$	0.034	0.073	0.058	-0.005	-0.005**	0.001
	(0.116)	(0.091)	(0.130)	(0.004)	(0.002)	(0.006)
$\eta \geq 4$	-0.083	-0.027	-0.071	-0.008*	-0.001	-0.002
	(0.122)	(0.044)	(0.135)	(0.004)	(0.001)	(0.005)
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-Sect FE	Yes	Yes	Yes	Yes	Yes	Yes
Adjusted R^2	0.77	0.77	–	0.94	0.94	–
# Observations	5,229	5,229	5,229	1,790	1,790	1,790
# Treated MNEs	169	169	169	116	116	116
# Non-Treated MNEs	310	310	310	59	59	59
# Sector-Year Bins	1,748	1,748	1,748	952	952	952

Notes: Table presents two-way fixed effects (TWFE), Sun and Abraham (2021) (SA) and IV estimates for the MNE-level version of the specification in equation (1). In columns 1–3, the outcome is the log sales of the MNE subsidiary in CR. In columns 4–6, the outcome is the fraction of sales of the CR subsidiary in total global sales of the MNE. The first-stage F-statistics for the IV columns exceed 50. Standard errors are clustered at the MNE level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. See Section 3.2 for discussion.

Table C11: Intensive Margin Effects of RS Rollouts on Sales Transactions to the RS-MNE

	(1)	(2)	(3)	(4)
	TWFE	TWFE	SA	IV
$\eta \leq -4$	-0.067*	-0.030	-0.049	-0.016
	(0.035)	(0.035)	(0.037)	(0.037)
$\eta = -4$	0.024	0.048	0.030	0.077**
	(0.033)	(0.034)	(0.034)	(0.035)
$\eta = -3$	-0.010	0.000	-0.012	0.020
	(0.030)	(0.030)	(0.030)	(0.031)
$\eta = -2$	0.001	0.008	0.003	0.026
	(0.022)	(0.022)	(0.022)	(0.023)
$\eta = 0$	0.018	0.010	0.011	0.003
	(0.018)	(0.018)	(0.018)	(0.019)
$\eta = 1$	0.000	-0.013	-0.008	-0.007
	(0.023)	(0.023)	(0.023)	(0.024)
$\eta = 2$	0.018	-0.001	0.007	-0.002
	(0.026)	(0.026)	(0.027)	(0.028)
$\eta = 3$	-0.015	-0.038	-0.032	-0.036
	(0.028)	(0.028)	(0.029)	(0.030)
$\eta = 4$	-0.031	-0.055*	-0.044	-0.054
	(0.030)	(0.031)	(0.032)	(0.033)
$\eta \geq 4$	-0.055*	-0.079**	-0.058	-0.076**
	(0.032)	(0.033)	(0.039)	(0.035)
Year-Sect FE	Yes	Yes	Yes	Yes
Year-BuyerSect FE	Yes	Yes	Yes	Yes
Supplier-Buyer FE	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes
Adjusted R^2	0.73	0.73	0.73	–
# Observations	515,797	515,797	515,797	515,797
# Treated	1,678	1,678	1,678	1,678
# Never Treated	13,079	13,079	13,079	13,079
# Sector-Year Bins	2,550	2,550	2,550	2,550

Notes: Table presents two-way fixed effects (TWFE), Sun and Abraham (2021) (SA) and IV estimates for the transaction-level version of the specification in equation (2). The outcome variable is the log transaction value between a supplier i and a buyer j. The first-stage F-statistic for the IV column exceeds 50. Columns with "Yes" in the "Controls" row include the full timeline of event-study indicators for having had positive sales to any MNE subsidiary in CR. Standard errors are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. See Section 3.2 for discussion.

Table C12: Supplier-Level Effects of RS Rollouts on Total Sales to Other (Non-RS) Buyers

	(1)	(2)	(3)	(4)
	TWFE	TWFE	SA	IV
$\eta \leq -4$	-0.039	0.031	0.069**	0.023
	(0.030)	(0.030)	(0.033)	(0.032)
$\eta = -4$	-0.012	0.016	0.050	0.018
	(0.030)	(0.031)	(0.032)	(0.032)
$\eta = -3$	0.003	0.001	0.034	0.010
	(0.027)	(0.027)	(0.028)	(0.028)
$\eta = -2$	0.012	-0.004	0.014	-0.002
	(0.021)	(0.021)	(0.021)	(0.022)
$\eta = 0$	0.024	-0.014	-0.018	-0.003
	(0.018)	(0.018)	(0.018)	(0.019)
$\eta = 1$	0.007	-0.025	-0.029	-0.027
	(0.022)	(0.023)	(0.023)	(0.024)
$\eta = 2$	-0.030	-0.050**	-0.051**	-0.041
	(0.023)	(0.024)	(0.024)	(0.025)
$\eta = 3$	-0.068**	-0.071***	-0.074***	-0.057*
	(0.027)	(0.028)	(0.028)	(0.030)
$\eta = 4$	-0.088***	-0.066**	-0.080**	-0.060*
	(0.030)	(0.032)	(0.032)	(0.034)
$\eta \geq 4$	-0.156***	-0.101***	-0.156***	-0.097***
	(0.031)	(0.033)	(0.035)	(0.036)
Firm FE	Yes	Yes	Yes	Yes
Year-Sect FE	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes
Adjusted R^2	0.78	0.78	0.78	–
# Observations	132,199	132,199	132,199	132,199
# Treated	2,098	2,098	2,098	2,098
# Never Treated	15,291	15,291	15,291	15,291
# Sector-Year Bins	2,630	2,630	2,630	2,630

Notes: The table presents two-way fixed effects (TWFE), Sun and Abraham (2021) (SA) and IV estimates for the supplier-level specification in equation (1). The outcome variable is the log total sales to other (non-RS) buyers. The first-stage F-statistic for the IV column exceeds 50. Columns with "Yes" in the "Controls" row include the full timeline of event-study indicators for having had positive sales to any MNE subsidiary in CR. Standard errors are clustered at the firm level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. See Section 3.2 for discussion.

Table C13: Sensitivity Analysis: Robustness to Violations of Parallel Trends

Exhibit	β	$M_{5\%}$	$M_{10\%}$	Exhibit	β	$M_{5\%}$	$M_{10\%}$
Figure 2A	-0.100	≥ 2	≥ 2	Figure 6B	-0.097	0.4	0.6
Figure 2B	-0.096	0.9	1.2	Appendix Figure B12A	-0.001	0.0	0.3
Figure 3B (Below)	-0.162	≥ 2	≥ 2	Appendix Figure B12D	0.018	0.0	≥ 2
Figure 3C (Above)	-0.108	0.6	0.6	Appendix Figure B1A	-0.100	≥ 2	≥ 2
Figure 3D (Above)	-0.146	0.8	1.1	Appendix Figure B1B	-0.096	0.9	1.2
Figure 3E (Above)	-0.114	1.8	≥ 2	Appendix Figure B7A (Services)	-0.177	0.4	0.6
Figure 3F (Above)	-0.110	1.3	1.7	Appendix Figure B7D (Above)	-0.112	0.4	0.5
Figure 4A	0.019	1.8	≥ 2	Appendix Figure B3A (Unrestricted)	-0.100	≥ 2	≥ 2
Figure 4B (Bottom Quartile)	0.053	0.9	0.9	Appendix Figure B3A (Restricted)	-0.123	≥ 2	≥ 2
Figure 4B (Middle)	0.021	1.5	1.5	Appendix Figure B3B (Unrestricted)	-0.096	0.9	1.2
Figure 4C (Bottom Quartile)	-0.091	0.3	0.6	Appendix Figure B3B (Restricted)	-0.089	0.6	0.9
Figure 6A	-0.076	0.0	0.6				

Notes: Table C13 presents a sensitivity analysis following Rambachan and Roth (2023) for every event-study estimate in the paper whose baseline $\eta \geq 4$ coefficient is statistically significant at the 5% level. The column "Exhibit" gives the figure number in the paper, and the column "β" gives the baseline IV point estimate for the $\eta \geq 4$ coefficient from the IV event-study specification. The table lists the estimates in two side-by-side blocks with the same four columns. A parenthetical after the figure number identifies the series of a multi-series figure to which the row refers. The columns "$M_{5\%}$" and "$M_{10\%}$" report, at the 5 and 10 percent significance levels, the breakdown value M: the largest Rambachan and Roth (2023) relative-magnitudes violation of parallel trends for which the robust confidence interval still excludes zero (among values from $M = 0$ to $M \geq 2$ in 0.1 intervals). The value of M bounds the post-treatment violation of parallel trends by M times the average pre-treatment slope (the average change from the earliest pre-period to $\eta = -1$); a value of $M = 1$ thus corresponds to a post-treatment violation equal to the average pre-treatment slope. "≥ 2" denotes that the estimate remains robust to a violation of at least twice the average pre-treatment slope.

Appendix D Theory appendix

Appendix D.1 Model derivations

In this appendix, we present the main equilibrium equations. Here, contrary to the main text, trade patterns are not restricted and Home firms can export to Foreign. This case nests our baseline specification.

We proceed in two steps. First, we characterize firm-level outcomes taking general equilibrium (GE) quantities as given, then we derive the complete GE solution of the model.

Notation Let $y^t_{ij,r}$ denote outcome y of an entity with RS status r and for workers of type t. When $i = H$, then $y^t_{ij,r}$ refers to a Home firm producing for the destination market $j = H, F, M$ (respectively: Home, exports to Foreign, or production of inputs for MNE subsidiaries).[i] When $i = M$, then $y^t_{M,r}$ refers to the production of MNE subsidiaries at Home. Market-level aggregates are denoted with capital letters. Finally, $y_{ij,r}$ without superscript t sums outcomes across worker types ($y_{ij,r} = \sum_{t=l,h} y^t_{ij,r}$) and $y^t_{i,r}$ without j subscript sums outcomes across all the production lines of a firm ($y^t_{i,r} = \sum_{j=H,F,M} y^t_{ij,r}$). Similarly, Y^t_{ij} without subscript r sums outcomes across firms of all r-statuses ($Y^t_{ij} = \sum_{r=R,N} Y^t_{ij,r}$). Numbers-only equations refer to equations in Section 4 in the main body of the paper.

Labor use Given the labor aggregator in equation (4), a firm or an MNE subsidiary facing wages $\{w^t_{i,r}\}$ chooses relative employment of low- and high-wage workers as follows:

$$\chi^t_{i,r} = \frac{w^t_{i,r}\ell^t_{i,r}}{W_{i,r}\ell_{i,r}} = \frac{(\alpha^t_i)^\rho \left(w^t_{i,r}\right)^{1-\rho}}{W^{1-\rho}_{i,r}},$$

where $\ell_{i,r}$ is the labor aggregate in equation (4) and $W_{i,r}$ is the corresponding labor cost index of the firm:

$$W_{i,r} = \left[(\alpha^l_i)^\rho \left(w^l_{i,r}\right)^{1-\rho} + (\alpha^h_i)^\rho \left(w^h_{i,r}\right)^{1-\rho}\right]^{\frac{1}{1-\rho}}, \text{ for } i = H, M, F.$$

In the baseline equilibrium without RS, we simply denote χ^t_H (χ^t_M, respectively) the share of t-workers in the wage bill of Home firms (MNE subsidiaries, respectively).

Non-MNE firms All firms with the same productivity z make the same choices and firm-level expressions are given as a function of their productivity. Denoting $\mu_{ij,r}$ the markup of firms of type r on market $j = H, F, M$, output prices are given by:

$$p_{ij,r} = \mu_{ij,r}\frac{\varrho_{ij}W_{i,r}}{z},$$

where $\varrho_{ij} = \varrho$ if $ij = HF, FH$ and $\varrho_{ij} = 1$ otherwise. In a baseline without RS, firms are monopolistically competitive on all markets so that

$$\mu_{ij,r} = \frac{\sigma}{\sigma - 1}.$$

When RS is implemented, we allow for the markups to adjust so that only a share β of the suppliers' cost increase is passed through to the input price paid by the MNE (on other markets, the pass-through is complete). We define

$$\beta = \frac{\partial \log p_{HM,R}}{\partial \chi^l_H \log \tau^l_R}, \tag{D1}$$

where $\chi^l_H \log \tau^l_R$ measures the size of the RS cost shock as described in the next section. Given CES demand and firms' linear production costs, firm sales $y_{ij,r}$, employment $\ell^t_{ij,r}$, and profits $\pi_{ij,r}$ (conditional on

[i] Notation is symmetric for Foreign, with fewer combinations of subscripts since, e.g., Foreign firms do not produce intermediates.

choosing to produce) are, respectively:

$$y_{ij,r} = \mu_{ij,r}^{1-\sigma} z^{\sigma-1} \varrho_{ij}^{1-\sigma} {W_{i,r}}^{1-\sigma} D_{j,r}, \tag{D2}$$

$$\ell_{ij,r}^{t} = \left(\alpha_i^t\right)^{\rho} \left(\mu_{ij,r}\right)^{-\sigma} z^{\sigma-1} \varrho_{ij}^{1-\sigma} \left(w_{i,r}^t\right)^{-\rho} {W_{i,r}}^{\rho-\sigma} D_{j,r} + f_{ij} \left(\alpha_i^t\right)^{\rho} \left(\frac{w_{i,r}^t}{W_{i,r}}\right)^{-\rho}, \quad t = l, h \tag{D3}$$

$$\pi_{ij,r} = \mu_{ij,r}^{-\sigma} \left(\mu_{ij,r} - 1\right) z^{\sigma-1} \varrho_{ij}^{1-\sigma} {W_{i,r}}^{1-\sigma} D_{j,r} - W_{i,r} f_{ij}, \tag{D4}$$

where $D_{j,r}$ corresponds to the aggregate demand shifter on market j for firms with status $r = R, N$:

$$D_{i,r} = P_i^{\sigma-1} X_i \quad \text{, for } i = H, F, \tag{D5}$$

$$D_{M,r} = R_r^{\sigma} M_r. \tag{D6}$$

In these expressions, we have used X_i to denote total expenditure in country $i = H, F$ and P_i to denote the ideal price index for consumption in i which, given the demand in equation (3), is:

$$P_i = \left(\int_{\Omega_i} d_{\omega,i} \, {p_{\omega,i}}^{1-\sigma} \, d\omega\right)^{\frac{1}{1-\sigma}}.$$

On the market for intermediate inputs sold to MNEs, R_r is the input cost index for an MNE subsidiary with RS status r, which, given the production function in equation (6), is:

$$R_r = \left(\int_{\Omega_x} {p_{HM,r}\left(\omega_x\right)}^{1-\sigma} d\omega_x + \left(\nu^{\ell}\right)^{\sigma} W_{M,r}^{1-\sigma}\right)^{\frac{1}{1-\sigma}}, \tag{D7}$$

and M_r is subsidiary output given in equation (6). Here ν^{ℓ} is the MNE's labor input shifter from equation (6), with ν^m normalized to one.

Profits on each destination market $j = H, M, F$ are increasing in productivity, so that only firms above a given productivity cutoff enter the market. Given the expression for profits in equation (D4), the selection cutoff corresponding to zero profit on market j for firms with RS status r is:

$$z_{ij,r}^{*} = \left(\frac{1}{\left(\mu_{ij,r} - 1\right)\mu_{ij,r}^{-\sigma}}\right)^{\frac{1}{\sigma-1}} \frac{f_{ij}^{\frac{1}{\sigma-1}} \varrho_{ij} {W_{i,r}}^{\frac{\sigma}{\sigma-1}}}{D_{j,r}^{\frac{1}{\sigma-1}}}. \tag{D8}$$

The total sales of firms on destination market $j = H, M, F$ are then given by $Y_{ij,r} = \int_{z_{ij,r}^{*}}^{\infty} y_{ij,r}\left(z\right) dG_i\left(z\right)$, which, given the assumption that productivity is Pareto distributed, can be written as:

$$Y_{ij,r} = \underline{z}_i^{\theta} \left(\frac{\theta}{\theta - \sigma + 1}\right) \left(\mu_{ij,r} \varrho_{ij} W_{i,r}\right)^{1-\sigma} D_{j,r} \left(z_{ij,r}^{*}\right)^{\sigma-1-\theta}. \tag{D9}$$

MNE headquarters and subsidiary We turn to describing the choices of the MNE headquarters in Foreign and subsidiary in Home. We have already defined above the cost index R_r of an MNE subsidiary with RS status r (see equation (D7)). The MNE headquarters in Foreign imports the good produced by its subsidiary in Home, subject to iceberg transport costs, so that the marginal cost of the MNE headquarters of type r is: $c_r = \varrho R_r$. The final goods markets in Foreign and at Home are monopolistically competitive, hence the MNE (headquarters) sells to final consumers in market $j = F, H$ at price:

$$p_{Mj,r} = \frac{\sigma}{\sigma - 1} \varrho_{Mj} c_r,$$

where $\varrho_{Mj} = 1$ if $j = F$ and $\varrho_{Mj} = \varrho$ if $j = H$. Given the CES final demand in equation (3), MNE sales in $j = F, H$ are:

$$p_{Mj,r} q_{Mj,r} = d_r \left(\frac{\sigma}{\sigma - 1} \varrho_{Mj} c_r \right)^{1-\sigma} D_{j,r}.$$

In turn, one can express the total MNE subsidiary output for an MNE of type r as:

$$M_r = d_r \left(\frac{\sigma}{\sigma - 1} \right)^{-\sigma} \varrho^{1-\sigma} R_r^{-\sigma} \sum_j \varrho_{Mj}^{1-\sigma} D_{j,r}. \tag{D10}$$

Finally, total sales of all MNEs across both markets H and F are:

$$Y_M = \frac{\sigma}{\sigma - 1} \varrho \sum_{r=N,R} N_{M,r} R_r \sum_{j=H,F} \varrho_{Mj} M_{j,r}.$$

Appendix D.2 Effects of RS (extension)

The RS policy described in the main text induces a net increase in labor costs for firms hiring low-wage workers. However, it is possible that RS policies incentivize firms to make their workers more productive, or that they are accompanied by transfers of technology or expertise by the MNE to its suppliers.[ii] In this more general model, we allow for RS to be potentially accompanied by such direct changes in labor productivity. We define labor productivity gains associated with RS as $T_R \geq 1$ and assume that they impact the productivity of all workers of the firm affected by RS and are paid to workers. Denoting $\tilde{w}^t_{H,R}$ the compensation paid to a type t worker by a supplier adopting RS policies, we therefore have:

$$\tilde{w}^t_{H,R} = T_R \tau^t_R w^t_{H,N}, \text{ for } t = l, h. \tag{D11}$$

From the point of view of suppliers, the net labor costs $w^t_{H,R}$ incurred for high- and low-wage labor per efficiency unit are still given by equation (7). That is, τ^l_R measures the pure labor cost increase on low-wage workers faced by an RS supplier, net of any labor productivity gains. The model of the main text is nested and corresponds to $T_R = 1$.

Appendix D.3 Derivations of the comparative statics

We compute here the first-order effect of RS. Hat notation $\hat{y} = d \log y$ denotes log changes in variable y.

In addition to the forces considered in the main text, we allow RS to directly impact the productivity of both low- and high-wage worker types at impacted suppliers, increasing it proportionately by $\hat{T}_R \geq 0$. Therefore, RS is summarized by $\left(\hat{\tau}^l_R, \hat{T}_R, \hat{d}_R, \beta \right)$, with $\hat{T}_R = 0$ in the main text.

Workers' compensation is impacted by RS due to the wage premium as well as the potential productivity gain that accompanies RS. Following the definition of RS in equations (7) and (D11), the relative effect of RS on workers' compensation is:

$$\hat{\tilde{w}}^l_{H,R} - \hat{\tilde{w}}^l_{H,N} = \hat{T}_R + \hat{\tau}^l_R,$$
$$\hat{\tilde{w}}^h_{H,R} - \hat{\tilde{w}}^h_{H,N} = \hat{T}_R.$$

Given the demand in equation (3) and monopolistic competition, markups are constant on the final goods markets (equal to $\frac{\sigma}{\sigma-1}$) so that $\hat{\mu}_{Hj,r} = 0$ for $j = H, F$. On the MNE market, we have defined the partial pass-through in equation (D1) so that:

$$\hat{\mu}_{HM,R} = (\beta - 1) \chi^l_H \hat{\tau}^l_R.$$

[ii] See, e.g., Verhoogen (2016) for a case in which labor standards improved productivity.

Putting together the impact of RS on costs and markups, the relative impact of RS on firms' prices is:

$$\hat{p}_{Hj,R} - \hat{p}_{Hj,N} = \chi_H^l \hat{\tau}_R^l, \quad \text{for } j = H, F,$$
$$\text{and } \hat{p}_{HM,R} - \hat{p}_{HM,N} = \beta \chi_H^l \hat{\tau}_R^l.$$

Turning to sales from equation (D2), notice that aggregate demand shifters on the final goods markets are the same for R and N firms, so that $\hat{D}_{j,R} - \hat{D}_{j,N} = 0$ for $j = H, F$. In contrast, aggregate demand shifters faced by R-suppliers depend on the demand for final goods produced by RS-MNEs, which is different from the one faced by non-RS MNEs. Specifically, using CES demand for MNE goods, one can show that $M_r = d_r \left(\frac{\sigma}{\sigma-1}\right)^{-\sigma} \varrho^{1-\sigma} R_r^{-\sigma} \sum_j \varrho_{Mj}^{1-\sigma} D_{j,r}$, so that $\hat{D}_{M,R} - \hat{D}_{M,N} = \hat{d}_R$.[iii] Combining the effect of RS on prices and on aggregate demand shifters yields the following differential sales response in each market:

$$\hat{y}_{Hj,R} - \hat{y}_{Hj,N} = (1-\sigma)\, \chi_H^l \hat{\tau}_R^l < 0, \quad \text{for } j = H, F,$$
$$\hat{y}_{HM,R} - \hat{y}_{HM,N} = (1-\sigma)\, \beta \chi_H^l \hat{\tau}_R^l + \hat{d}_R.$$

Turning to RS-exposed firms, log-differentiating the cutoffs in equation (D8) yields:

$$\hat{z}^*_{Hj,R} - \hat{z}^*_{Hj,N} = 0 \text{ for } j = H, F, \tag{D12}$$
$$\hat{z}^*_{HM,R} - \hat{z}^*_{HM,N} = \frac{\sigma}{\sigma-1} \chi_H^l \hat{\tau}_R^l - \frac{1}{\sigma-1} \hat{d}_R. \tag{D13}$$

The expressions (14) and (13) stem from log-differentiating equation (D9) combined with the results above on the cutoff. Finally, log-differentiating equation (D10) yields:

$$\widehat{R_R M_R} - \widehat{R_N M_N} = \hat{d}_R + (1-\sigma)\left(\hat{R}_R - \hat{R}_N\right). \tag{D14}$$

To compute the change in the MNE subsidiary output price (also its cost, given the monopolistic competition assumption), note that equation (D7) integrated over the MNE suppliers yields:

$$R_r^{1-\sigma} = \mu_{HM,r}^{1-\sigma} W_{H,r}^{1-\sigma} \underline{z}_H^{\theta} \left(\frac{\theta}{\theta-\sigma+1}\right) \left(\left[z^*_{HM,r}\right]^{\sigma-1-\theta}\right) + \left(\nu^{\ell}\right)^{\sigma} W_{M,r}^{1-\sigma}. \tag{D15}$$

Log-differentiating this expression yields:

$$\hat{R}_R - \hat{R}_N = \left[\Xi\beta + \Xi\left(\frac{\sigma-1-\theta}{1-\sigma}\right)\frac{\sigma}{\sigma-1}\right] \chi_H^l \hat{\tau}_R^l - \Xi\left(\frac{\sigma-1-\theta}{1-\sigma}\right)\frac{1}{\sigma-1}\hat{d}_R \tag{D16}$$

which, combined with equation (D14), yields the desired comparative statics reported in the main text (see equation (15)).

Appendix D.4 Welfare effects: full model

To compute the welfare effects of RS, we start by writing down the equations that define the general equilibrium of the model. To close the model and solve for general equilibrium quantities, we write down the income in each country, labor market clearing and trade balance. These steps are detailed next.

[iii] The demand shifter on the MNE input market can be computed from the MNE output demand. Given equation (3), MNE sales in $j = F, H$ are $p_{Mj,r} q_{Mj,r} = d_r \left(\frac{\sigma}{\sigma-1}\varrho_{Mj} c_r\right)^{1-\sigma} D_{j,r}$, so that total output for MNE subsidiary of type r is: $M_r = d_r \left(\frac{\sigma}{\sigma-1}\right)^{-\sigma} \varrho^{1-\sigma} R_r^{-\sigma} \sum_j \varrho_{Mj}^{1-\sigma} D_{j,r}$.

General equilibrium Denote $L^t_{Hj,r}$ the total number of type t workers in Home firms working in a production line corresponding to goods sold on market $j = H, F, M$, and $L^t_{M,r}$ the total labor directly employed at MNE subsidiaries of type r at Home. Labor market clearing at Home yields:

$$\bar{L}^t_H = \sum_{r=R,N} \left\{ \sum_{j=H,M,F} L^t_{Hj,r} + L^t_{M,r} \right\}.$$

Denote X^t_j the income of type t workers in country j. Workers derive income from their labor, as well as their share in local (non-MNE) firms' profits. The total income of workers of type t at Home is:

$$X^t_H = \sum_{r=R,N} \left\{ \sum_{j=H,M,F} \tilde{w}^t_{H,r} L^t_{Hj,r} + \tilde{w}^t_{M,r} L^t_{M,r} \right\} + \sum_{r=R,N} \sum_{j=H,M,F} \Pi^t_{Hj,r}, \tag{D17}$$

where $\Pi^t_{Hj,r}$ is the share of Home firm profits generated on market j for firms of type r that are apportioned (proportionally to labor income) to type t workers. Given the CES-Pareto setup, profits are a constant fraction of sales and wage bill in the equilibrium without RS. In particular, one can write:

$$\Pi^t_{Hj,r} = (a-1)\, \tilde{w}^t_{H,r} L^t_{Hj,r},$$

where a denotes the ratio of sales to the wage bill (including fixed-cost labor), equal to $\frac{\theta\sigma}{\theta\sigma-(\sigma-1)}$ in the initial equilibrium. Finally, total income in H is:

$$X_H = \sum_{t=l,h} X^t_H.$$

Throughout, $\tilde{w}^t_{i,r}$ denotes the compensation received by a type-t worker, inclusive of any RS or MNE wage premium; for MNE subsidiaries the premium is specified in Appendix D.6.4.

In Foreign, total income is made of labor income, Foreign firm profits and profits made by multinationals (which are a constant fraction of their sales $Y_M = X_{MF} + X_{MH}$):

$$X_F = aW_F L_F + \frac{1}{\sigma}\left(X_{MF} + X_{MH}\right).$$

In both countries, income equals expenditures: $X_j = \sum_{i=H,F,M} X_{ij}$. Equivalently, trade balance is given by:

$$X_{FH} + X_{MH} = \frac{\sigma-1}{\sigma}\left(X_{MF} + X_{MH}\right) + X_{HF},$$

where X_{ij} denotes expenditure of country $j = H, F$ on final goods produced by firms in country $i = H, F$ or final goods sold by MNEs' headquarters in Foreign when $i = M$. This expression accounts for the fact that MNE subsidiary production is imported by F from H at cost and then sold by the MNE headquarters with a constant markup $\frac{\sigma}{\sigma-1}$ to final consumers in $j = H, F$.

An equilibrium of this economy is a set of wages $\left\{w^t_{j,N}\right\}_{j,t}$ and labor allocations $\left\{L^t_{ij,r}\right\}_{ij,r,t}$ such that (i) consumers maximize utility; (ii) firms make profit-maximizing decisions, as summarized in the main text; (iii) labor markets clear in both countries; and (iv) trade is balanced as described above.

Effect of RS on welfare We consider the first-order effect of a small RS policy $\left(\hat{\tau}^l_R, \hat{T}_R, \hat{d}_R, \beta\right)$ implemented by a fraction γ of MNEs, which are otherwise identical to other MNEs. We take the Foreign wage index as the numeraire so that $\hat{W}_F = 0$. We are interested in the change in welfare for workers of type t,

$$\hat{U}^t_H = \hat{X}^t_H - \hat{P}_H. \tag{D18}$$

Throughout this appendix, $\hat{\tau}_R^l$ denotes the RS-induced log change in the cost of low-wage labor at Home suppliers; we write $\hat{\tau}_{H,R}^l$ where the country of the supplier matters, and $\hat{\tau}_R^t$ for type-t workers, with $\hat{\tau}_R^h = 0$. A general country index is added ($\hat{\tau}_{i,R}^l$) only in Appendix D.6.5, where the shock differs across sourcing countries.

Taking log-differentials of equation (D17) yields:

$$\begin{aligned}\hat{X}_H^t = \Phi^t \left[\hat{w}_{H,N}^t + \phi_{H,R}^t \left(\hat{\tau}_R^t + \hat{T}_R\right) + \left(1 - \phi_M^t\right) \hat{L}_H^t + \phi_M^t \hat{L}_M^t\right] \\ + \left(1 - \Phi^t\right) \left[\hat{w}_{H,N}^t + \hat{L}_H^t + \varphi_{H,R} \left(\hat{\tau}_R^t + \hat{T}_R\right) + \varphi_{HM,R} \hat{a} \frac{a}{a-1}\right],\end{aligned} \tag{D19}$$

where we recognize that markups can change after RS, given the partial pass-through of the policy—see below eq. (D20); Φ^t is the share of the labor income in the income of type t in the initial equilibrium:

$$\Phi^t \equiv \frac{\sum_{j=H,M,F} \chi_H^t X_{Hj}/a + \chi_M^t W_M L_M}{\sum_{j=H,M,F} \chi_H^t X_{Hj} + \chi_M^t W_M L_M}.$$

$\phi_{H,R}^t$ is the share of total Home wage bill for type t corresponding to suppliers that are subject to RS:

$$\phi_{H,R}^t \equiv \frac{\sum_{j=H,M,F} \gamma \phi_{Hj,R} \chi_H^t X_{Hj}/a}{\sum_{j=H,M,F} \chi_H^t X_{Hj}/a + \chi_M^t W L_M}.$$

ϕ_M^t is the share of total Home wage bill for type t corresponding to MNE subsidiaries M:

$$\phi_M^t \equiv \frac{\chi_M^t W L_M}{\sum_{j=H,M,F} \chi_H^t X_{Hj}/a + \chi_M^t W L_M}.$$

$\varphi_{H,R}$ is the share of total Home firm profits corresponding to firms that are subject to RS requirements:

$$\varphi_{H,R} \equiv \frac{\sum_{j=H,M,F} \gamma \phi_{Hj,R} X_{Hj}}{\sum_{j=H,M,F} X_{Hj}}.$$

Finally, $\varphi_{HM,R}$ is the share of Home firm profits corresponding to the sales to MNE subsidiaries that will implement RS:

$$\varphi_{HM,R} \equiv \frac{\gamma X_{HM}}{\sum_{j=H,M,F} X_{Hj}}.$$

Firm profits will be differentially affected between RS and non-RS firms, and within RS firms between production lines, since RS can be accompanied by a reduction in the suppliers' markup, through the imperfect pass-through β of the policy. Specifically,

$$\hat{a} = \frac{(\sigma - 1)^2}{\theta\sigma - (\sigma - 1)} (\beta - 1) \chi_H^l \hat{\tau}_R^l. \tag{D20}$$

Summing up over both worker types, the change in aggregate total income is:

$$\hat{X}_H = \sum_t \frac{X_H^t}{X_H} \hat{X}_H^t.$$

Denote X_{ij} the expenditure of country $j = H, F$ on final goods produced by firms in country $i = H, F$ or final goods sold by MNEs' headquarters in Foreign when $i = M$. Let P_{ij} denote the corresponding price

index. Given the CES demand in equation (3), changes in expenditure are given by:

$$\hat{X}_{ij} = (1-\sigma)\left(\hat{P}_{ij} - \hat{P}_j\right) + \hat{X}_j \qquad \text{for } i,j = H,F, \tag{D21}$$

$$\hat{X}_{Mj} = (1-\sigma)\left(\hat{P}_{Mj} - \hat{P}_j\right) + \hat{X}_j \qquad \text{for } j = H,F, \tag{D22}$$

$$\hat{X}_{HM,r} = (1-\sigma)\left(\hat{P}_{HM,r} - \hat{R}_r\right) + \widehat{R_r M}_r \qquad \text{for } r = N,R, \tag{D23}$$

$$\hat{X}_{M,r} = \widehat{R_r M}_r = \hat{d}_r + \sum_{j=H,F} \frac{X_{Mj}}{X_{MH}+X_{MF}}\left[(1-\sigma)\left(\hat{R}_r - \hat{P}_j\right) + \hat{X}_j\right] \qquad \text{for } r = N,R, \tag{D24}$$

$$\hat{Y}_M = \gamma\widehat{R_R M}_R + (1-\gamma)\,\widehat{R_N M}_N. \tag{D25}$$

Price index changes are given by:

$$\hat{P}_j = \sum_{i=H,F,M} \lambda_{ij}\hat{P}_{ij} \qquad \text{for } j = H,F, \tag{D26}$$

$$\hat{P}_{ij} = \hat{W}_{i,N} + \gamma\phi_{ij,R}\chi_i^l\hat{\tau}_R^l + \frac{\theta-(\sigma-1)}{\sigma-1}\hat{z}_{ij} \qquad \text{for } i,j = H,F, \tag{D27}$$

$$\hat{P}_{HM,r} = \hat{W}_{H,N} + \beta\chi_H^l\hat{\tau}_{H,r}^l + \frac{\theta-(\sigma-1)}{\sigma-1}\hat{z}_{HM,r} \qquad \text{for } r = N,R, \tag{D28}$$

$$\hat{R}_r = \Xi\hat{P}_{HM,r} + (1-\Xi)\,\hat{W}_{M,r} \qquad \text{for } r = N,R, \tag{D29}$$

$$\hat{P}_{Mj} = (1-\gamma)\,\hat{R}_N + \gamma\left(\hat{R}_R - \frac{1}{\sigma-1}\hat{d}_R\right) \qquad \text{for } j = H,F, \tag{D30}$$

where we have used the usual notation for trade shares, $\lambda_{ij} = \frac{X_{ij}}{X_j}$, and $\phi_{Fj,R} = 0$, since RS costs fall only on Home suppliers; $\phi_{Hj,R}$ denotes the share of Home firms' sales on market j made by suppliers of MNEs before RS (so that $\phi_{HM,R} = 1$ and $\gamma\phi_{Hj,R}$ is the share made by suppliers of RS-MNEs). The threshold changes are:

$$\hat{z}_{ij} = \frac{\sigma}{\sigma-1}\hat{W}_{i,N} - \hat{P}_j - \frac{1}{\sigma-1}\hat{X}_j \qquad \text{for } i = H; j = H,F, \tag{D31}$$

$$\hat{z}_{Fj} = -\hat{P}_j - \frac{1}{\sigma-1}\hat{X}_j, \qquad \text{for } j = H,F, \tag{D32}$$

$$\hat{z}_{HM,r} = \frac{\sigma}{\sigma-1}\hat{W}_{H,r} - \frac{1}{\sigma-1}\hat{d}_r - \sum_{j=H,F}\frac{X_{Mj}}{X_{MH}+X_{MF}}\left[\hat{P}_j + \frac{1}{\sigma-1}\hat{X}_j\right] \qquad \text{for } r = N,R. \tag{D33}$$

Log-differentiating Foreign income and the trade balance yields:

$$\hat{X}_F = \frac{aW_F L_F}{aW_F L_F + \frac{1}{\sigma}Y_M}\hat{W}_F + \frac{\frac{1}{\sigma}Y_M}{aW_F L_F + \frac{1}{\sigma}Y_M}\hat{Y}_M, \tag{D34}$$

$$\frac{X_{FH}}{X_{FH}+\frac{1}{\sigma}X_{MH}}\hat{X}_{FH} + \frac{\frac{1}{\sigma}X_{MH}}{X_{FH}+\frac{1}{\sigma}X_{MH}}\hat{X}_{MH} = \frac{\frac{\sigma-1}{\sigma}X_{MF}}{X_{HF}+\frac{\sigma-1}{\sigma}X_{MF}}\hat{X}_{MF} + \frac{X_{HF}}{X_{HF}+\frac{\sigma-1}{\sigma}X_{MF}}\hat{X}_{HF}. \tag{D35}$$

Changes in wage indexes are given by:

$$\hat{W}_{H,R} - \hat{W}_{H,N} = \chi_H^l\hat{\tau}_R^l, \tag{D36}$$

$$\hat{W}_{M,R} - \hat{W}_{M,N} = 0, \tag{D37}$$

as workers directly working at MNEs are not impacted by the RS shock – they are already enjoying better labor conditions before the MNE implements RS labor standards for its suppliers.

Changes in labor allocations $\hat{L}_H^t$ and $\hat{L}_M^t$ needed in equation (D19) are given by noting that $\hat{L}_H^t = -\hat{L}_M^t \frac{L_M^t}{L_H^t}$, and by using the MNE's change in labor demand, given CES combination of high- and low-wage workers in production:

$$\widehat{WL}_{M,r} = (1-\sigma)\left(\hat{W}_{M,r} - \hat{R}_r\right) + \hat{X}_{M,r} \qquad \text{for } r = R, N, \quad \text{(D38)}$$

$$\widehat{wL}_{M,r}^t = (1-\rho)\left(\hat{w}_H^t - \hat{W}_M\right) + \widehat{WL}_{M,r} \qquad \text{for } t = l, h; r = R, N, \quad \text{(D39)}$$

$$\widehat{wL}_M^t = (1-\rho)\left(\hat{w}_H^t - \hat{W}_M\right) + \gamma(1-\sigma)\left(\hat{W}_M - \hat{R}_R\right) \quad \text{(D40)}$$
$$+ (1-\gamma)(1-\sigma)\left(\hat{W}_M - \hat{R}_N\right) + \gamma\hat{X}_{M,R} + (1-\gamma)\hat{X}_{M,N} \text{ for } t = l, h,$$

so that overall:

$$\hat{L}_M^t = (1-\gamma)\widehat{wL}_{M,N}^t + \gamma\widehat{wL}_{M,R}^t - \hat{w}_{H,N}^t \text{ for } t = l, h. \quad \text{(D41)}$$

Finally, log-differentiating the labor market clearing equation leads to:

$$\hat{w}_{H,N}^h - \hat{w}_{H,N}^l = \frac{\varphi_{H,R}\Phi_{labor}^h \hat{\tau}_{H,R}^l + \frac{1}{\rho}\left(1-\tau_M^{-\rho}\right)\left(1-\Phi_{labor}^h\right)\hat{L}_M^h}{1-\left(1-\tau_M^{-\rho}\right)\left(1-\Phi_{labor}^h\right)}, \quad \text{(D42)}$$

where τ_M^t is the fixed wage premium that MNE subsidiaries pay type-t workers, with $\tau_M \equiv \tau_M^l > 1$ and $\tau_M^h = 1$, and Φ_{labor}^t is the share of workers of type t hired by firms, rather than by MNE subsidiaries:

$$\Phi_{labor}^t \equiv \frac{\sum_{j=H,M,F} X_{Hj}/a}{\sum_{j=H,M,F} X_{Hj}/a + \frac{\chi_M^t}{\chi_H^t}\frac{W_M L_M}{\tau_M^t}}.$$

The change in welfare for Home workers of type t is given by the solution of the system of equations (D18)–(D42).

Distributional implications To compute the welfare impact of RS on exposed workers, we write that

$$X_H^{t,E} = \left(1+\Pi^0\right) w_H^t L_H^{t,E},$$

where

$$\Pi^0 = \frac{\frac{a-1}{a}\sum_{j=H,M,F} X_{Hj}}{W_H L_H + W_M L_M}$$

measures the aggregate domestic profits that are redistributed to workers, proportional to their wage. The number of exposed workers is fixed (corresponds to those working initially for exposed suppliers). Therefore,

$$\hat{X}_H^{t,E} = \hat{w}_H^t + \hat{\tau}^t + \hat{T}_R + \hat{\Pi}^0\left(1-\varpi\right),$$

where ϖ is the share of labor income in total income of domestic workers, $\varpi = \frac{\sum_{j=H,M,F} X_{Hj}/a + W_M L_M}{\sum_{j=H,M,F} X_{Hj} + W_M L_M}$.

Changes in aggregate profits depend on changes in markups corresponding to suppliers selling to RS-MNEs, as well as changes in the composition of domestic production. Indeed, when MNE subsidiaries expand, the corresponding profit is not redistributed to domestic workers, so aggregate profits decrease when $\hat{L}_M^t$ increases. Specifically, we find:

$$\hat{\Pi}^0 = \hat{\Pi}_H - (1-\phi_M)\widehat{W_H L_H} - \phi_M \widehat{W_M L_M},$$

where Π_H denotes aggregate Home firm profits (the numerator of Π^0), which in the limit $\theta \to \sigma - 1$ satisfy $\hat{\Pi}_H = \widehat{W_H L_H} + \frac{a}{a-1}\varphi_{HM,R}\hat{a}$, and where ϕ_M is the share of total Home wage bill corresponding to MNE

subsidiaries:

$$\phi_M \equiv \frac{WL_M}{\sum_{j=H,M,F} X_{Hj}/a + WL_M},$$

and $\varphi_{HM,R}$ is the share of Home firm profits from sales to RS-MNEs defined above, so that overall, the welfare impact of RS on exposed workers is:

$$\hat{U}_H^{t,E} = \hat{w}_H^t + \hat{\tau}^t + \hat{T}_R + (1-\varpi)\,\hat{\Pi}^0 - \hat{P}_H.$$

Appendix D.5 Welfare effects: sufficient statistics

With some simplifications, the effect of RS on welfare can be derived in closed form. First, in our baseline model $X_{HF} = X_{MH} = 0$ and MNEs are owned by absentee capitalists so that $\hat{X}_F = \hat{W}_F$. Second, we consider for this derivation the special case where $\Xi = 1$, $\sigma - 1 \to \theta$ and $\gamma = 1$ (all MNEs implement RS). The system of equations (D18)–(D42) simplifies as follows. The changes in incomes are given by:

$$\begin{aligned}
\hat{X}_H &= \hat{W}_{H,N} + \Lambda\lambda_{HH}\chi_H^l\hat{\tau}_{H,R}^l + \lambda_{FH}\beta\chi_H^l\hat{\tau}_{H,R}^l + (\Lambda\lambda_{HH} + \lambda_{FH})\,\hat{T}_R,\\
\hat{X}_F &= 0.
\end{aligned}$$

The change in the Home price index is simply:

$$\hat{P}_H = \lambda_{HH}\left(\hat{W}_{H,N} + \Lambda\chi_H^l\hat{\tau}_{H,R}^l\right), \tag{D43}$$

so that the change in aggregate welfare is:

$$\hat{U}_H = (1-\lambda_{HH})\,\hat{W}_{H,N} + \beta\lambda_{FH}\chi_H^l\hat{\tau}_{H,R}^l + (\Lambda\lambda_{HH} + \lambda_{FH})\,\hat{T}_R.$$

Trade balance finally pins down the change in the Home wage index. Given simple trade patterns,

$$\hat{\lambda}_{FH} + \hat{X}_H = \hat{\lambda}_{MF},$$

where the relevant changes in expenditures are given through CES demand by:

$$\begin{aligned}
\hat{\lambda}_{MF} &= \lambda_{FF}\hat{d}_R + \lambda_{FF}\,(1-\sigma)\left(\hat{W}_{H,N} + \beta\chi_H^l\hat{\tau}_{H,R}^l\right) \quad \text{and}\\
\hat{\lambda}_{FH} &= (\sigma-1)\,\lambda_{HH}\left(\hat{W}_{H,N} + \Lambda\chi_H^l\hat{\tau}_{H,R}^l\right).
\end{aligned}$$

Solving for the Home wage index yields:

$$\hat{W}_{H,N} = -\frac{\left[\lambda_{FF}\,(\sigma-1)\,\beta + \sigma\lambda_{HH}\Lambda + \lambda_{FH}\beta\right]\chi_H^l\hat{\tau}_{H,R}^l + (\Lambda\lambda_{HH} + \lambda_{FH})\,\hat{T}_R - \lambda_{FF}\hat{d}_R}{1 + (\lambda_{FF} + \lambda_{HH})\,(\sigma-1)},$$

so that, finally, the change in aggregate welfare is:

$$\begin{aligned}
\hat{U}_H = (\beta-\Lambda)\underbrace{\frac{\lambda_{HH}\lambda_{FH}\sigma}{1 + (\lambda_{FF} + \lambda_{HH})\,(\sigma-1)}}_{\equiv W^{tax}}\chi_H^l\hat{\tau}_{H,R}^l + (\Lambda\lambda_{HH} + \lambda_{FH})\underbrace{\frac{(\sigma\lambda_{HH} + \lambda_{FF}\,(\sigma-1))}{1 + (\lambda_{FF} + \lambda_{HH})\,(\sigma-1)}}_{\equiv W^{prod}}\hat{T}_R\\
+ \lambda_{FH}\underbrace{\frac{\lambda_{FF}}{1 + (\lambda_{FF} + \lambda_{HH})\,(\sigma-1)}}_{\equiv W^{d}}\hat{d}_R.
\end{aligned} \tag{D44}$$

Distributional effects for low- vs. high-wage workers To disentangle the effect of the policy on low- and high-wage workers, note that labor market clearing yields:

$$\hat{w}^l_{H,N} - \hat{w}^h_{H,N} = -\left(\Lambda\lambda_{HH} + \lambda_{FH}\right)\hat{\tau}^l_{H,R},$$

while the change in income for worker type t is given by:

$$\hat{X}^t_H = \hat{w}^t_{H,N} + \left(\Lambda\lambda_{HH} + \lambda_{FH}\right)\left(\hat{T}_R + \hat{\tau}^t_{H,R}\right) + \lambda_{FH}\left(\beta - 1\right)\chi^l_H\hat{\tau}^l_{H,R},$$

as profits are apportioned to the wage bill. Overall,

$$\hat{U}^l_H - \hat{U}^h_H = \hat{X}^l_H - \hat{X}^h_H = \left(\hat{w}^l_{H,N} - \hat{w}^h_{H,N}\right) + \left(\Lambda\lambda_{HH} + \lambda_{FH}\right)\left(\hat{\tau}^l_{H,R}\right) = 0,$$

so that low- and high-wage workers benefit on average from the exact same welfare gains:

$$\hat{U}^l_H = \hat{U}^h_H = \hat{U}_H. \tag{D45}$$

Distributional effects for exposed vs. non-exposed workers Exposed low- or high-wage workers are defined as those who were working at suppliers to an MNE before that MNE rolls out its first RS policy. The exposed group is indexed by superscript E. We have:

$$\begin{aligned} X_H &= X_{HH} + X_{HM}, \\ X^E_H &= \Lambda X_{HH} + X_{HM}, \end{aligned}$$

where X^E_H is exposed income (and for type t: $X^t_H = \chi^t_H X_H$, $X^{t,E}_H = \chi^{t,E}_H X^E_H$). Using derivations similar to the ones in the main model, we have:

$$\hat{X}^{t,E}_H = \hat{w}^t_{H,N} + \hat{T}_R + \hat{\tau}^t_{H,R} + \lambda_{FH}\hat{a},$$

while

$$\hat{X}^{t,NE}_H = \hat{w}^t_{H,N} + \lambda_{FH}\hat{a},$$

where $\lambda_{FH}\hat{a}$ is the change in per-capita profit, redistributed to all workers irrespective of whether they are exposed or not. Since both groups face the same change in the price index (given by equation (D43)), we have, in relative terms:

$$\Delta\hat{U}^t_H \equiv \hat{U}^{t,E}_H - \hat{U}^{t,NE}_H = \hat{X}^{t,E}_H - \hat{X}^{t,NE}_H = \hat{T}_R + \hat{\tau}^t_{H,R},$$

which is different for high- vs. low-wage workers because of the term $\hat{\tau}^t_{H,R}$.

Using that $\Gamma^{t,E}\hat{U}^{t,E}_H + \left(1 - \Gamma^{t,E}\right)\hat{U}^{t,NE}_H = \hat{U}^t_H$ where $\hat{U}^t_H$ is given by equation (D45) and $\Gamma^{t,E} \equiv \frac{X^{t,E}_H}{X^t_H}$, we find:

$$\begin{aligned} \hat{U}^{t,E}_H &= \hat{U}_H + \lambda_{HH}\left(1 - \Lambda\right)\Delta\hat{U}^t_H, \\ \hat{U}^{t,NE}_H &= \hat{U}_H - \left(\Lambda\lambda_{HH} + \lambda_{FH}\right)\Delta\hat{U}^t_H. \end{aligned}$$

That is, using equation (D44):

$$\hat{U}^{l,NE}_H = \left[\left(\beta - \Lambda\right)W^{tax}\chi^l_H - \left(\lambda_{FH} + \Lambda\lambda_{HH}\right)\right]\hat{\tau}^l_R \;+\; \left(\lambda_{FH} + \Lambda\lambda_{HH}\right)\left(W^{prod} - 1\right)\hat{T}_R + \lambda_{FH}W^d\hat{d}_R.$$

It is immediately clear that $W^{prod} - 1 = \frac{-1+\lambda_{HH}}{1+(\lambda_{FF}+\lambda_{HH})(\sigma-1)} < 0$, so that the second term is negative.

One can also show that for $\beta \leq 1$, $(\beta - \Lambda) W^{tax} < (\lambda_{FH} + \Lambda\lambda_{HH}) W^{prod}$. To that end, note that

$$\lambda_{HH}\lambda_{FH} < \lambda_{FH}\left[\lambda_{HH} + \lambda_{FF}\left(\frac{\sigma-1}{\sigma}\right)\right],$$

therefore for any $\Lambda \geq 0$, the following is also true:

$$\lambda_{HH}\lambda_{FH}(1-\Lambda) < (\Lambda\lambda_{HH} + \lambda_{FH})\left[\lambda_{HH} + \lambda_{FF}\left(\frac{\sigma-1}{\sigma}\right)\right].$$

Hence

$$\lambda_{HH}\lambda_{FH}\sigma(1-\Lambda) < (\Lambda\lambda_{HH} + \lambda_{FH})\left[\sigma\lambda_{HH} + \lambda_{FF}(\sigma-1)\right],$$

which means that

$$(1-\Lambda) W^{tax} < (\lambda_{FH} + \Lambda\lambda_{HH}) W^{prod},$$

and, in turn, for any $\beta \leq 1$ and $\chi_H^l \leq 1$:

$$(\beta - \Lambda) W^{tax}\chi_H^l < (\lambda_{FH} + \Lambda\lambda_{HH}) W^{prod}.$$

Therefore,

$$(\beta - \Lambda) W^{tax}\chi_H^l - (\lambda_{FH} + \Lambda\lambda_{HH}) < (\lambda_{FH} + \Lambda\lambda_{HH}) W^{prod} - (\lambda_{FH} + \Lambda\lambda_{HH}) < 0.$$

To conclude, the first and second terms in $\hat{U}_H^{l,NE}$ are unambiguously negative while only the last term is positive.

Appendix D.6 Model extensions

Appendix D.6.1 Labor market power

A natural question is whether RS policies are implemented in a context where—from an efficiency perspective—wages are initially too low. In the baseline model presented in the main text, wages are those that clear the market—they are not too low from an efficiency perspective, and raising them introduces, *a priori*, a distortion. Alternatively, it could be that wages are set too low compared to an efficient benchmark. Capturing this possibility requires Home firms to exert labor market power on the Home labor market. To entertain the possibility of pre-existing wage markdowns, we now extend the model to feature an upward-sloping labor supply curve that Home firms are facing. To generate this feature in the most tractable way, we assume that, in addition to their preferences over a CES consumption bundle, workers have heterogeneous preferences for jobs. Utility of worker i working on production line ω is:

$$U^i = C\varepsilon^i(\omega),$$

where $C = \left(\int_{\Omega^k} d_\omega^{1/\sigma} q_\omega^{\frac{\sigma-1}{\sigma}} d\omega\right)^{\frac{\sigma}{\sigma-1}}$ as above, and idiosyncratic preferences $\varepsilon^i(\omega)$ are drawn i.i.d. across workers and production lines, according to a Fréchet distribution with shape parameter κ. Workers are, therefore, ex-ante homogeneous but ex-post heterogeneous. Production of firms and MNEs is otherwise unchanged, with, for simplicity, only one ex-ante worker type, whose exogenous aggregate supply is L_k in country k. That is, workers are perfect substitutes in production and ℓ_ω is the number of workers hired on production line ω. With this setup, firms face an upward-sloping labor supply curve when hiring on their production line producing good ω:

$$\frac{\ell_\omega}{L_H} = \left(\frac{w_\omega}{\Phi}\right)^\kappa, \text{with } \Phi = \left(\int_{\Omega_{HH}\cup\Omega_{HF}\cup\Omega_x} {w_\omega}^\kappa d\omega\right)^{\frac{1}{\kappa}}.$$

Notice that when $\kappa \to \infty$, the model collapses to a familiar setup in which all workers are identical and firms face a perfectly elastic labor supply, as in our baseline model with one type of worker (nested in the main one). Importantly, we assume here that firms set wages according to monopsonistic competition. Because they face a firm-specific upward-sloping labor supply curve, firms restrict hiring to keep the wages of all their workers low. Formally, taking the first-order condition for profit maximization of the supplier leads to the following wage profile across heterogeneous firms and across production lines:

$$w_{Hj,r} = \frac{\sigma-1}{\sigma}\frac{\kappa}{\kappa+1} z p_{Hj,r}.$$

Firms optimally set wages at a markdown $\frac{\kappa}{\kappa+1}$ relative to their marginal revenue product of labor. Using the product market clearing on the output markets pins down the scale of production of each firm on each production line, given this wage-price schedule. In equilibrium, a firm with productivity z on production line $j = H, F, M$ optimally offers wages:

$$w^*_{Hj,r} = z^{\frac{\sigma-1}{\kappa+\sigma}} \Phi^{\frac{\kappa}{\kappa+\sigma}} D_{j,r}^{\frac{1}{\kappa+\sigma}} L_H^{-\frac{1}{\kappa+\sigma}} \left(\frac{\sigma-1}{\sigma}\frac{\kappa}{\kappa+1}\right)^{\frac{\sigma}{\kappa+\sigma}}, \tag{D46}$$

where the demand shifters $D_{j,r}$ on market j are defined in equations (D5) and (D6). When wages are optimally chosen, the sales of a firm with productivity z on market j are given by:

$$y_{Hj,r} = z^{\frac{(\sigma-1)(\kappa+1)}{\kappa+\sigma}} \Phi^{\frac{\kappa(1-\sigma)}{\kappa+\sigma}} (D_{j,r})^{\frac{1+\kappa}{\kappa+\sigma}} L_H^{\frac{\sigma-1}{\kappa+\sigma}} \left(\frac{\sigma-1}{\sigma}\frac{\kappa}{\kappa+1}\right)^{\frac{\kappa(\sigma-1)}{\kappa+\sigma}}.$$

Note that if a wage $\underline{w}$ is imposed on the firm through RS, rather than being chosen optimally by the firm, firm sales depend on whether hiring is determined by the labor supply curve (which is the case when labor supply $\leq$ labor demand), or whether it is determined by the labor demand curve (when labor supply $\geq$ labor demand). In the former case, we have:

$$y_{Hj,R} = \frac{\sigma}{\sigma-1}\underline{w}^{\kappa}\Phi^{1-\kappa}\Gamma\left(1-\frac{1}{\kappa}\right)L_H, \tag{D47}$$

where Γ is the Gamma function. In the latter case, we have:

$$y_{Hj,R} = \left(\frac{\sigma}{\sigma-1}\right)^{1-\sigma} z^{\sigma-1}\underline{w}^{1-\sigma} D_{j,r}. \tag{D48}$$

Comparative statics Next, we examine how RS impacts firm sales under the hypothesis that labor markets are monopsonistic. Our strategy is still to compare firms with similar productivity, some being exposed to RS, and others not. Given that wages are heterogeneous, we model the RS policy as a wage floor that stipulates that $w_{i,R} \geq \underline{w}$. Given that the RS policy $\underline{w}$ is binding only for firms at which wages are low, several cases arise, depending on where the firm wage, pre-RS, lies compared to the wage floor $\underline{w}$ imposed by the RS policy. To that end, we denote $w^*_k(z)$ the monopsony wage level of a firm with productivity z on production line $k = H, F, M$. Three main cases arise.

First, if $w^*_k(z) \geq \underline{w}$, that is, given equation (D46), when firm productivity is high enough, RS is not binding. There is no relative effect of RS on suppliers that adopt it vs. those with equivalent productivity that do not.

Second, when $w^*_k(z) < \underline{w} \leq \frac{\kappa+1}{\kappa} w^*_k(z)$, RS is now binding and corresponds to a wage increase from $w^*_k(z)$ to $\underline{w}$ for impacted firms. In this case, the sales of compliers go up, both on the final goods markets and on the intermediate goods market. This sales increase comes from the following mechanism: higher wages make the firm hire more employees compared to the monopsonistic case where the firm voluntarily restricted its hiring. This leads to higher production and higher sales, given that the wage (hence price)

increase is moderate—but of course, to lower profits.[iv]

Third, if $\underline{w} > \frac{\kappa+1}{\kappa} w_k^*(z)$, which could be the case for the lowest productivity firms, these firms see their sales decrease. The wage increase is too high to sustain higher sales.

Overall, for compliers, we have the following comparative statics for $k = H, F, M$:

$$\begin{cases} \hat{y}_{k,R} - \hat{y}_{k,N} = 0 & \text{if } w_k^*(z) \geq \underline{w} \\ \hat{y}_{k,R} - \hat{y}_{k,N} \geq 0 & \text{if } w_k^*(z) \leq \underline{w} \leq \frac{\kappa+1}{\kappa} w_k^*(z) \\ \hat{y}_{k,R} - \hat{y}_{k,N} \leq 0 & \text{if } w_k^*(z) < \frac{\kappa}{\kappa+1} \underline{w} \end{cases}$$

In practice, note that this third case is likely to be of limited empirical relevance. First, because these lower productivity firms are likely to exit the market. Second, because the RS wage is unlikely to be high enough to trigger a wage increase of more than $\frac{\kappa+1}{\kappa}$, which corresponds to a 20% increase in wages for typical values of the parameter κ.[v] Therefore, we expect that on average for complying firms,

$$\hat{y}_{k,R} - \hat{y}_{k,N} \geq 0,$$

both on the final goods market and on the MNE input market.

Turning to the effect on exposed firms, we need to take into account the extensive margin effect of the RS policy, in addition to this positive effect of RS on the intensive margin. Because the RS policy reduces profits for all firms for which RS is binding, the policy is accompanied by exits of pre-existing suppliers that were close to the selection cutoff. Therefore, the effect of the RS policy on exposed firms is overall ambiguous:

$$\hat{Y}_{tot,R} - \hat{Y}_{tot,N} \text{ has an ambiguous sign.}$$

Appendix D.6.2 Unemployment

We recompute the welfare gains of a representative worker $\hat{U}_H$, now assuming that there is unemployment in the economy. Specifically, workers choose whether to be unemployed and get a fixed utility u_0, or work and get utility $\frac{\bar{w}}{P_H}$. Each worker has idiosyncratic preferences for either option, assumed to follow a Fréchet distribution with mean 1 and shape κ. Formally,

$$U(\omega) = \max\left\{ \frac{\bar{w}}{P_H} \epsilon_w(\omega), u_0 \epsilon_u(\omega) \right\},$$

where $\bar{w} = \frac{X_H}{L_{H,N}+L_{H,R}} = \frac{w_H L_{H,N} + \tilde{w}_{H,R} L_{H,R} + \Pi_{H,N} + \Pi_{H,R}}{L_{H,N}+L_{H,R}}$ is the income per capita in the employed sector. Labor market clearing writes:

$$L_{H,N} + L_{H,R} + L_{H,U} = L_H.$$

Given the properties of the Fréchet distribution, the share of unemployed workers λ^U is:

$$\lambda^U = \frac{L_{H,U}}{L_H} = \frac{u_0^\kappa}{u_0^\kappa + \left(\frac{\bar{w}}{P_H}\right)^\kappa}.$$

[iv] These qualitative patterns mask two different subcases: one where firm hiring is set by the labor supply curve, hence sales are given by (D47). This happens so long as $\underline{w} \leq w_k^{eq}$, where w_k^{eq} is a firm-specific equilibrium wage for which labor supply equals labor demand on production line k. In the other subcase, the wage increase is high enough that the labor supply is higher than labor demand, hence sales are pinned down by equation (D48), but the wage increase is not too high, so that labor hired is still above the monopsonistic level.

[v] For instance, Berger et al. (2022) find values of the labor supply elasticity ranging from $\kappa \in (3, 7)$, which leads to $\frac{1}{\kappa} \in (14\%, 33\%)$.

We now compute how the expected welfare in the economy,

$$U_H = \mathbb{E}\left(U\right) \propto \left(u_0^{\kappa} + \left(\frac{\bar{w}}{P_H}\right)^{\kappa}\right)^{\frac{1}{\kappa}}, \tag{D49}$$

will change following RS. Given equation (D49), a small shock to the economy yields:

$$\hat{U}_H = \left(1 - \lambda^U\right)\left(\hat{\bar{w}} - \hat{P}_H\right),$$

with

$$\widehat{L}_{H,E} = \left(\hat{\bar{w}} - \hat{P}_H\right)\kappa\lambda^U,$$

where $L_{H,E} = L_{H,N} + L_{H,R}$ (E stands for employed). First, we compute $\hat{\bar{w}} - \hat{P}_H$. As in the baseline case, income per capita changes according to:

$$\hat{\bar{w}} = \hat{W}_{H,N} + \Lambda\lambda_{HH}\chi_H^l\hat{\tau}_{H,R}^l + \lambda_{FH}\beta\chi_H^l\hat{\tau}_{H,R}^l + \left(\Lambda\lambda_{HH} + \lambda_{FH}\right)\hat{T}_R,$$

while the expression for the change in the price index is unchanged compared to the baseline case so that:

$$\hat{\bar{w}} - \hat{P}_H = \lambda_{FH}\hat{W}_{H,N} + \lambda_{FH}\beta\chi_H^l\hat{\tau}_{H,R}^l + \left(\Lambda\lambda_{HH} + \lambda_{FH}\right)\hat{T}_R. \tag{D50}$$

Second, we use the trade balance to solve for $\hat{W}_{H,N}$ as a function of $\hat{\bar{w}} - \hat{P}_H$:

$$\begin{aligned}
\hat{X}_H &= \hat{\lambda}_{MF} - \hat{\lambda}_{FH}, \\
\hat{X}_H - \hat{P}_H &= \hat{\lambda}_{MF} - \hat{\lambda}_{FH} - \hat{P}_H, \\
\left(\hat{\bar{w}} - \hat{P}_H\right)\left(1 + \kappa\lambda^U\right) &= \hat{\lambda}_{MF} - \hat{\lambda}_{FH} - \hat{P}_H.
\end{aligned}$$

Using, as in the baseline case:

$$\begin{aligned}
\hat{\lambda}_{MF} &= \lambda_{FF}\hat{d}_R + \left(1 - \lambda_{MF}\right)\left(1 - \sigma\right)\left(\hat{W}_{H,N} + \beta\chi_H^l\hat{\tau}_{H,R}^l\right), \\
\hat{\lambda}_{FH} &= \left(\sigma - 1\right)\lambda_{HH}\left(\hat{W}_{H,N} + \Lambda\chi_H^l\hat{\tau}_{H,R}^l\right),
\end{aligned}$$

we get:

$$\begin{aligned}
\hat{W}_{H,N} = \left(\hat{\bar{w}} - \hat{P}_H\right)\frac{\left(1 + \kappa\lambda^U\right)}{\left(\lambda_{FF} + \lambda_{HH}\right)\left(1 - \sigma\right) - \lambda_{HH}} - \left[\frac{\left(1 - \sigma\right)\lambda_{FF}\beta - \sigma\lambda_{HH}\Lambda}{\left(\lambda_{FF} + \lambda_{HH}\right)\left(1 - \sigma\right) - \lambda_{HH}}\right]\chi_H^l\hat{\tau}_{H,R}^l \\
- \frac{\lambda_{FF}}{\left(\lambda_{FF} + \lambda_{HH}\right)\left(1 - \sigma\right) - \lambda_{HH}}\hat{d}_R.
\end{aligned}$$

Therefore, using equation (D50), we have:

$$\begin{aligned}
\hat{\bar{w}} - \hat{P}_H = \frac{\left(\lambda_{FF} + \lambda_{HH}\right)\left(1 - \sigma\right) - 1}{\left(\lambda_{FF} + \lambda_{HH}\right)\left(1 - \sigma\right) - 1 - \lambda_{FH}\kappa\lambda^U}\Big\{\left(\beta - \Lambda\right)W^{tax}\chi_H^l\hat{\tau}_{H,R}^l \\
+ \left(\Lambda\lambda_{HH} + \lambda_{FH}\right)W^{prod}\hat{T}_R + \lambda_{FH}W^d\hat{d}_R\Big\},
\end{aligned}$$

so that overall:

$$\hat{U}_H = \frac{1-\lambda^U}{1+\frac{\lambda_{FH}\kappa\lambda^U}{1+(\lambda_{FF}+\lambda_{HH})(\sigma-1)}} \left\{ \underbrace{(\beta-\Lambda)\, W^{tax}\chi_H^l\hat{\tau}_{H,R}^l + (\Lambda\lambda_{HH}+\lambda_{FH})\, W^{prod}\hat{T}_R + \lambda_{FH}W^d\hat{d}_R}_{\text{baseline } \hat{U}_H} \right\}.$$

We see that the aggregate welfare gains of the baseline case are dampened by the term:

$$\frac{1-\lambda^U}{1+\frac{\lambda_{FH}\kappa\lambda^U}{1+(\lambda_{FF}+\lambda_{HH})(\sigma-1)}}$$

in the presence of unemployment. Overall, the welfare effect of RS is only very slightly attenuated by the presence of unemployment in our empirical setting (Appendix Figure E3).

Appendix D.6.3 Extension with a more general demand system

We allow here for a more flexible demand system faced by CR firms, with different elasticities of substitution between the final goods and the intermediate input markets. Specifically, we assume that:

$$U_i = \left(\int_{\Omega_i} d_\omega^{\frac{1}{\sigma_i}} q_\omega^{\frac{\sigma_i-1}{\sigma_i}} d\omega \right)^{\frac{\sigma_i}{\sigma_i-1}}, \text{ for } i = H, F,$$

$$M = \left(\int_{\Omega_x} m_{\omega(x)}^{\frac{\sigma_M-1}{\sigma_M}} d\omega\,(x) + \nu^\ell \ell_M^{\frac{\sigma_M-1}{\sigma_M}} \right)^{\frac{\sigma_M}{\sigma_M-1}},$$

with $\sigma_H = \sigma_F$ but σ_M is possibly different.

Comparative statics The comparative statics are modified as follows. The relative impact of RS on firms' output prices is unchanged, but firm sales on destination market $j = H, F, M$ for a firm with RS status $r = R$ or N are now given by:

$$y_{Hj,r} = p_{Hj,r}^{1-\sigma_j} D_{j,r},$$

where

$$D_{j,r} = P_j^{\sigma_j-1} X_j \quad \text{for } j = H, F,$$
$$\text{and } D_{M,r} = N_r R_r^{\sigma_M} M_r.$$

Therefore, computations for the sales on the final goods market (where the relevant elasticity of substitution is σ_H) are unchanged and, with N_r the mass of MNEs with RS status r and M_r the output of each of them, the sales to the MNE become:

$$y_{HM,r} = p_{HM,r}^{1-\sigma_M} N_r R_r^{\sigma_M-\sigma_H} d_r \left(\frac{\sigma_H}{\sigma_H-1} \right)^{-\sigma_H} (\varrho)^{1-\sigma_H} \sum_j \varrho_{Mj}^{1-\sigma_H} D_{j,r}.$$

The comparative statics on the sales of compliers are:

$$\hat{y}_{Hj,R} - \hat{y}_{Hj,N} = (1-\sigma_H)\,\chi_H^l\hat{\tau}_R^l < 0, \quad \text{for } j = H, F,$$

$$\hat{y}_{HM,R} - \hat{y}_{HM,N} = (1-\sigma_M)\,\beta\chi_H^l\hat{\tau}_R^l + (\sigma_M-\sigma_H)\,\Xi \left(\frac{\beta + \left(\frac{\sigma_M-1-\theta}{1-\sigma_M}\right)\frac{\sigma_M}{\sigma_M-1}}{1+\Xi\left(\frac{\sigma_M-1-\theta}{1-\sigma_M}\right)\frac{\sigma_M-\sigma_H}{\sigma_M-1}} \right) \chi_H^l\hat{\tau}_R^l$$
$$+ \hat{d}_R \left(1 - (\sigma_M-\sigma_H) \left(\frac{\Xi\frac{\theta-\sigma_M+1}{\sigma_M-1}\frac{1}{\sigma_M-1}}{1+\Xi\left(\frac{\theta-\sigma_M+1}{\sigma_M-1}\right)\frac{\sigma_M-\sigma_H}{\sigma_M-1}} \right) \right).$$

Turning to exposed firms, the relative change in the productivity cutoff for serving the MNE market is:

$$\hat{z}^*_{HM,R} - \hat{z}^*_{HM,N} = \frac{\sigma_M}{\sigma_M - 1}\chi^l_H \hat{\tau}^l_R - \frac{\sigma_M - \sigma_H}{\sigma_M - 1}\left(\frac{\Xi\beta\chi^l_H\hat{\tau}^l_R + \Xi\left(\frac{\sigma_M-1-\theta}{1-\sigma_M}\right)\frac{\sigma_M}{\sigma_M-1}\chi^l_H\hat{\tau}^l_R - \Xi\frac{\sigma_M-1-\theta}{1-\sigma_M}\frac{1}{\sigma_M-1}\hat{d}_R}{1+\Xi\left(\frac{\sigma_M-1-\theta}{1-\sigma_M}\right)\frac{\sigma_M-\sigma_H}{\sigma_M-1}}\right) - \frac{1}{\sigma_M-1}\hat{d}_R,$$

so that the impact of RS on exposed firms (including total sales) is:

$$\hat{Y}_{Hj,R} - \hat{Y}_{Hj,N} = (1-\sigma_H)\ \chi^l_H\hat{\tau}^l_R < 0, \quad \text{for } j = H, F,$$

$$\hat{Y}_{HM,R} - \hat{Y}_{HM,N} = \left\{(1-\sigma_M)\beta - (\theta-\sigma_M+1)\frac{\sigma_M}{\sigma_M-1} + \frac{\theta(\sigma_M-\sigma_H)}{\sigma_M-1}\left(\frac{\beta + \frac{\theta-\sigma_M+1}{\sigma_M-1}\frac{\sigma_M}{\sigma_M-1}}{1+\Xi\frac{\theta-\sigma_M+1}{\sigma_M-1}\frac{\sigma_M-\sigma_H}{\sigma_M-1}}\right)\Xi\right\}\chi^l_H\hat{\tau}^l_R + \left(\frac{\theta}{\sigma_M-1} - (\sigma_M-\sigma_H)\frac{\theta}{\sigma_M-1}\left(\frac{\frac{\theta-\sigma_M+1}{\sigma_M-1}\frac{1}{\sigma_M-1}}{1+\Xi\left(\frac{\theta-\sigma_M+1}{\sigma_M-1}\right)\frac{\sigma_M-\sigma_H}{\sigma_M-1}}\right)\Xi\right)\hat{d}_R,$$

$$\hat{Y}_{Htot,R} - \hat{Y}_{Htot,N} = (1-\zeta)(1-\sigma_H)\ \chi^l_H\hat{\tau}^l_R + \zeta\left\{(1-\sigma_M)\beta - (\theta-\sigma_M+1)\frac{\sigma_M}{\sigma_M-1} + \frac{\theta(\sigma_M-\sigma_H)}{\sigma_M-1}\left(\frac{\beta + \frac{\theta-\sigma_M+1}{\sigma_M-1}\frac{\sigma_M}{\sigma_M-1}}{1+\Xi\frac{\theta-\sigma_M+1}{\sigma_M-1}\frac{\sigma_M-\sigma_H}{\sigma_M-1}}\right)\Xi\right\}\chi^l_H\hat{\tau}^l_R + \zeta\frac{\theta}{\sigma_M-1}\left(1 - (\sigma_M-\sigma_H)\left(\frac{\frac{\theta-\sigma_M+1}{\sigma_M-1}\frac{1}{\sigma_M-1}}{1+\Xi\left(\frac{\theta-\sigma_M+1}{\sigma_M-1}\right)\frac{\sigma_M-\sigma_H}{\sigma_M-1}}\right)\Xi\right)\hat{d}_R. \quad \text{(D51)}$$

Finally, turning to the impact of RS on the sales of the MNE subsidiary, we get:

$$\widehat{R_R M_R} - \widehat{R_N M_N} = (1-\sigma_H)\frac{\beta + \frac{\theta-\sigma_M+1}{\sigma_M-1}\frac{\sigma_M}{\sigma_M-1}}{1+\Xi\frac{\theta-\sigma_M+1}{\sigma_M-1}\frac{\sigma_M-\sigma_H}{\sigma_M-1}}\Xi\chi^l_H\hat{\tau}^l_R + \left(1+\Xi\frac{\frac{\theta-\sigma_M+1}{\sigma_M-1}\frac{\sigma_H-1}{\sigma_M-1}}{1+\Xi\frac{\theta-\sigma_M+1}{\sigma_M-1}\frac{\sigma_M-\sigma_H}{\sigma_M-1}}\right)\hat{d}_R.$$

Welfare computations We are interested in the change in welfare for workers of type t,

$$\hat{U}^t_H = \hat{X}^t_H - \hat{P}_H. \quad \text{(D52)}$$

The income of type t is given by:

$$X^t_H = \sum_{r=R,N}\left\{\sum_{j=H,M,F}\tilde{w}^t_{H,r}L^t_{Hj,r} + \tilde{w}^t_{M,r}L^t_{M,r}\right\} + \sum_{r=R,N}\sum_{j=H,M,F}(a_{Hj,r}-1)\tilde{w}^t_{H,r}L^t_{Hj,r},$$

where $a_{H,N} = \frac{\theta\sigma_H}{\theta\sigma_H-(\sigma_H-1)} = a_{F,N}$ and $a_{M,N} \equiv \frac{\theta\sigma_M}{\theta\sigma_M-(\sigma_M-1)}$. Throughout this subsection we write $a_j \equiv a_{Hj,N}$ for $j = H, F$ and $a_M \equiv a_{M,N}$. Log-differentiating this equation leads to:

$$\hat{X}^t_H = \Phi^t\left[\hat{w}^t_{H,N} + \phi^t_{H,R}\left(\hat{\tau}^t_R + \hat{T}_R\right) + \left(1-\phi^t_M\right)\hat{L}^t_H + \phi^t_M\hat{L}^t_M\right] + \left(1-\Phi^t\right)\left[\hat{w}^t_{H,N} + \sum_{j=H,F,M}\Psi_j\hat{L}^t_{Hj} + \Psi^t_R\left(\hat{\tau}^t_R + \hat{T}_R\right) + \Psi_{HM,R}\hat{a}_{M,R}\frac{a_{M,N}}{a_{M,N}-1}\right], \quad \text{(D53)}$$

where Φ^t is the share of the labor income in the income of type t in the initial equilibrium:

$$\Phi^t \equiv \frac{\sum_{j=H,M,F}\chi^t_H X_{Hj}/a_j + \chi^t_M W_M L_M}{\sum_{j=H,M,F}\chi^t_H X_{Hj} + \chi^t_M W_M L_M},$$

$\phi^t_{H,R}$ is the share of total Home wage bill for type t corresponding to firms that will implement RS labor standards:

$$\phi^t_{H,R} \equiv \frac{\sum_{j=H,M,F} \gamma \phi_{Hj,R} \chi^t_H X_{Hj}/a_j}{\sum_{j=H,M,F} \chi^t_H X_{Hj}/a_j + \chi^t_M W L_M},$$

ϕ^t_M is the share of total Home wage bill for type t corresponding to MNE subsidiaries M:

$$\phi^t_M \equiv \frac{\chi^t_M W L_M}{\sum_{j=H,M,F} \chi^t_H X_{Hj}/a_j + \chi^t_M W L_M},$$

Ψ_j is the share of total Home firm profits coming from each market j:

$$\Psi_j = \frac{\Pi^t_{Hj}}{\sum_{j=H,F,M} \Pi^t_{Hj}},$$

Ψ_R is the share of total Home firm profits corresponding to firms that will implement RS labor standards, that is:

$$\Psi^t_R = \frac{\sum_{j=H,M,F} \Pi^t_{Hj,R}}{\sum_{j=H,F,M} \Pi^t_{Hj}},$$

and finally $\Psi_{HM,R}$ is the share of Home firm profits corresponding to sales to MNE subsidiaries that will implement RS:

$$\Psi_{HM,R} \equiv \frac{\gamma \Pi_{HM}}{\sum_{j=H,M,F} \Pi_{Hj}}.$$

These latter profits will be differentially affected between production lines since RS can be accompanied by a reduction in the supplier's markup, through the imperfect pass-through β of the policy. Specifically,

$$\hat{a}_{M,R} = \frac{(\sigma_M - 1)^2}{\theta \sigma_M - (\sigma_M - 1)} (\beta - 1) \chi^l_H \hat{\tau}^l_R.$$

Summing up over both worker types, the change in aggregate total income is:

$$\hat{X}_H = \sum_t \frac{X^t_H}{X_H} \hat{X}^t_H.$$

Denote X_{ij} the expenditure of country $j = H, F$ on final goods produced by firms in country $i = H, F$ or final goods sold by MNEs' headquarters in Foreign when $i = M$. Let P_{ij} denote the corresponding price index. Given the CES demand in equation (3), the changes in expenditure are given by:

$$\hat{X}_{ij} = (1 - \sigma_j) \left(\hat{P}_{ij} - \hat{P}_j \right) + \hat{X}_j \qquad \text{for } i, j = H, F, \tag{D54}$$

$$\hat{X}_{Mj} = (1 - \sigma_j) \left(\hat{P}_{Mj} - \hat{P}_j \right) + \hat{X}_j \qquad \text{for } j = H, F, \tag{D55}$$

$$\hat{X}_{HM,r} = (1 - \sigma_M) \left(\hat{P}_{HM,r} - \hat{R}_r \right) + \widehat{R_r M}_r \qquad \text{for } r = N, R, \tag{D56}$$

$$\widehat{R_r M}_r = \hat{d}_r + \sum_{j=H,F} \frac{X_{Mj}}{X_{MH} + X_{MF}} \left[(1 - \sigma_j) \left(\hat{R}_r - \hat{P}_j \right) + \hat{X}_j \right], \tag{D57}$$

$$\hat{Y}_M = \gamma \widehat{R_R M}_R + (1 - \gamma) \widehat{R_N M}_N. \tag{D58}$$

Price index changes are given by:

$$\hat{P}_j = \sum_{i=H,F,M} \lambda_{ij} \hat{P}_{ij} \qquad \text{for } j = H, F, \tag{D59}$$

$$\hat{P}_{ij} = \hat{W}_{i,N} + \gamma\phi_{ij,R}\chi_i^l\hat{\tau}_R^l + \frac{\theta-(\sigma_j-1)}{\sigma_j-1}\hat{z}_{ij} \qquad \text{for } i,j = H,F, \tag{D60}$$

$$\hat{P}_{HM,r} = \hat{W}_{H,N} + \beta\chi_H^l\hat{\tau}_{H,r}^l + \frac{\theta-(\sigma_M-1)}{\sigma_M-1}\hat{z}_{HM,r} \qquad \text{for } r = N,R, \tag{D61}$$

$$\hat{R}_r = \Xi\hat{P}_{HM,r} + (1-\Xi)\hat{W}_{M,r} \qquad \text{for } r = N,R, \tag{D62}$$

$$\hat{P}_{Mj} = (1-\gamma)\hat{R}_N + \gamma\left(\hat{R}_R - \frac{1}{\sigma_j-1}\hat{d}_R\right) \qquad \text{for } j = H,F, \tag{D63}$$

where we have used the usual notation for trade shares, $\lambda_{ij} = \frac{X_{ij}}{X_j}$, and $\phi_{Fj,R} = 0$, since RS costs fall only on Home suppliers; $\phi_{Hj,R}$ denotes the share of Home firms' sales on market j made by suppliers of MNEs before RS (so that $\phi_{HM,R} = 1$ and $\gamma\phi_{Hj,R}$ is the share made by suppliers of RS-MNEs). The threshold changes are:

$$\hat{z}_{ij} = \frac{\sigma_j}{\sigma_j-1}\hat{W}_{i,N} - \hat{P}_j - \frac{1}{\sigma_j-1}\hat{X}_j \qquad \text{for } i = H; j = H,F, \tag{D64}$$

$$\hat{z}_{Fj} = -\hat{P}_j - \frac{1}{\sigma_j-1}\hat{X}_j, \tag{D65}$$

$$\begin{aligned}\hat{z}_{HM,r} = {} & \frac{\sigma_M}{\sigma_M-1}\hat{W}_{H,r} - \frac{\sigma_M-\sigma_H}{\sigma_M-1}\hat{R}_r - \frac{1}{\sigma_M-1}\hat{d}_r \\ & - \sum_{j=H,F}\frac{X_{Mj}}{X_{MH}+X_{MF}}\left[\frac{\sigma_H-1}{\sigma_M-1}\hat{P}_j + \frac{1}{\sigma_M-1}\hat{X}_j\right] \qquad \text{for } r = N,R.\end{aligned} \tag{D66}$$

Log-differentiating Foreign income and the trade balance yields:

$$\hat{X}_F = \frac{a_H W_F L_F}{a_H W_F L_F + \frac{1}{\sigma_H}Y_M}\hat{W}_F + \frac{\frac{1}{\sigma_H}Y_M}{a_H W_F L_F + \frac{1}{\sigma_H}Y_M}\hat{Y}_M, \tag{D67}$$

$$\frac{X_{FH}}{X_{FH}+\frac{1}{\sigma_H}X_{MH}}\hat{X}_{FH} + \frac{\frac{1}{\sigma_H}X_{MH}}{X_{FH}+\frac{1}{\sigma_H}X_{MH}}\hat{X}_{MH} = \frac{\frac{\sigma_H-1}{\sigma_H}X_{MF}}{X_{HF}+\frac{\sigma_H-1}{\sigma_H}X_{MF}}\hat{X}_{MF} + \frac{X_{HF}}{X_{HF}+\frac{\sigma_H-1}{\sigma_H}X_{MF}}\hat{X}_{HF}. \tag{D68}$$

Changes in labor allocations $\hat{L}_H^t$ and $\hat{L}_M^t$ needed in equation (D53) are given by noting that $\hat{L}_H^t = -\hat{L}_M^t\frac{L_M^t}{L_H^t}$ and using the MNE's change in labor demand, given the CES combination of high- and low-wage workers:

$$\widehat{WL}_{M,r} = (1-\sigma_M)\left(\hat{W}_{M,r} - \hat{R}_r\right) + \hat{X}_{M,r} \text{ for } r = R,N, \tag{D69}$$

$$\widehat{wL}_{M,r}^t = (1-\rho)\left(\hat{w}_H^t - \hat{W}_M\right) + \widehat{WL}_{M,r} \text{ for } t = l,h; r = R,N, \tag{D70}$$

$$\begin{aligned}\widehat{wL}_M^t = {} & (1-\rho)\left(\hat{w}_H^t - \hat{W}_M\right) + \gamma(1-\sigma_M)\left(\hat{W}_M - \hat{R}_R\right) + (1-\gamma)(1-\sigma_M)\left(\hat{W}_M - \hat{R}_N\right) \\ & + \gamma\hat{X}_{M,R} + (1-\gamma)\hat{X}_{M,N} \text{ for } t = l,h,\end{aligned} \tag{D71}$$

so that overall:

$$\hat{L}_M^t = (1-\gamma)\widehat{wL}_{M,N}^t + \gamma\widehat{wL}_{M,R}^t - \hat{w}_{H,N}^t \text{ for } t = l,h. \tag{D72}$$

Finally, log-differentiating the labor market clearing leads to:

$$\hat{w}_{H,N}^h - \hat{w}_{H,N}^l = \frac{\frac{\phi_{H,R}^h}{1-\phi_M^h}\Phi_{labor}^h\hat{\tau}_{H,R}^l + \frac{1}{\rho}\left(1-\tau_M^{-\rho}\right)\left(1-\Phi_{labor}^h\right)\hat{L}_M^h}{1-\left(1-\tau_M^{-\rho}\right)\left(1-\Phi_{labor}^h\right)}, \tag{D73}$$

where Φ^t_{labor} is the share of workers of type t employed by firms rather than by MNE subsidiaries:

$$\Phi^t_{labor} \equiv \frac{\sum_{j=H,M,F} X_{Hj}/a_j}{\sum_{j=H,M,F} X_{Hj}/a_j + \frac{\chi^t_M}{\chi^t_H}\frac{W_M L_M}{\tau^t_M}}.$$

The change in the welfare for Home workers of type t is given by the solution of the system of equations (D52)–(D73).

Appendix D.6.4 MNEs also implement RS-like labor standards at their subsidiaries

In our baseline specification, we assume that all MNEs pay a wage premium, whether or not they implement RS, and that this premium is equal to the RS premium. Formally, we assume $\tau^t_{M,R} = \tau^t_{M,N} = \tau^t_{H,R}$ for $t = l, h$. Here, we consider an alternative specification in which MNEs adopt the same labor conditions as their suppliers: standard labor conditions for MNEs without RS and RS-like labor conditions for MNEs with RS. Under this assumption, we have:

$$\tau^t_{M,r} = \tau^t_{H,r}, \text{for } t = l, h; r = N, R.$$

In this case, the first-order effect of a small RS policy $\left(\hat{\tau}^l_R, \hat{T}_R, \hat{d}_R\right)$ implemented by a fraction γ of MNEs is different from the one computed in the baseline, as the policy also applies to workers directly employed by MNE subsidiaries. We report here the equations that differ from the welfare equations detailed in Appendix D.4. First, the change in wage indexes is now:

$$\hat{W}_{H,R} - \hat{W}_{H,N} = \chi^l_H \hat{\tau}^l_R,$$
$$\hat{W}_{M,R} - \hat{W}_{M,N} = \chi^l_M \hat{\tau}^l_R,$$

which replace equations (D36) and (D37).

Second, the change in income for a representative worker of type t is:

$$\hat{X}^t_H = \Phi^t \left[\hat{w}^t_{H,N} + \tilde{\phi}^t_{H,R}\left(\hat{\tau}^t_R + \hat{T}_R\right)\right] + \left(1-\Phi^t\right)\left[\hat{w}^t_{H,N} + \hat{L}^t_H + \varphi_{H,R}\left(\hat{\tau}^t_R + \hat{T}_R\right) + \varphi_{HM,R}\hat{a}\frac{a}{a-1}\right],$$

instead of equation (D19), where Φ^t, $\varphi_{H,R}$ and $\varphi_{HM,R}$ are like in the baseline, but the share of total Home wage bill for type t corresponding to entities (suppliers and MNE subsidiaries) that will be subject to RS is now $\tilde{\phi}^t_{H,R}$, with

$$\tilde{\phi}^t_{H,R} \equiv \frac{\sum_{j=H,M,F} X_{Hj,R} + aWL_{M,R}}{\sum_{r=R,N}\left\{\sum_{j=H,M,F} X_{Hj,r} + aWL_{M,r}\right\}}.$$

Finally, log-differentiating the labor market clearing condition leads to:

$$\hat{w}^h_{H,N} - \hat{w}^l_{H,N} = \frac{\sum_{j=H,M,F} X_{Hj,R} + aWL_{M,R}}{\sum_{r=R,N}\left\{\sum_{j=H,M,F} X_{Hj,r} + aWL_{M,r}\right\}}\hat{\tau}^l_R,$$

which replaces equation (D42). Because RS-MNEs now pay the RS premium to their own workers, the MNE labor-allocation equations also change: $\widehat{wL}^t_{M,r} = (1-\rho)\left(\hat{w}^t_{H,N} + \eta_r\hat{\tau}^t_R - \hat{W}_{M,r}\right) + \widehat{WL}_{M,r}$, with $\eta_R = 1$ and $\eta_N = 0$, and $\hat{L}^t_M = (1-\gamma)\,\widehat{wL}^t_{M,N} + \gamma\widehat{wL}^t_{M,R} - \hat{w}^t_{H,N} - \gamma\left(\hat{\tau}^t_R + \hat{T}_R\right)$. The other equations in (D18)–(D42) are unchanged. Together with the substitutions above, these yield the change in welfare for Home workers of type t.

Appendix D.6.5 Extension with multiple sourcing countries

We lay out here the case where the MNE sources from many countries indexed by $i \in \mathcal{I}$, among which Costa Rica (that is, $H \in \mathcal{I}$). The MNE headquarters aggregates input from subsidiaries in a range of countries, with CES aggregator:

$$\mathcal{M} = \left(\sum_{i \in \mathcal{I}} M_i^{\frac{\alpha-1}{\alpha}} \right)^{\frac{\alpha}{\alpha-1}}. \tag{D74}$$

Its cost index is therefore:

$$P_M = \left(\sum_{i \in \mathcal{I}} \varrho_{iF}^{1-\alpha} R_i^{1-\alpha} \right)^{\frac{1}{1-\alpha}},$$

where ϱ_{iF} is the iceberg trade cost from i to F and R_i is the cost index of production in country i. Every country $i \in \mathcal{I}$ is modeled like the Home country of the main text. The RS policy is implemented everywhere, but we assume that it may be more or less easy to implement across countries. That is, countries may already be close to, or far from, the labor standards imposed by RS. To that end, we allow $\tau_{i,R}^l$, the size of the RS cost shock, to be different for different countries $i \in \mathcal{I}$. Other RS characteristics (T_R, d_R, β) are symmetric across countries.

We now recompute the comparative statics for outcomes measured at Home under these assumptions, and compare them to the baseline case. On the final goods market, it is easy to see that the comparative statics remain unchanged, but they are now different on the MNE input market as this market is impacted by RS shocks happening in all sourcing countries. For simplicity of exposition (but without loss of generality for what we aim to establish here), we consider the case of the simplified model described in Section 4. That is, we assume that sourcing countries i do not trade with one another directly ($X_{iF} = X_{Mi} = 0$, $X_{ij} = 0$ for $i \neq j, \{i,j\} \in \mathcal{I}^2$), that MNEs only use local inputs but not local labor directly ($\Xi = 1$) and that extensive margin effects are turned off ($\sigma - 1 \to \theta$).

Following RS, an MNE of type $r = N, R$, sees its final good price change by:

$$\hat{P}_{M,r} = \sum_{i \in \mathcal{I}} s_i^o \hat{R}_{i,r}, \tag{D75}$$

where s_i^o is the share of MNE sourcing done in country i, $s_i^o = \varrho_{iF}^{1-\alpha} R_i^{1-\alpha} / \sum_{j \in \mathcal{I}} \varrho_{jF}^{1-\alpha} R_j^{1-\alpha}$. Similar to our derivation in the baseline, for an MNE of type r, input costs change in each sourcing country i according to:

$$\hat{p}_{iM,r} = \hat{R}_{i,r} = \hat{W}_{i,N} + \beta \chi_i^l \hat{\tau}_{i,r}^l,$$

where $\hat{\tau}_{i,N}^l = 0$ but $\hat{\tau}_{i,R}^l$ can be heterogeneous across countries. Suppliers in country i supply the MNE market and have corresponding sales:

$$y_{iM,r} = p_{iM,r}^{1-\sigma} N_r R_{i,r}^{\sigma} M_{i,r},$$

where N_r is the mass of MNEs with RS status r, and each MNE's production in sourcing country i is given by:

$$R_{i,r} M_{i,r} = \left(\frac{\varrho_{iF} R_{i,r}}{P_{M,r}} \right)^{1-\alpha} \left(\frac{\sigma}{\sigma - 1} \right)^{-\sigma} P_{M,r}^{1-\sigma} d_r D_M, \tag{D76}$$

where D_M is a demand shifter measuring demand for the final good produced by the MNE worldwide ($D_M = \sum_j \varrho_{Fj}^{1-\sigma} X_j P_j^{\sigma-1}$). Combining equations (D75) and (D76), the relative change in the sales of suppliers and MNE subsidiary at Home, relative to those not impacted by RS, are:

$$\hat{y}_{HM,R} - \hat{y}_{HM,N} = (1-\alpha)\, \beta \chi_H^l \hat{\tau}_{H,R}^l + (\alpha - \sigma) \left(\sum_{j \in \mathcal{I}} s_j^o \beta \chi_j^l \hat{\tau}_{j,R}^l \right) + \hat{d}_R,$$

$$\widehat{R_{H,R}M_{H,R}} - \widehat{R_{H,N}M_{H,N}} = (1-\alpha)\,\beta\chi^l_H\hat{\tau}^l_{H,R} + (\alpha-\sigma)\left(\sum_{j\in\mathcal{I}} s^o_j\beta\chi^l_j\hat{\tau}^l_{j,R}\right) + \hat{d}_R,$$

while in the baseline case, we had:

$$\begin{aligned}\hat{y}_{HM,R} - \hat{y}_{HM,N} &= (1-\sigma)\,\beta\chi^l_H\hat{\tau}^l_{H,R} + \hat{d}_R,\\ \widehat{R_{H,R}M_{H,R}} - \widehat{R_{H,N}M_{H,N}} &= (1-\sigma)\,\beta\chi^l_H\hat{\tau}^l_{H,R} + \hat{d}_R.\end{aligned}$$

Comparing the two comparative statics, we see, first, that if $\alpha = \sigma$, then the two models yield exactly the same comparative statics. Second, assume now that $\alpha \neq \sigma$, and that a researcher employs the empirical strategy outlined in this paper, assuming Costa Rica is the sole sourcing country, while the actual data-generating process involves multiple sourcing countries. That is, we assume that from the event study on sales in our empirical strategy, the researcher backs out a residual "demand shock" $\hat{D}_R$:

$$\hat{D}_R \equiv \hat{y}_{HM,R} - \hat{y}_{HM,N} - (1-\sigma)\,\beta\chi^l_H\hat{\tau}^l_{H,R}. \tag{D77}$$

Under the multi-country sourcing model, what this shifter captures is, in fact:

$$\hat{D}_R = (\alpha-\sigma)\left(\sum_{j\in\mathcal{I}} s^o_j\beta\chi^l_j\hat{\tau}^l_{j,R} - \beta\chi^l_H\hat{\tau}^l_{H,R}\right) + \hat{d}_R. \tag{D78}$$

In this model with multiple sourcing countries, the residual shock $\hat{D}_R$ captures both the true RS demand shock $\hat{d}_R$, as well as the fact that the MNE reallocates production between sourcing countries when RS is differently costly across countries. This second effect is captured by the reallocation term $\sum_{j\in\mathcal{I}} s^o_j\beta\chi^l_j\hat{\tau}^l_{j,R} - \beta\chi^l_H\hat{\tau}^l_{H,R}$, which measures the cost change in the average sourcing country relative to Home. First, note that if the RS cost shock $\chi^l_i\hat{\tau}^l_{i,R}$ is not heterogeneous across countries, that term is equal to zero and the empirical strategy corresponding to the baseline model correctly identifies $\hat{D}_R = \hat{d}_R$. Second, the reallocation term is non-zero when Home's cost shock differs from the sourcing-weighted average; heterogeneous shocks across countries do not by themselves imply this. To gauge whether, in our context, the effect of RS is likely very heterogeneous between Home (CR) and other countries on average, we turn to the MNE-level event study. First, return to the equation defining the share of MNE costs coming from MNE subsidiary i (equation (D76)) and define the share of inputs sourced in country i as $s_{i,r} = \frac{R_{i,r}M_{i,r}}{\sum_j R_{j,r}M_{j,r}} = \left(\frac{\varrho_{iF}R_{i,r}}{P_{M,r}}\right)^{1-\alpha}$. Then, after the RS shock, for an RS-MNE compared to a baseline non-RS-MNE subsidiary at Home, we have:

$$\hat{s}_{H,R} - \hat{s}_{H,N} = (1-\alpha)\left(\beta\chi^l_H\hat{\tau}^l_{H,R} - \sum_{j\in\mathcal{I}} s^o_j\beta\chi^l_j\hat{\tau}^l_{j,R}\right).$$

Recall that in our empirical results, we find that this regression estimates a statistical zero: we do not find evidence that the RS event affects subsidiary sales in CR differently from MNE-level sales in the Orbis data (Panel B of Figure 5). This indicates that, on average (across the RS rollouts we consider in our sample), CR is neither more nor less affected than other sourcing countries by the RS shock. Overall, this suggests that CR is close to the average country in terms of how binding RS is, i.e., there is no reallocation of production in or out of CR due to a particularly strong or weak RS policy in CR compared to the other sourcing countries. In this context, we can now come back to the estimation of the demand shock. Given that $\alpha \neq 1$ and $(1-\alpha)\left(\beta\chi^l_H\hat{\tau}^l_{H,R} - \sum_{j\in\mathcal{I}} s^o_j\beta\chi^l_j\hat{\tau}^l_{j,R}\right) = 0$, we find that running the regression corresponding to equation (D77) to estimate the demand shock $\hat{d}_R$ is valid (see equation (D78)) and

recovers:

$$\hat{D}_R = \hat{d}_R,$$

as in the baseline model. To complete this discussion (including for contexts in which this margin may feature more strongly), we move on to the welfare equations to evaluate the corresponding welfare consequences in this extended model with multiple sourcing countries.

Welfare We compute the welfare effects of all MNEs implementing RS in all of their sourcing countries, extending the simplified baseline of equation (16) to a multi-country setting. The expressions for the changes in income and price index, and therefore, the aggregate welfare in Home are unchanged:

$$\hat{U}_H = (1-\lambda_{HH})\,\hat{W}_{H,N} + \beta\lambda_{FH}\chi_H^l\hat{\tau}_{H,R}^l + (\Lambda\lambda_{HH}+\lambda_{FH})\,\hat{T}_R.$$

What changes compared to the baseline model is the trade balance expression that pins down the Home wage index. Given the simple trade patterns we assume, we have:

$$\hat{\lambda}_{FH} + \hat{X}_H = \hat{\lambda}_{M_H F},$$

where the expression for changes in expenditures $\hat{\lambda}_{FH}$ is unchanged and, using equations (D74) and (D76), the total exports from H to F are given by:

$$X_{M_H F} = \left(\frac{\sigma}{\sigma-1}\right)^{-\sigma} \frac{\varrho_{HF}^{1-\alpha} R_H^{1-\alpha}}{P_M^{1-\alpha}} \frac{P_M^{1-\sigma}}{P_F^{1-\sigma}} d_M X_F,$$

so that, in changes, after an RS shock:

$$\hat{\lambda}_{M_H F} = (1-\sigma)\left(\hat{W}_{H,N} + \beta\chi_H^l\hat{\tau}_{H,R}^l\right) - (1-\sigma)\,\hat{P}_F + \underbrace{\hat{d}_R - (\sigma-\alpha)\left(\sum_{i\in\mathcal{I}} s_i^o \beta\chi_i^l\hat{\tau}_{i,R}^l - \beta\chi_H^l\hat{\tau}_{H,R}^l\right)}_{=\hat{D}_R} \tag{D79}$$

$$- (\sigma-\alpha)\left(\sum_{i\in\mathcal{I}} s_i^o\hat{W}_{i,N} - \hat{W}_{H,N}\right).$$

Using that $\hat{D}_R = \hat{d}_R$ in our empirical context, we get:

$$\hat{\lambda}_{M_H F} = (1-\sigma)\left(\hat{W}_{H,N} + \beta\chi_H^l\hat{\tau}_{H,R}^l\right) - (1-\sigma)\,\hat{P}_F + \hat{d}_R - (\sigma-\alpha)\left(\sum_{i\in\mathcal{I}} s_i^o\hat{W}_{i,N} - \hat{W}_{H,N}\right), \tag{D80}$$

while in the baseline case we had:

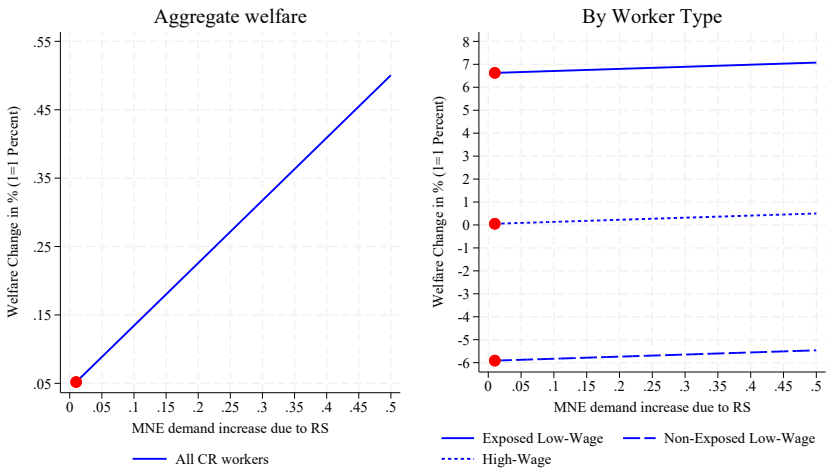
$$\hat{\lambda}_{M_H F} = (1-\sigma)\left(\hat{W}_{H,N} + \beta\chi_H^l\hat{\tau}_{H,R}^l\right) - (1-\sigma)\,\hat{P}_F + \hat{d}_R. \tag{D81}$$

Comparing equations (D80) and (D81), after assuming that $(\sigma-\alpha)\left(\sum_{i\in\mathcal{I}} s_i^o\hat{W}_{i,N} - \hat{W}_{H,N}\right) \sim 0$,[vi] the two models (single-sourcing-country and multiple-sourcing-countries) lead to the same welfare expressions for Home in our empirical context, and the baseline approach correctly backs out the welfare effect of RS, even in a multi-country setting.

[vi] $(\sigma-\alpha)\left(\sum_{i\in\mathcal{I}} s_i^o\hat{W}_{i,N} - \hat{W}_{H,N}\right) \sim 0$ implies that the cross-country reallocation we can observe due to a given RS rollout by an MNE also holds in GE. That is, if we do not observe cross-country reallocation by MNEs due to their own RS rollouts, then the assumption is that those same MNEs also do not reallocate due to broader GE effects of RS.

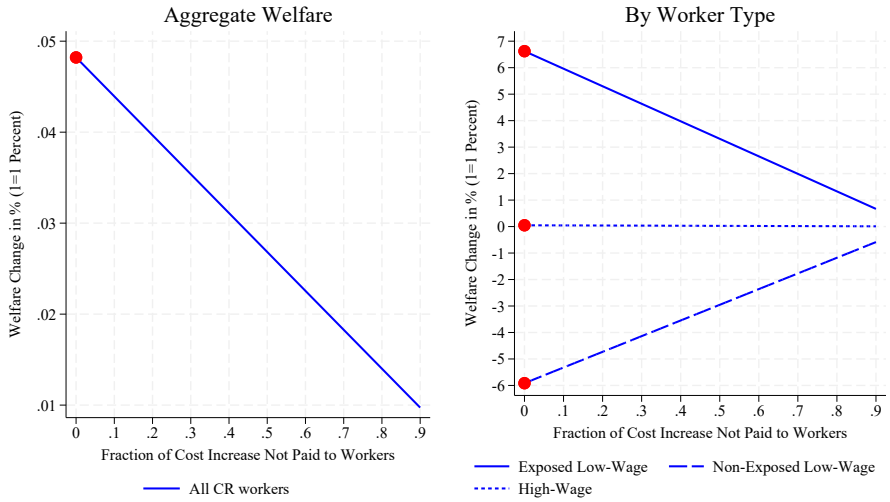
Appendix E Additional counterfactual results

Figure E1: Alternative values of the RS-induced demand shock $\left(\hat{d}_R\right)$



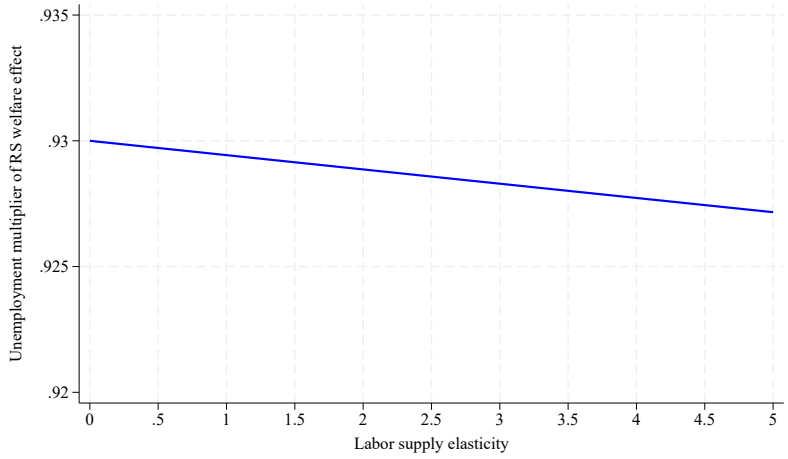
Notes: Red dots indicate our baseline parameter values. As the demand shock increases, MNE production in CR increases. Through the lens of the welfare expressions (16)–(18), this increases domestic welfare of all groups proportionately. See Section 5 for discussion.

Figure E2: Alternative fractions of $\hat{\tau}_R^l$ captured by workers



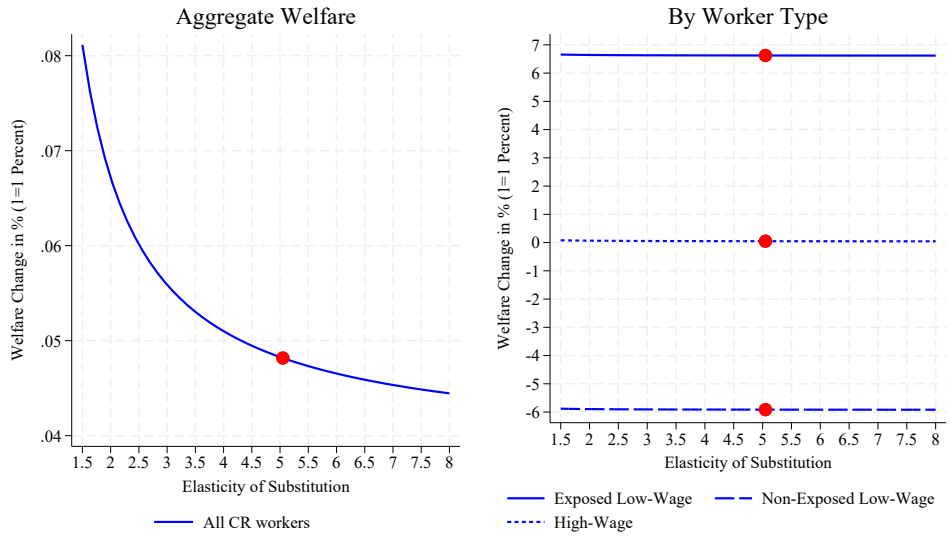
Notes: Red dots indicate our baseline assumption that the increase in labor-related costs is captured by workers through either direct pay or improved workplace amenities. The graphs depict results from the extended welfare expression in the presence of "red tape" in equation (19). See Sections 4 and 5 for discussion.

Figure E3: Welfare implications of RS in a setting with unemployment for different values of the labor supply elasticity (κ)



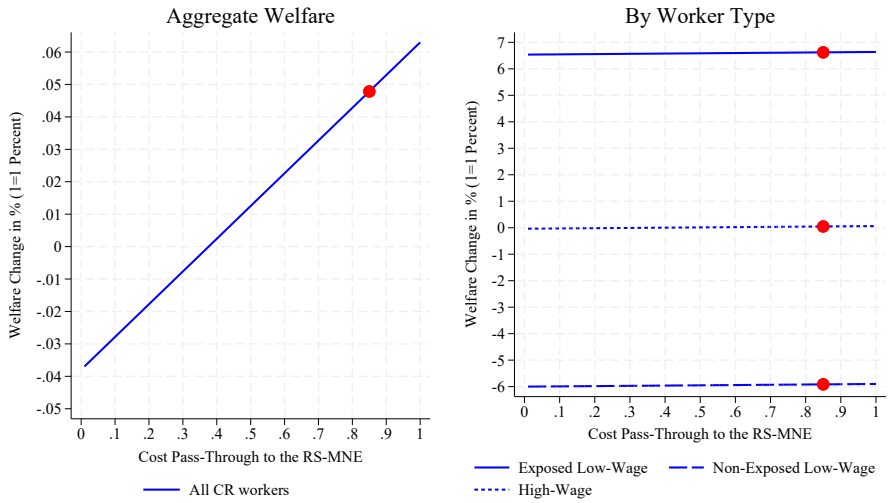
Notes: Figure plots the attenuating factor of the welfare effect of RS in a model with unemployment across different values of the labor supply elasticity κ. The level of unemployment at the beginning of our sample is 7%. See Section 5 for discussion and Appendix D.6.2 for a detailed exposition.

Figure E4: Alternative values of the elasticity of substitution between varieties (σ)



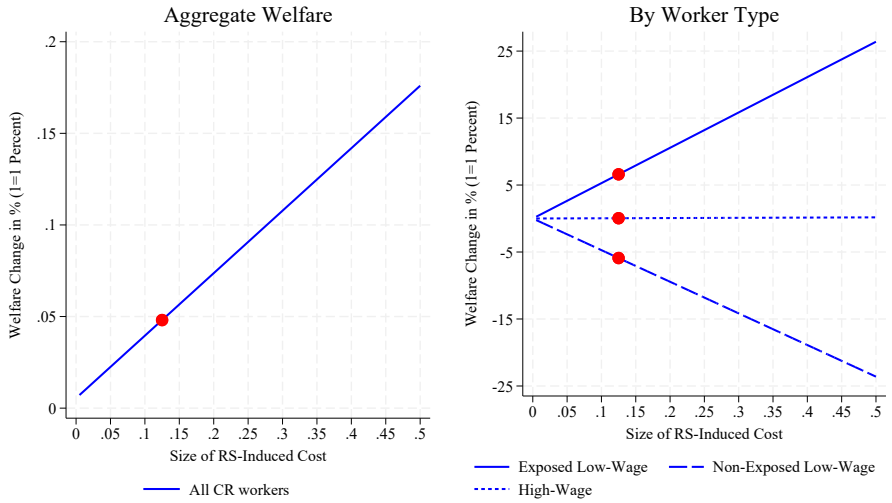
Notes: Figure plots how the welfare effect of RS changes for a reasonable range of values for the elasticity of substitution between varieties. The elasticity σ governs the price elasticity of demand for inputs and final goods, and it matters to welfare in the baseline model through its effect on W^{tax} and W^{d} defined in equation (16). Red dots indicate our baseline parameter values. See Section 5 for discussion.

Figure E5: Alternative values of the pass-through rate (β) of the RS cost increase



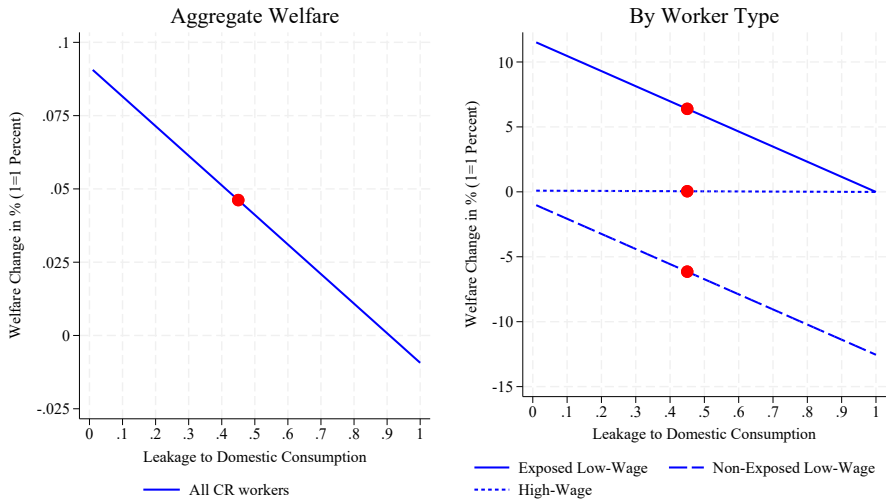
Notes: Figure plots how the welfare effect of RS changes for the full range of possible values for the pass-through rate. When $\beta = 0$, Home suppliers bear the full cost increase from RS policies; when $\beta = 1$, RS cost increases are fully passed through to the input price paid by the MNEs. Red dots indicate our baseline parameter values. See Section 5 for discussion.

Figure E6: Alternative values of the RS wage increase for low-wage workers $\left(\hat{\tau}_R^l\right)$



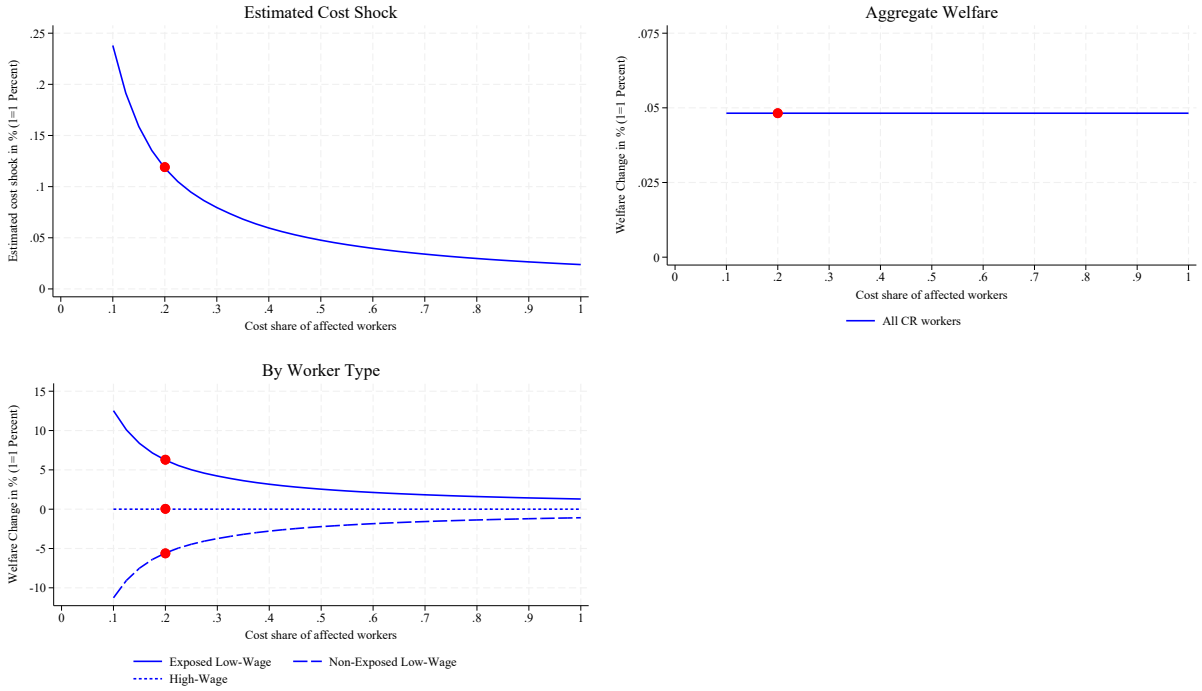
Notes: Figure plots how the welfare effect of RS changes for different values of the wage increase for low-wage workers $\left(\hat{\tau}_R^l\right)$, defined in equation (7). Red dots indicate our baseline parameter values. See Section 5 for discussion.

Figure E7: Alternative values of leakage (Λ)



Notes: Figure plots how the welfare effect of RS changes for the full range of possible values for the share of expenditure on domestic goods that is spent on goods produced by RS-compliant firms (Λ). This parameter measures the degree of "leakage" of RS policies into the domestic price index, and is defined in footnote 35. Red dots indicate our baseline parameter values. See Section 5 for discussion.

Figure E8: Alternative values of the cost share of workers affected by RS $\left(\chi_H^l\right)$



Notes: Figure plots how the welfare effect of RS changes across alternative cost shares of affected workers in suppliers' production $\left(\chi_H^l\right)$. Red dots indicate our baseline parameter values ($\chi_H^l = 0.19$). See Section 5 for discussion.

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